

VECTOR SPACES (CHAPTER 1)

$\mathbb{R}^n$ AND $\mathbb{C}^n$ (SECTION 1A)

LISTS (SUB-SECTION 1A1)

DEF 1.8 (*list, length*):

A list of length n is the same as an n -tuple. We write a list of vectors of length m , $(v_1, \dots, v_m)$, as $v_1, \dots, v_m$ throughout this document without closing parentheses. Since lists are the same as tuples, two lists are only equal if they have the same length and the same elements in the same order.

DEF 1.11 ($\mathbb{F}^n$, *coordinate*):

$$\mathbb{F}^n := \left\{ (x_1, \dots, x_n) : x_k \in \mathbb{F}, k \in \{1, \dots, n\} \right\}$$

DEFINITION OF A VECTOR SPACE (SECTION 1B)

DEF 1.20 (*vector space*):

A “vector space” is a set V along with an addition on V and a scalar multiplication on V such that the following properties hold.

- **COMMUTATIVITY:** $u + v = v + u$, $\forall u, v \in V$
- **ASSOCIATIVITY:** $(u + v) + w = u + (v + w)$ and $(ab)v = a(bv)$, $\forall u, v, w \in V$ and $\forall a, b \in \mathbb{F}$
- **ADDITIVE IDENTITY:** $\exists 0 \in V : v + 0 = v$, $\forall v \in V$.
 $\exists!$ is not a requirement, but a property of vector spaces. See Theorem 1.26.
- **ADDITIVE INVERSE:** $\forall v \in V$, $\exists w \in V : v + w = 0$
- **MULTIPLICATIVE IDENTITY:** $1v = v$, $\forall v \in V$
- **DISTRIBUTIVE PROPERTIES:** $a(u + v) = au + av$, $\forall u, v \in V$ and $(a + b)v = av + bv$, $\forall a, b \in \mathbb{F}$ and $v \in V$.

DEF 1.24 (*notation $\mathbb{F}^S$*):

If S is a set, then $\mathbb{F}^S$ denotes the set of functions from S to $\mathbb{F}$. Let $f, g \in \mathbb{F}^S$ and $\lambda \in \mathbb{F}$:

- The “sum” $f + g \in \mathbb{F}^S$ is defined as follows: $(f + g)(x) := f(x) + g(x)$, $\forall x \in S$
- The “product” $\lambda f \in \mathbb{F}^S$ is defined as follows: $(\lambda f)(x) := \lambda f(x)$, $\forall x \in S$.

The vector space $\mathbb{F}^n := \mathbb{F}^{\{1, 2, \dots, n\}}$ is a special case, because each $(x_1, \dots, x_n) \in \mathbb{F}^n$ can be thought of as a function x from the set $\{1, 2, \dots, n\}$ to $\mathbb{F}$ by writing $x(k)$ instead of x_k for the k^{th} coordinate of $(x_1, \dots, x_n)$.

THM 1.26 (*unique additive identity*):

A vector space has a unique additive identity.

PROOF: Suppose 0 and $0'$ are both additive identities. Then $0' = 0' + 0 = 0 + 0' = 0$. So, $0' = 0$. $\square$

THM 1.27 (*unique additive inverse*):

Every element in a vector space has a unique additive inverse.

PROOF: Let $v \in V$. Suppose $u, w \in V$ are additive inverses of v . Using associativity and the definition of the additive inverse, we get

$$u = u + 0 = u + (v + w) = (u + v) + w = 0 + w = w.$$

Thus, $u = w$, as desired. $\square$

DEF 1.28:

For $v, w \in V$, $-v$ denotes the additive inverse of v and we define subtraction as $w - v := w + (-v)$.

DEF 1.29:

For the rest of this summary, V denotes a vector space over $\mathbb{F}$ most of the time.

THM 1.30 (*the number 0 times a vector*):

$$0v = 0 \quad \forall v \in V.$$

PROOF: Let $v \in V$. We have $0v = (0 + 0)v = 0v + 0v$ by the distributive property. Adding the additive inverse of $0v$ to both sides yields $0 = 0v$, as desired. $\square$

THM 1.31 (*a number times the vector 0*):

$$a0 = 0 \quad \forall a \in \mathbb{F}.$$

PROOF: For every $a \in \mathbb{F}$, we have $a0 = a(0 + 0) = a0 + a0$. Adding the additive inverse of $a0$ to both sides yields $0 = a0$, as desired. $\square$

THM 1.32 (*the number -1 times a vector*):

$$(-1)v = -v \quad \forall v \in V.$$

PROOF: Let $v \in V$. Hence, $v + (-1)v = 1v + (-1)v = (1 + (-1))v = 0v = 0$, using the definition of the multiplicative identity, the distributive property, and Theorem 1.30. This shows that $(-1)v$ is an additive inverse of v , and therefore, $(-1)v = -v$ by the uniqueness of the additive inverse. $\square$

EXERCISES ABOUT DEFINITION OF A VECTOR SPACE 1B

- ① Prove that $-(-v) = v \quad \forall v \in V$.

PROOF: Let $v \in V$. By definition, $v + (-v) = 0$. This means that v is an additive inverse of $-v$. Since additive inverses are unique by ??, and the unique additive inverse of $-v$ is denoted by $-(-v)$, it follows that $-(-v) = v$. $\square$

- ② Suppose that $a \in \mathbb{F}$ and $v \in V$ satisfy $av = 0$. Prove that $a = 0$ or $v = 0$.

PROOF: There are two cases for a . If $a = 0$, we are done. If $a \neq 0$, then multiplying by $\frac{1}{a} \in \mathbb{F}$ gives:

$$av = 0 \implies \frac{1}{a}(av) = \frac{1}{a}0 \implies \left(\frac{1}{a}a\right)v = 0 \implies 1v = 0 \implies v = 0.$$

So, either $a = 0$ or $v = 0$. $\square$

- ③ Suppose $v, w \in V$. Explain why there exists a unique $x \in V$ such that $v + 3x = w$.

SOLUTION: To solve this equation for x , we add the unique additive inverse $-v$ to both sides and multiply by $\frac{1}{3} \in \mathbb{F}$, yielding $x = \frac{1}{3}(w - v)$. Since addition, additive inverses, and scalar multiplication produce unique results, this solution $x \in V$ is unique.

- ④ Show that the empty set is not a vector space.

SOLUTION: A vector space must contain at least one element, namely the additive identity 0 . The empty set contains no elements, and therefore, it cannot be a vector space.

- ⑤ Show that in the definition of a vector space, the additive inverse condition can be replaced with the condition that

$$0v = 0 \quad \forall v \in V.$$

PROOF: Let $v \in V$. Assuming $0v = 0$, the distributive property and multiplicative identity give:

$$\begin{aligned} 0 &= 0v = (1 + (-1))v \\ &= 1v + (-1)v \\ &= v + (-1)v. \end{aligned}$$

Thus, for every $v \in V$, the element $(-1)v$ is an additive inverse of v . □

SUBSPACES (SECTION 1C)**DEF 1.33** (*subspace*):

A subset U of V ($U \subseteq V$) is called a “*subspace*” of V if U is also a vector space with the same additive identity, addition, and scalar multiplication as on V .

THM 1.34 (*conditions for a subspace*):

A subset U of V is a subspace of V if and only if U satisfies the following three conditions:

- **ADDITIVE IDENTITY:** $0 \in U$
- **CLOSED UNDER ADDITION:** $u, w \in U \implies u + w \in U$
- **CLOSED UNDER SCALAR MULTIPLICATION:** $a \in \mathbb{F}$ and $u \in U \implies au \in U$

The additive identity condition above could be replaced by the condition that U is nonempty (because taking $u \in U$ and multiplying it by 0 implies that $0 \in U$).

PROOF: We need to show the proposition in two directions.

$\Rightarrow$ **DIRECTION:** If U is a subspace of V , then U satisfies the three conditions above by the definition of a vector space.

$\Leftarrow$ **DIRECTION:** The first condition ensures that the additive identity 0 of V is in U . The other two conditions ensure that addition and scalar multiplication are well-defined operations on U . By Theorem 1.32 and closure under scalar multiplication, every $u \in U$ has an additive inverse $-u = (-1)u \in U$. The other axioms of a vector space (associativity, commutativity, and the distributive properties) are automatically inherited by U from V . Thus, U is a vector space, and hence, a subspace of V . $\square$

SUM OF SUBSPACES (SUB-SECTION 1C1)**DEF 1.36** (*sum of subspaces*):

If $V_1, \dots, V_m$ are subspaces of V , the “*sum*” of $V_1, \dots, V_m$, denoted by $V_1 + \dots + V_m$, is the set of all possible sums of elements of $V_1, \dots, V_m$. More precisely,

$$V_1 + \dots + V_m := \{v_1 + \dots + v_m \mid v_1 \in V_1, \dots, v_m \in V_m\}.$$

THM 1.40 (*sum of subspaces is the smallest containing subspace*):

If $V_1, \dots, V_m$ are subspaces of V , then $V_1 + \dots + V_m$ is the smallest subspace of V containing $V_1, \dots, V_m$.

PROOF: We first prove that $W := \sum_{k=1}^m V_k$ is a subspace of V , and then show that it is the smallest such subspace containing each V_k . Since $0 \in V_k$ for every $k \in \{1, \dots, m\}$, we have $0 = 0 + \dots + 0 \in W$. If

$u = \sum_{k=1}^m v_k \in W$ and $w = \sum_{k=1}^m v'_k \in W$ with $v_k, v'_k \in V_k$, then

$$u + w = \sum_{k=1}^m (v_k + v'_k) \in W, \quad \text{since each } V_k \text{ is closed under addition.}$$

For any $\lambda \in \mathbb{F}$,

$$\lambda u = \lambda \sum_{k=1}^m v_k = \sum_{k=1}^m \underbrace{\lambda v_k}_{\in V_k} \in W, \quad \text{as each } V_k \text{ is closed under scalar multiplication.}$$

Thus, W contains 0, is closed under addition and scalar multiplication, and hence, is a subspace of V by Theorem 1.34. Now, we show that W is contained in every subspace of V containing $V_1, \dots, V_m$. The subspaces $V_1, \dots, V_m$ are all contained in W , i.e., $V_k \subseteq W$ for every $k \in \{1, \dots, m\}$, because every $v \in V_k$ can be written as $v = 0 + \dots + v + \dots + 0$.

Conversely, every subspace of V containing $V_1, \dots, V_m$ must contain all sums of their elements because subspaces are closed under addition. Therefore, every such subspace contains W ,

showing that W is the smallest subspace of V containing $V_1, \dots, V_m$. $\square$

DIRECT SUMS (SUB-SECTION 1C2)

DEF 1.41 (*direct sum, $\oplus$*):

$V_1 + \dots + V_m$ is called a “direct sum” if each element of $V_1 + \dots + V_m$ can be written in only one way as a sum $v_1 + \dots + v_m$, where each $v_k \in V_k$. In this case, we write

$$V_1 + \dots + V_m = V_1 \oplus \dots \oplus V_m$$

to denote that $V_1 + \dots + V_m$ is a direct sum.

EXAMPLE 1.42:

$$\mathbb{F}^3 = \left\{ (x, y, 0) \in \mathbb{F}^3 \mid x, y \in \mathbb{F} \right\} \oplus \left\{ (0, 0, z) \in \mathbb{F}^3 \mid z \in \mathbb{F} \right\}.$$

THM 1.45 (*condition for a direct sum*):

$\sum_{k=1}^m V_k$ is a direct sum (i.e., $\sum_{k=1}^m V_k = \bigoplus_{k=1}^m V_k$) if and only if the only way to express 0 as $v_1 + \dots + v_m$ with $v_k \in V_k$ is $v_1 = \dots = v_m = 0$.

PROOF: $\Rightarrow$ **DIRECTION:** First, suppose $\sum_{k=1}^m V_k$ is a direct sum. By the uniqueness of decompositions, the only way to write $0 = v_1 + \dots + v_m$ with $v_k \in V_k$ is $v_1 = \dots = v_m = 0$.

$\Leftarrow$ **DIRECTION:** Second, suppose that the only way to write 0 as a sum $w_1 + \dots + w_m$ with $w_k \in V_k$ is by taking each $w_k = 0$. Let $v \in \sum_{k=1}^m V_k$, with

$$v = v_1 + \dots + v_m = u_1 + \dots + u_m, \quad v_k, u_k \in V_k \text{ (for } k = 1, \dots, m).$$

Subtracting gives

$$0 = \underbrace{(v_1 - u_1)}_{\in V_1} + \dots + \underbrace{(v_m - u_m)}_{\in V_m}.$$

By hypothesis, each summand must be zero, so $v_1 = u_1, \dots, v_m = u_m$. Hence, every element has a unique decomposition, so $\sum_{k=1}^m V_k$ is direct. $\square$

THM 1.46 (*sum and intersection of two subspaces*):

$U + W$ is a direct sum (i.e., $U + W = U \oplus W$) if and only if $U \cap W = \{0\}$.

PROOF: Let U and W be subspaces of V .

$\Rightarrow$ **DIRECTION:** First, suppose $U + W$ is a direct sum. If $v \in U \cap W$, then $v \in U$ and $-v \in W$. We also have

$$0 = v + (-v), \text{ where } v \in U \text{ and } (-v) \in W.$$

Because $U + W$ is a direct sum and $0 = 0 + 0$ with $0 \in U$ and $0 \in W$, the uniqueness of representation implies that $v = 0$. Therefore, $U \cap W = \{0\}$.

$\Leftarrow$ **DIRECTION:** Suppose $U \cap W = \{0\}$. Let $u \in U$ and $w \in W$ be such that

$$0 = u + w. \tag{1.1}$$

By Theorem 1.45, it suffices to show that $u = w = 0$. The equation (1.1) implies that $u = -w \in W$ (since W is a subspace) and $w = -u \in U$ (since U is a subspace). Thus, $u, w \in U \cap W$. Since $U \cap W = \{0\}$, it follows that $u = w = 0$. Therefore, $U + W$ is a direct sum. $\square$

EXERCISES ABOUT SUBSPACES 1C

① For each of the following subsets of $\mathbb{F}^3$, determine whether it is a subspace of $\mathbb{F}^3$.

(a) $\left\{ (x_1, x_2, x_3) \in \mathbb{F}^3 : x_1 + 2x_2 + 3x_3 = 0 \right\}$

- (b) $\{(x_1, x_2, x_3) \in \mathbb{F}^3 : x_1 + 2x_2 + 3x_3 = 4\}$
 (c) $\{(x_1, x_2, x_3) \in \mathbb{F}^3 : x_1 x_2 x_3 = 0\}$
 (d) $\{(x_1, x_2, x_3) \in \mathbb{F}^3 : x_1 = 5x_3\}$

SOLUTION:

- (a) Let $E := \{(x_1, x_2, x_3) \in \mathbb{F}^3 \mid x_1 + 2x_2 + 3x_3 = 0\}$. We check the subspace criteria: The additive identity $(0, 0, 0) \in E$ since $0 + 2 \cdot 0 + 3 \cdot 0 = 0$. For $u, v \in E$, we have $u_1 + 2u_2 + 3u_3 = 0$ and $v_1 + 2v_2 + 3v_3 = 0$, so $(u_1 + v_1) + 2(u_2 + v_2) + 3(u_3 + v_3) = 0$, hence $u + v \in E$. For $\lambda \in \mathbb{F}$ and $u \in E$, $\lambda u_1 + 2\lambda u_2 + 3\lambda u_3 = \lambda(u_1 + 2u_2 + 3u_3) = 0$, hence $\lambda u \in E$. Therefore, E is a subspace of $\mathbb{F}^3$ by Theorem 1.34.
- (b) It is not a subspace because it does not contain the additive identity $(0, 0, 0)$.
- (c) Let $U := \{(x_1, x_2, x_3) \in \mathbb{F}^3 \mid x_1 x_2 x_3 = 0\}$. Then U is not closed under addition:
 $(1, 1, 0) \in U$ and $(0, 0, 1) \in U$, but $(1, 1, 0) + (0, 0, 1) = (1, 1, 1) \notin U$.
 Therefore, U is not a subspace of $\mathbb{F}^3$.
- (d) Let $U := \{(x_1, x_2, x_3) \in \mathbb{F}^3 \mid x_1 = 5x_3\}$. Then $(0, 0, 0) \in U$. If $u, v \in U$, then $(u_1 + v_1) = 5u_3 + 5v_3 = 5(u_3 + v_3)$, so $u + v \in U$. If $\lambda \in \mathbb{F}$, then $\lambda u_1 = \lambda(5u_3) = 5(\lambda u_3)$, so $\lambda u \in U$. Hence, U is a subspace of $\mathbb{F}^3$.

② Verify all assertions about subspaces in example 1.35.

- (a) If $b \in \mathbb{F}$, then $U := \{(x_1, x_2, x_3, x_4) \in \mathbb{F}^4 \mid x_3 = 5x_4 + b\}$ is a subspace of $\mathbb{F}^4$ if and only if $b = 0$.

SOLUTION: If U is a subspace of $\mathbb{F}^4$, then $(0, 0, 0, 0) \in U$, which requires $0 = 5(0) + b \implies b = 0$. Conversely, if $b = 0$, then $(0, 0, 0, 0) \in U$. For $u, v \in U$, $(u_3 + v_3) = 5u_4 + 5v_4 = 5(u_4 + v_4)$, so $u + v \in U$. For $\lambda \in \mathbb{F}$, $\lambda u_3 = \lambda(5u_4) = 5(\lambda u_4)$, so $\lambda u \in U$. Thus, U is a subspace of $\mathbb{F}^4$ if and only if $b = 0$.

- (b) The set of continuous real-valued functions on the interval $[0, 1]$ is a subspace of $\mathbb{R}^{[0,1]}$.

SOLUTION: The constant function 0 is continuous. The sum of two continuous functions is continuous, and any scalar multiple of a continuous function is continuous. Therefore, the set of continuous real-valued functions on $[0, 1]$ is a subspace of $\mathbb{R}^{[0,1]}$.

- (c) The set of differentiable real-valued functions on $\mathbb{R}$ is a subspace of $\mathbb{R}^{\mathbb{R}}$.

SOLUTION: We check the three subspace properties:

ADDITIVE IDENTITY: The zero function $f(x) = 0$ for all $x \in \mathbb{R}$ is differentiable with derivative $f'(x) = 0$.

CLOSEDNESS UNDER ADDITION: If f and g are differentiable, then $(f + g)'(x) = f'(x) + g'(x)$ for all $x \in \mathbb{R}$, so $f + g$ is differentiable.

CLOSEDNESS UNDER SCALAR MULTIPLICATION: If f is differentiable and $c \in \mathbb{R}$, then $(cf)'(x) = cf'(x)$ for all $x \in \mathbb{R}$, so cf is differentiable.

- (d) The set S of differentiable real-valued functions f on the interval $(0, 3)$ such that $f'(2) = b$ is a subspace of $\mathbb{R}^{(0,3)}$ if and only if $b = 0$.

SOLUTION: For the zero function $h(x) = 0$, we have $h'(x) = 0$. The condition $h'(2) = 0 = b$ holds if and only if $b = 0$. If $b = 0$, then $0 \in S$. For $f, g \in S$, $(f + g)'(2) = f'(2) + g'(2) = 0 + 0 = 0$, so $f + g \in S$. For $c \in \mathbb{R}$, $(cf)'(2) = cf'(2) = c \cdot 0 = 0$, so $cf \in S$. Thus, S is a subspace if and only if $b = 0$.

③ Show that the set S of differentiable real-valued functions f on the interval $(-4, 4)$ such that $f'(-1) = 3f(2)$ is a subspace of $\mathbb{R}^{(-4,4)}$.

SOLUTION: We check all three subspace properties:

ADDITIVE IDENTITY: For the zero function $f(x) = 0$, we have $f'(x) = 0$, so $f'(-1) = 0 = 3(0) = 3f(2)$, which implies $0 \in S$.

CLOSED UNDER ADDITION: If $f, g \in S$ and $h := f + g$, then $h'(-1) = f'(-1) + g'(-1) = 3f(2) + 3g(2) = 3(f(2) + g(2)) = 3h(2)$. Therefore, $h \in S$.

CLOSED UNDER SCALAR MULTIPLICATION: If $f \in S$, $c \in \mathbb{R}$, and $h := cf$, then $h'(-1) = cf'(-1) = c(3f(2)) = 3(cf(2)) = 3h(2)$. Therefore, $h \in S$.

Hence, S is a subspace of $\mathbb{R}^{(-4,4)}$.

- 8 Prove or give a counterexample: If U is a nonempty subset of $\mathbb{R}^2$ such that U is closed under addition and under taking additive inverses (meaning $-u \in U$ whenever $u \in U$), then U is a subspace of $\mathbb{R}^2$.

SOLUTION: The statement is false. Consider $U := \mathbb{Z}^2 = \{(x, y) \in \mathbb{R}^2 \mid x, y \in \mathbb{Z}\} \subset \mathbb{R}^2$. Then U is nonempty, closed under addition, and closed under additive inverses. However, if $\lambda \in \mathbb{R} \setminus \mathbb{Z}$, then for any non-zero $(a, b) \in U$,

$$\lambda(a, b) = (\lambda a, \lambda b) \notin U.$$

For example, $(1, 1) \in U$ and $\frac{1}{2} \in \mathbb{R}$, but $\frac{1}{2}(1, 1) = (\frac{1}{2}, \frac{1}{2}) \notin U$. Therefore, U is not a subspace of $\mathbb{R}^2$.

- 12 Prove that the union $U \cup W$ of two subspaces U, W of V is a subspace of V if and only if one of the subspaces is contained in the other (i.e., $U \subseteq W$ or $W \subseteq U$).

PROOF: $\Leftarrow$ **DIRECTION:** Let U and W be two subspaces of V . If $U \subseteq W$, then $U \cup W = W$, which is a subspace of V . Similarly, if $W \subseteq U$, then $U \cup W = U$, which is a subspace of V .

$\Rightarrow$ **DIRECTION:** Assume that $U \cup W$ is a subspace of V , where U and W are subspaces of V .

PROOF BY CONTRADICTION: Suppose $U \not\subseteq W$ and $W \not\subseteq U$. Therefore,

$$\exists u \in U \text{ with } u \notin W, \quad \text{and} \quad \exists w \in W \text{ with } w \notin U.$$

Since $u, w \in U \cup W$ and $U \cup W$ is a subspace, we have $u + w \in U \cup W$. Thus, $u + w \in U$ or $u + w \in W$.

- **CASE** $u + w \in U$: Then $w = (u + w) - u \in U$, contradicting $w \notin U$.
- **CASE** $u + w \in W$: Then $u = (u + w) - w \in W$, contradicting $u \notin W$.

In both cases, we obtain a contradiction. Therefore, $U \subseteq W$ or $W \subseteq U$.

ALTERNATIVE DIRECT PROOF: If $U \subseteq W$, the disjunction is immediately true. So suppose $U \not\subseteq W$. Then there exists $u \in U$ such that $u \notin W$. Let $w \in W$ be arbitrary. Since $u, w \in U \cup W$ and $U \cup W$ is a subspace, $u + w \in U \cup W$. Hence, $u + w \in U$ or $u + w \in W$. If $u + w \in U$, then $w = (u + w) - u \in U$, which implies $w \in U$. If $u + w \in W$, then $u = (u + w) - w \in W$, which contradicts $u \notin W$. Therefore, it must be that $u + w \in U$, which forces $w \in U$. Since $w \in W$ was arbitrary, $W \subseteq U$. Thus, $U \subseteq W$ or $W \subseteq U$. $\square$

- 23 Prove or give a counterexample: If V_1, V_2, U are subspaces of V such that $V = V_1 \oplus U$ and $V = V_2 \oplus U$, then $V_1 = V_2$.

SOLUTION: The statement is false. As a counterexample, consider $V := \mathbb{R}^2$ and

$$U := \{(x, 0) \in \mathbb{R}^2 \mid x \in \mathbb{R}\},$$

which is the x -axis. Let V_1 and V_2 be defined as follows:

$$V_1 := \{(0, y) \in \mathbb{R}^2 \mid y \in \mathbb{R}\} \quad \text{and} \quad V_2 := \{(x, x) \in \mathbb{R}^2 \mid x \in \mathbb{R}\}.$$

Then $V_1 \cap U = \{(0, 0)\}$ and $V_2 \cap U = \{(0, 0)\}$. Furthermore, every $(x, y) \in \mathbb{R}^2$ can be uniquely decomposed as:

$$(x, y) = (x, 0) + (0, y) \in U \oplus V_1 \quad \text{and} \quad (x, y) = (x - y, 0) + (y, y) \in U \oplus V_2.$$

Thus, $V = V_1 \oplus U = V_2 \oplus U$, but $V_1 \neq V_2$.

FINITE DIMENSIONAL VECTOR SPACES (CHAPTER 2)

SPAN AND LINEAR INDEPENDENCE (SECTION 2A)

LINEAR COMBINATIONS AND SPAN (SUB-SECTION 2A1)

DEF 2.2 (*linear combination*):

A “linear combination” of a list of vectors $v_1, \dots, v_m$ in V is a vector of the form $a_1v_1 + \dots + a_mv_m$, where $a_1, \dots, a_m \in \mathbb{F}$.

DEF 2.4 (*span of a list of vectors*):

The “span” of a list of vectors $v_1, \dots, v_m$ in V is defined as follows:

$$\text{span}(v_1, \dots, v_m) := \{a_1v_1 + \dots + a_mv_m \mid a_1, \dots, a_m \in \mathbb{F}\},$$

$$\text{span}() := \{0\} \text{ (the span of the empty list } () \text{ is defined to be } \{0\}) .$$

The definition also implies that $\text{span}(0) = \{0\}$.

THM 2.6 (*the span of a list of vectors is a subspace*):

The span of a list of vectors in V is the smallest subspace of V containing all vectors in the list.

PROOF: Let $v_1, \dots, v_m \in V$. We have

$$0 = 0v_1 + \dots + 0v_m \in \text{span}(v_1, \dots, v_m),$$

and hence, the additive identity is in $\text{span}(v_1, \dots, v_m)$.

$\text{span}(v_1, \dots, v_m)$ is closed under addition, because for all $a_1, \dots, a_m, c_1, \dots, c_m \in \mathbb{F}$, we have

$$(a_1v_1 + \dots + a_mv_m) + (c_1v_1 + \dots + c_mv_m) = (a_1 + c_1)v_1 + \dots + (a_m + c_m)v_m \in \text{span}(v_1, \dots, v_m).$$

Furthermore, $\text{span}(v_1, \dots, v_m)$ is closed under scalar multiplication because for any $\lambda \in \mathbb{F}$,

$$\lambda(a_1v_1 + \dots + a_mv_m) = (\lambda a_1)v_1 + \dots + (\lambda a_m)v_m \in \text{span}(v_1, \dots, v_m).$$

Thus, $\text{span}(v_1, \dots, v_m)$ is a subspace of V by Theorem 1.34. Every subspace of V that contains each v_k is closed under scalar multiplication and addition, and therefore, contains $\text{span}(v_1, \dots, v_m)$.

Hence, $\text{span}(v_1, \dots, v_m)$ is the smallest subspace of V containing each $v_1, \dots, v_m$. $\square$

DEF 2.7 (*spanning a vector space*):

If $V = \text{span}(v_1, \dots, v_m)$, we say $v_1, \dots, v_m$ “spans” V .

DEF 2.9 (*finite-dimensional vector space*):

A vector space V is called “finite-dimensional” if some list of vectors in V spans V .

DEF 2.10 (*polynomial, $\mathcal{P}(\mathbb{F})$*):

A function $p : \mathbb{F} \rightarrow \mathbb{F}$ is called a “polynomial” with coefficients in $\mathbb{F}$ if there exist $a_0, \dots, a_m \in \mathbb{F}$ such that

$$p(z) = a_0 + a_1z + a_2z^2 + \dots + a_mz^m \quad \forall z \in \mathbb{F}.$$

The set of all polynomials with coefficients in $\mathbb{F}$ is denoted by $\mathcal{P}(\mathbb{F})$. It is a subspace of $\mathbb{F}^{\mathbb{F}}$.

DEF 2.11 (*degree of a polynomial, $\deg p$*):

A polynomial $p \in \mathcal{P}(\mathbb{F})$ is said to have “degree” m , written $\deg p = m$, if there exist $a_0, \dots, a_m \in \mathbb{F}$ with $a_m \neq 0$ such that

$$p(z) = a_0 + a_1z + \dots + a_mz^m \quad \forall z \in \mathbb{F}.$$

If p is the zero polynomial, then we define $\deg 0 := -\infty$.

DEF 2.12 (*the set of polynomials of degree at most m*):

For $m \geq 0$, $\mathcal{P}_m(\mathbb{F}) := \text{span}(1, z, \dots, z^m)$ denotes the subspace of $\mathcal{P}(\mathbb{F})$ consisting of all polynomials with degree at most m .

DEF 2.13 (*infinite-dimensional*):

A vector space is called “infinite-dimensional” if it is not finite-dimensional.

LINEAR INDEPENDENCE (SUB-SECTION 2A2)

DEF 2.15 (*linear independence*):

A list of vectors $v_1, \dots, v_m$ in V is called “linearly independent” if the only choice of scalars $a_1, \dots, a_m \in \mathbb{F}$ that satisfies

$$a_1 v_1 + \dots + a_m v_m = 0$$

is $a_1 = \dots = a_m = 0$.

The empty list $()$ is also declared to be linearly independent.

DEF 2.16 (*linear dependence*):

A list of vectors $v_1, \dots, v_m$ in V is called “linearly dependent” if it is not linearly independent. That is, there exist scalars $a_1, \dots, a_m \in \mathbb{F}$, not all zero, such that

$$a_1 v_1 + \dots + a_m v_m = 0.$$

THM 2.19 (*linear dependence lemma*):

Suppose the list $v_1, \dots, v_m \in V$ is linearly dependent. Then there is an index $k \in \{1, \dots, m\}$ such that:

- (a) $v_k \in \text{span}(v_1, \dots, v_{k-1})$.
- (b) If the k^{th} term is removed from $v_1, \dots, v_m$, then the span of the remaining list equals $\text{span}(v_1, \dots, v_m)$:

$$\text{span}(v_1, \dots, v_{k-1}, v_{k+1}, \dots, v_m) = \text{span}(v_1, \dots, v_m).$$

PROOF: Because the list $v_1, \dots, v_m$ is linearly dependent, there exist $a_1, \dots, a_m \in \mathbb{F}$, not all zero, such that

$$a_1 v_1 + \dots + a_m v_m = 0.$$

Let k be the largest element of $\{1, \dots, m\}$ such that $a_k \neq 0$. Then

$$v_k = -\frac{a_1}{a_k} v_1 - \dots - \frac{a_{k-1}}{a_k} v_{k-1},$$

which proves that $v_k \in \text{span}(v_1, \dots, v_{k-1})$.

To prove the second assertion, suppose $u \in \text{span}(v_1, \dots, v_m)$. Then there exist $c_1, \dots, c_m \in \mathbb{F}$ such that

$$u = c_1 v_1 + \dots + c_m v_m. \tag{2.1}$$

Since $v_k \in \text{span}(v_1, \dots, v_{k-1})$, we can write $v_k = b_1 v_1 + \dots + b_{k-1} v_{k-1}$ for some $b_1, \dots, b_{k-1} \in \mathbb{F}$.

Substituting this into (2.1) yields:

$$\begin{aligned} u &= c_1 v_1 + \dots + c_k (b_1 v_1 + \dots + b_{k-1} v_{k-1}) + \dots + c_m v_m \\ &= (c_1 + c_k b_1) v_1 + \dots + (c_{k-1} + c_k b_{k-1}) v_{k-1} + c_{k+1} v_{k+1} + \dots + c_m v_m, \end{aligned}$$

which shows that $u \in \text{span}(v_1, \dots, v_{k-1}, v_{k+1}, \dots, v_m)$. Therefore,

$$\text{span}(v_1, \dots, v_{k-1}, v_{k+1}, \dots, v_m) = \text{span}(v_1, \dots, v_m),$$

as desired. □

THM 2.22 (*length of linearly independent list $\leq$ length of spanning list*):

In a finite-dimensional vector space, the length of every linearly independent list of vectors is less

than or equal to the length of every spanning list of vectors.

PROOF: Let $u_1, \dots, u_m \in V$ be a linearly independent list and let $w_1, \dots, w_n \in V$ be a list that spans V . We need to prove that $m \leq n$. We construct, in m steps, a spanning list of size n that incorporates $u_1, \dots, u_m$. Start with

$$B_0 := (w_1, \dots, w_n), \quad V = \text{span}(B_0).$$

STEP 1: Add u_1 to B_0 , obtaining the list $(u_1, w_1, \dots, w_n)$. Since $u_1 \in V = \text{span}(B_0)$, this enlarged list is linearly dependent. By Theorem 2.19, one of the vectors is in the span of the previous vectors; since $u_1 \neq 0$ (as the u 's are linearly independent), this vector must be one of the w 's, say w_j . Removing w_j gives a new list B_1 of length n with $u_1 \in B_1$ and $V = \text{span}(B_1)$.

STEP k (FOR $k = 2, \dots, m$): The list B_{k-1} from the previous step spans V and contains $u_1, \dots, u_{k-1}$ along with $n - (k - 1)$ of the original w 's. Adding u_k to the front of the w 's gives a list of length $n + 1$:

$$B_k^* := (u_1, \dots, u_k, \tilde{w}_1, \dots, \tilde{w}_{n-(k-1)}), \quad V = \text{span}(B_k^*).$$

Since $u_k \in V = \text{span}(B_{k-1})$, the list B_k^* is linearly dependent. By Theorem 2.19, some vector in B_k^* is in the span of the previous vectors. Because $u_1, \dots, u_k$ are linearly independent, this vector cannot be one of the u 's, so it must be one of the $\tilde{w}$'s, say $\tilde{w}_j$. Removing $\tilde{w}_j$ leaves a list B_k of length n containing $u_1, \dots, u_k$ such that $V = \text{span}(B_k)$.

CONCLUSION: After m steps, we have a list B_m of length n containing all m independent vectors $u_1, \dots, u_m$. Therefore, $m \leq n$. $\square$

THM 2.25 (finite-dimensional subspace):

Every subspace of a finite-dimensional vector space is finite-dimensional.

PROOF: Suppose V is finite-dimensional, and let U be a subspace of V . Since V is finite-dimensional, there is a spanning list of length n for V .

STEP 1: If $U = \{0\}$, then U is finite-dimensional, and we are done. Otherwise, choose a non-zero vector $u_1 \in U$.

STEP k ($k \geq 2$): If $U = \text{span}(u_1, \dots, u_{k-1})$, then U is finite-dimensional, and we are done. If $U \neq \text{span}(u_1, \dots, u_{k-1})$, choose a vector $u_k \in U$ such that

$$u_k \notin \text{span}(u_1, \dots, u_{k-1}).$$

After each step, the list $u_1, \dots, u_k$ is linearly independent by Theorem 2.19. By Theorem 2.22, the length of any linearly independent list in V cannot exceed n . Thus, the process must terminate in at most n steps, which means $U = \text{span}(u_1, \dots, u_k)$ for some $k \leq n$. Hence, U is finite-dimensional. $\square$

EXERCISES ABOUT SPAN AND LINEAR INDEPENDENCE 2A

- ① Find a list of four distinct vectors in $\mathbb{F}^3$ whose span equals

$$U := \{(x, y, z) \in \mathbb{F}^3 \mid x + y + z = 0\}.$$

SOLUTION: The list $(1, -1, 0), (0, 1, -1), (1, 0, -1), (2, -1, -1)$ consists of four distinct vectors in U . The first two vectors span U because any $(x, y, z) \in U$ satisfies $z = -x - y$, so $(x, y, z) = x(1, -1, 0) + (x + y)(0, 1, -1)$. Since the other two vectors also belong to U , the span of all four vectors is precisely U .

- ② Prove that if v_1, v_2, v_3, v_4 spans V , then the list $v_1 - v_2, v_2 - v_3, v_3 - v_4, v_4$ also spans V .

PROOF: Notice that

$$\begin{aligned} v_1 &= (v_1 - v_2) + (v_2 - v_3) + (v_3 - v_4) + v_4, \\ v_2 &= (v_2 - v_3) + (v_3 - v_4) + v_4, \\ v_3 &= (v_3 - v_4) + v_4, \\ v_4 &= v_4. \end{aligned}$$

Thus, each $v_k \in \text{span}(v_1 - v_2, v_2 - v_3, v_3 - v_4, v_4)$ for $k = 1, 2, 3, 4$. Since v_1, v_2, v_3, v_4 spans V , it follows that

$$V = \text{span}(v_1, v_2, v_3, v_4) \subseteq \text{span}(v_1 - v_2, v_2 - v_3, v_3 - v_4, v_4) \subseteq V.$$

Hence, $v_1 - v_2, v_2 - v_3, v_3 - v_4, v_4$ spans V . □

- ③ Prove that if $v_1, \dots, v_m \in V$ and

$$w_k := v_1 + \dots + v_k \quad \text{for } k = 1, \dots, m,$$

then $\text{span}(v_1, \dots, v_m) = \text{span}(w_1, \dots, w_m)$.

PROOF: By definition, each $w_k = v_1 + \dots + v_k$ is a linear combination of $v_1, \dots, v_m$, and therefore, $w_k \in \text{span}(v_1, \dots, v_m)$. Consequently,

$$\text{span}(w_1, \dots, w_m) \subseteq \text{span}(v_1, \dots, v_m). \quad (2.2)$$

Conversely, observe that

$$\begin{aligned} v_1 &= w_1, \\ v_2 &= w_2 - w_1, \\ v_3 &= w_3 - w_2, \\ &\vdots \\ v_m &= w_m - w_{m-1}. \end{aligned}$$

Thus, each $v_k \in \text{span}(w_1, \dots, w_m)$ for $k = 1, \dots, m$. It follows that

$$\text{span}(v_1, \dots, v_m) \subseteq \text{span}(w_1, \dots, w_m). \quad (2.3)$$

Combining (2.2) and (2.3), we conclude that $\text{span}(w_1, \dots, w_m) = \text{span}(v_1, \dots, v_m)$. □

- ①⑥ Explain why no list of four polynomials spans $\mathcal{P}_4(\mathbb{F})$.

SOLUTION: The list $1, z, z^2, z^3, z^4$ in $\mathcal{P}_4(\mathbb{F})$ is linearly independent and has length 5. By Theorem 2.22, the length of every linearly independent list is less than or equal to the length of every spanning list. Therefore, every spanning list of $\mathcal{P}_4(\mathbb{F})$ must contain at least 5 polynomials, so no list of four polynomials can span $\mathcal{P}_4(\mathbb{F})$.

- ②⑩ Suppose $p_0, p_1, \dots, p_m$ are polynomials in $\mathcal{P}_m(\mathbb{F})$ such that $p_k(2) = 0$ for $0 \leq k \leq m$. Prove that $p_0, p_1, \dots, p_m$ is not linearly independent in $\mathcal{P}_m(\mathbb{F})$.

PROOF:

Let $U := \{p \in \mathcal{P}_m(\mathbb{F}) \mid p(2) = 0\}$. Every $p \in U$ can be factored as $p(z) = (z - 2)q(z)$ for some $q \in \mathcal{P}_{m-1}(\mathbb{F})$. Since $\mathcal{P}_{m-1}(\mathbb{F}) = \text{span}(1, z, \dots, z^{m-1})$, the list

$$(z - 2), (z - 2)z, \dots, (z - 2)z^{m-1}$$

spans U and has length m . By Theorem 2.22, the length of any linearly independent list in U cannot exceed the length of a spanning list, which is m . Since the list $p_0, p_1, \dots, p_m$ consists of $m + 1$ polynomials in U , it cannot be linearly independent. □

BASES (SECTION 2B)

DEF 2.26 (basis):

A “basis” of V is a list of vectors in V that is linearly independent and spans V .

THM 2.28 (criterion for a basis):

A list $v_1, \dots, v_n$ of vectors in V is a basis of V if and only if every $v \in V$ can be written uniquely in the form

$$v = a_1 v_1 + \dots + a_n v_n,$$

where $a_1, \dots, a_n \in \mathbb{F}$.

THM 2.30 (every spanning list contains a basis):

Every spanning list of vectors in a vector space V can be reduced to a basis of V .

PROOF: Suppose $V = \text{span}(v_1, \dots, v_n)$. Let $B_0 := (v_1, \dots, v_n)$.

STEP 1: If $v_1 = 0$, delete v_1 from B_0 , obtaining $B_1 := (v_2, \dots, v_n)$. If $v_1 \neq 0$, leave B_0 unchanged, so $B_1 := B_0 = (v_1, \dots, v_n)$.

STEP k ($k = 2, \dots, n$): Let B_{k-1} be the list obtained after step $k-1$. If v_k is in the span of the previous vectors in B_{k-1} , delete v_k from B_{k-1} to form B_k . Otherwise, leave B_{k-1} unchanged, setting $B_k := B_{k-1}$.

After step n , we obtain a list B_n . The list B_n spans V because our original list B_0 spanned V , and we only discarded vectors that were already in the span of the previous vectors. Furthermore, the construction ensures that no vector in B_n is in the span of the previous ones. By Theorem 2.19, B_n is linearly independent. Hence, B_n is a basis of V . $\square$

THM 2.31 (basis of a finite-dimensional vector space):

Every finite-dimensional vector space has a basis.

PROOF: By definition, every finite-dimensional vector space has a spanning list. By Theorem 2.30, this spanning list can be reduced to a basis. $\square$

THM 2.32 (linearly independent list extends to a basis):

Every linearly independent list of vectors in a finite-dimensional vector space can be extended to a basis of the vector space.

PROOF: Let $u_1, \dots, u_m \in V$ be linearly independent. Since V is finite-dimensional, there is a spanning list $w_1, \dots, w_n$ of V . Then

$$V = \text{span}(u_1, \dots, u_m, w_1, \dots, w_n).$$

Applying the reduction procedure of Theorem 2.30 to this list produces a basis of V . Because $u_1, \dots, u_m$ is linearly independent, none of the u_j are deleted during the first m steps. Thus, the resulting basis contains all the vectors $u_1, \dots, u_m$. $\square$

THM 2.33 (every subspace of V is part of a direct sum equal to V):

Suppose V is finite-dimensional and U is a subspace of V . Then there is a subspace W of V such that $V = U \oplus W$.

PROOF: Suppose U is a subspace of a finite-dimensional vector space V . By Theorem 2.25, U is finite-dimensional, and therefore, U has a basis $u_1, \dots, u_m$ by Theorem 2.31. By Theorem 2.32, we can extend this to a basis $u_1, \dots, u_m, w_1, \dots, w_n$ of V . Set

$$W := \text{span}(w_1, \dots, w_n).$$

By Theorem 1.46, to show that $V = U \oplus W$, it suffices to prove that $V = U + W$ and $U \cap W = \{0\}$.

PROOF OF $V = U + W$: Let $v \in V$. Since $u_1, \dots, u_m, w_1, \dots, w_n$ is a basis of V , there exist scalars $a_1, \dots, a_m, b_1, \dots, b_n \in \mathbb{F}$ such that

$$v = \underbrace{(a_1 u_1 + \dots + a_m u_m)}_{\in U} + \underbrace{(b_1 w_1 + \dots + b_n w_n)}_{\in W}.$$

Hence, $v \in U + W$, so $V \subseteq U + W$. Conversely, since $U, W \subseteq V$, we have $U + W \subseteq V$.

Therefore, $V = U + W$.

PROOF OF $U \cap W = \{0\}$: Let $v \in U \cap W$. Then there exist scalars $a_1, \dots, a_m, b_1, \dots, b_n \in \mathbb{F}$ such that

$$v = a_1 u_1 + \dots + a_m u_m \quad \text{and} \quad v = b_1 w_1 + \dots + b_n w_n.$$

Thus,

$$a_1 u_1 + \dots + a_m u_m - b_1 w_1 - \dots - b_n w_n = 0.$$

Because $u_1, \dots, u_m, w_1, \dots, w_n$ is linearly independent, all scalars a_i and b_j must be zero.

Therefore, $v = 0$, which proves that $U \cap W = \{0\}$.

Hence, $V = U \oplus W$, as desired. $\square$

EXERCISES ABOUT BASES (SUB-SECTION 2B0)

- ⑥ Prove that if v_1, v_2, v_3, v_4 is a basis of V , then

$$v_1 + v_2, v_2 + v_3, v_3 + v_4, v_4$$

is also a basis of V .

PROOF: STEP 1: We first show linear independence. Let $a_1, a_2, a_3, a_4 \in \mathbb{F}$ such that

$$a_1(v_1 + v_2) + a_2(v_2 + v_3) + a_3(v_3 + v_4) + a_4 v_4 = 0.$$

Rearranging terms gives:

$$a_1 v_1 + (a_1 + a_2)v_2 + (a_2 + a_3)v_3 + (a_3 + a_4)v_4 = 0.$$

Since v_1, v_2, v_3, v_4 is linearly independent, all coefficients must be zero:

$$a_1 = 0, \quad a_1 + a_2 = 0, \quad a_2 + a_3 = 0, \quad a_3 + a_4 = 0.$$

Substituting $a_1 = 0$ successively yields $a_2 = 0$, $a_3 = 0$, and $a_4 = 0$. Thus, the list $v_1 + v_2, v_2 + v_3, v_3 + v_4, v_4$ is linearly independent.

STEP 2: Next, we show that $v_1 + v_2, v_2 + v_3, v_3 + v_4, v_4$ spans V . Notice that:

$$v_4 = v_4,$$

$$v_3 = (v_3 + v_4) - v_4,$$

$$v_2 = (v_2 + v_3) - (v_3 + v_4) + v_4,$$

$$v_1 = (v_1 + v_2) - (v_2 + v_3) + (v_3 + v_4) - v_4.$$

Thus, each $v_k \in \text{span}(v_1 + v_2, v_2 + v_3, v_3 + v_4, v_4)$ for $k = 1, 2, 3, 4$. Since v_1, v_2, v_3, v_4 spans V , it follows that $v_1 + v_2, v_2 + v_3, v_3 + v_4, v_4$ spans V .

Therefore, $v_1 + v_2, v_2 + v_3, v_3 + v_4, v_4$ is a basis of V . $\square$

- ⑦ Prove or give a counterexample: If v_1, v_2, v_3, v_4 is a basis of V and U is a subspace of V such that $v_1, v_2 \in U$ and $v_3 \notin U$ and $v_4 \notin U$, then v_1, v_2 is a basis of U .

SOLUTION:

The statement is **false**. Here is a counterexample:

Let $V = \mathbb{F}^4$ with standard basis e_1, e_2, e_3, e_4 , and let

$$U := \text{span}(e_1, e_2, e_3 + e_4).$$

Then $e_1, e_2 \in U$, but $e_3 \notin U$ and $e_4 \notin U$. However, e_1, e_2 does not span U because $e_3 + e_4 \in U \setminus \text{span}(e_1, e_2)$. Therefore, e_1, e_2 is not a basis of U .

⑧ Suppose $v_1, \dots, v_m$ is a list of vectors in V . For $k \in \{1, \dots, m\}$, let

$$w_k := v_1 + \dots + v_k.$$

Show that $v_1, \dots, v_m$ is a basis of V if and only if $w_1, \dots, w_m$ is a basis of V .

PROOF: From Exercise 2A.3, we already know that $\text{span}(w_1, \dots, w_m) = \text{span}(v_1, \dots, v_m)$. Thus, $w_1, \dots, w_m$ spans V if and only if $v_1, \dots, v_m$ spans V .

It remains to show that $w_1, \dots, w_m$ is linearly independent if and only if $v_1, \dots, v_m$ is linearly independent.

$\Rightarrow$ **DIRECTION:** Suppose $v_1, \dots, v_m$ is linearly independent. Let $c_1, \dots, c_m \in \mathbb{F}$ satisfy

$$c_1 w_1 + c_2 w_2 + \dots + c_m w_m = 0.$$

Substituting $w_k = v_1 + \dots + v_k$ and collecting coefficients yields:

$$(c_1 + c_2 + \dots + c_m)v_1 + (c_2 + \dots + c_m)v_2 + \dots + c_m v_m = 0.$$

Since $v_1, \dots, v_m$ is linearly independent, all coefficients are zero:

$$c_m = 0, \quad c_{m-1} + c_m = 0 \implies c_{m-1} = 0, \quad \dots, \quad c_1 = 0.$$

Thus, $w_1, \dots, w_m$ is linearly independent.

$\Leftarrow$ **DIRECTION:** Suppose $w_1, \dots, w_m$ is linearly independent. Let $d_1, \dots, d_m \in \mathbb{F}$ satisfy

$$d_1 v_1 + d_2 v_2 + \dots + d_m v_m = 0.$$

Substituting $v_1 = w_1$ and $v_k = w_k - w_{k-1}$ for $k \geq 2$ gives:

$$(d_1 - d_2)w_1 + (d_2 - d_3)w_2 + \dots + (d_{m-1} - d_m)w_{m-1} + d_m w_m = 0.$$

Since $w_1, \dots, w_m$ is linearly independent, $d_m = 0$, $d_{m-1} - d_m = 0 \implies d_{m-1} = 0, \dots, d_1 = 0$.

Thus, $v_1, \dots, v_m$ is linearly independent.

Therefore, $v_1, \dots, v_m$ is a basis of V if and only if $w_1, \dots, w_m$ is a basis of V . □

DIMENSION (SECTION 2C)

THM 2.34 (*basis length*):

Any two bases of a finite-dimensional vector space have the same length.

PROOF: Let B_1 and B_2 be two bases of a finite-dimensional vector space V . Since B_1 is linearly independent and B_2 spans V , we have $|B_1| \leq |B_2|$ by Theorem 2.22. Reversing roles, since B_2 is linearly independent and B_1 spans V , we obtain $|B_2| \leq |B_1|$. Therefore, $|B_1| = |B_2|$. $\square$

DEF 2.35 (*dimension*):

The “*dimension*” of a finite-dimensional vector space V , denoted by $\dim V$, is the length of any basis of V . We also set $\dim\{0\} := 0$.

THM 2.37 (*dimension of a subspace*):

If U is a subspace of V , then

$$\dim U \leq \dim V.$$

PROOF: By Theorem 2.25, U is finite-dimensional, so it has a basis. Any basis of U is a linearly independent list in V , and any basis of V is a spanning list of V . By Theorem 2.22, the length of a basis of U is less than or equal to the length of a basis of V . Thus, $\dim U \leq \dim V$. $\square$

THM 2.38 (*linearly independent list of the right length is a basis*):

Suppose V is finite-dimensional. Every linearly independent list of vectors in V of length $\dim V$ is a basis of V .

PROOF: Let $\dim V = n$, and let $v_1, \dots, v_n$ be a linearly independent list in V . This list can be extended to a basis of V by Theorem 2.32. However, every basis of V has length n by Theorem 2.34, so no additional vectors can be added. Hence, $v_1, \dots, v_n$ is already a basis of V . $\square$

THM 2.39 (*subspace of full dimension*):

If U is a subspace of a finite-dimensional vector space V with $\dim U = \dim V$, then $U = V$.

PROOF: Let $u_1, \dots, u_n$ be a basis of U . Thus, $n = \dim U$, and by hypothesis, $n = \dim V$. Hence, $u_1, \dots, u_n$ is a linearly independent list in V of length $\dim V$. By Theorem 2.38, $u_1, \dots, u_n$ is a basis of V . In particular, every vector in V is a linear combination of $u_1, \dots, u_n$. Therefore, $U = V$. $\square$

THM 2.42 (*spanning list of the right length is a basis*):

Suppose V is finite-dimensional. Every spanning list of vectors in V of length $\dim V$ is a basis of V .

PROOF: Suppose $\dim V = n$, and let $v_1, \dots, v_n$ be a list that spans V . By Theorem 2.30, this list can be reduced to a basis of V . Since every basis of V has length n by Theorem 2.34, no vector can be removed during the reduction process. Thus, $v_1, \dots, v_n$ is a basis of V . $\square$

THM 2.43 (*dimension of a sum of subspaces*):

If V_1 and V_2 are subspaces of a finite-dimensional vector space V , then

$$\dim(V_1 + V_2) = \dim V_1 + \dim V_2 - \dim(V_1 \cap V_2).$$

PROOF: Let $r_1, \dots, r_m$ be a basis of $V_1 \cap V_2$, so $\dim(V_1 \cap V_2) = m$. (If $V_1 \cap V_2 = \{0\}$, the list is empty and $m = 0$).

Since $r_1, \dots, r_m$ is linearly independent in V_1 and in V_2 , it can be extended to a basis

$$r_1, \dots, r_m, u_1, \dots, u_j \quad \text{of } V_1,$$

and to a basis

$$r_1, \dots, r_m, w_1, \dots, w_k \quad \text{of } V_2.$$

Thus, $\dim V_1 = m + j$ and $\dim V_2 = m + k$. We will show that the list

$$r_1, \dots, r_m, u_1, \dots, u_j, w_1, \dots, w_k \quad (2.4)$$

is a basis of $V_1 + V_2$. Once this is established, we have:

$$\begin{aligned} \dim(V_1 + V_2) &= m + j + k \\ &= (m + j) + (m + k) - m \\ &= \dim V_1 + \dim V_2 - \dim(V_1 \cap V_2). \end{aligned}$$

The span of the list (2.4) contains V_1 and V_2 , and therefore, equals $V_1 + V_2$. To show that it is a basis, it remains only to prove linear independence.

Suppose scalars $a_1, \dots, a_m, b_1, \dots, b_j, c_1, \dots, c_k \in \mathbb{F}$ satisfy:

$$a_1 r_1 + \dots + a_m r_m + b_1 u_1 + \dots + b_j u_j + c_1 w_1 + \dots + c_k w_k = 0. \quad (2.5)$$

Rearranging gives:

$$c_1 w_1 + \dots + c_k w_k = -a_1 r_1 - \dots - a_m r_m - b_1 u_1 - \dots - b_j u_j.$$

The right side belongs to V_1 . Since the left side belongs to V_2 , it follows that

$$c_1 w_1 + \dots + c_k w_k \in V_1 \cap V_2.$$

Because $r_1, \dots, r_m$ is a basis of $V_1 \cap V_2$, there exist scalars $d_1, \dots, d_m \in \mathbb{F}$ such that

$$c_1 w_1 + \dots + c_k w_k = d_1 r_1 + \dots + d_m r_m,$$

or equivalently,

$$d_1 r_1 + \dots + d_m r_m - c_1 w_1 - \dots - c_k w_k = 0.$$

Because $r_1, \dots, r_m, w_1, \dots, w_k$ is a basis of V_2 , it is linearly independent, which forces all $c_i = 0$ (and $d_i = 0$). Substituting $c_1 = \dots = c_k = 0$ into (2.5) leaves:

$$a_1 r_1 + \dots + a_m r_m + b_1 u_1 + \dots + b_j u_j = 0.$$

Since $r_1, \dots, r_m, u_1, \dots, u_j$ is a basis of V_1 , it is linearly independent, so $a_1 = \dots = a_m = 0$ and $b_1 = \dots = b_j = 0$.

Thus, all coefficients in (2.5) are zero, proving linear independence and completing the proof.

□

EXERCISES ABOUT DIMENSION 2C (SUB-SECTION 2C0)

- 20 Explain why you might guess, motivated by analogy with the formula for the number of elements in the union of three finite sets, that if V_1, V_2, V_3 are subspaces of a finite-dimensional vector space V , then

$$\begin{aligned} \dim(V_1 + V_2 + V_3) &= \dim(V_1) + \dim(V_2) + \dim(V_3) \\ &\quad - \dim(V_1 \cap V_2) - \dim(V_1 \cap V_3) - \dim(V_2 \cap V_3) \\ &\quad + \dim(V_1 \cap V_2 \cap V_3). \end{aligned}$$

Then either prove the formula above or give a counterexample.

SOLUTION: The formula is **wrong**. Consider the following subspaces of $\mathbb{R}^2$:

- $V_1 := \{(x, 0) \mid x \in \mathbb{R}\}$ (the x -axis)
- $V_2 := \{(x, x) \mid x \in \mathbb{R}\}$ (the line $y = x$)
- $V_3 := \{(0, y) \mid y \in \mathbb{R}\}$ (the y -axis)

The right-hand side of the formula gives: $1 + 1 + 1 - 0 - 0 - 0 + 0 = 3$, whereas the left-hand side is $\dim(V_1 + V_2 + V_3) = 2$. So the formula **DOES NOT HOLD** for this example.

However, **IT DOES HOLD** in every case where one of the subspaces is contained in another; hence, without loss of generality, assume that $V_1 \subseteq V_3$. For this situation, we have the following lemma.

LEMMA: If V_1, V_2, V_3 are finite-dimensional subspaces of a vector space, and $V_1 \subseteq V_3$, then

$$(V_1 + V_2) \cap V_3 = (V_1 \cap V_3) + (V_2 \cap V_3)$$

PROOF: If $v \in (V_1 \cap V_3) + (V_2 \cap V_3)$, then we can write $v = u + w$, where $u \in V_1 \cap V_3$ and $w \in V_2 \cap V_3$. Because V_3 is closed under addition and both u and w are in V_3 , so is v . Hence, $v \in V_3$ and $v = u + w$, such that $u \in V_1$ and $w \in V_2$. Therefore, $v \in (V_1 + V_2) \cap V_3$. We conclude:

$$(V_1 \cap V_3) + (V_2 \cap V_3) \subseteq (V_1 + V_2) \cap V_3 \quad (2.6)$$

Conversely, if $v \in (V_1 + V_2) \cap V_3$, then we can write $v = u + w$, where $u \in V_1$, $w \in V_2$, and $v \in V_3$. Since $V_1 \subseteq V_3$ (by our assumption), we can rewrite

$$w = \underbrace{v}_{\in V_3} - \underbrace{u}_{\in V_3}.$$

Since both u and v belong to V_3 , so does w . Therefore, $v = u + w$ such that $u \in V_1 \cap V_3$ and $w \in V_2 \cap V_3$. Hence, $v \in (V_1 \cap V_3) + (V_2 \cap V_3)$. We conclude:

$$(V_1 + V_2) \cap V_3 \subseteq (V_1 \cap V_3) + (V_2 \cap V_3) \quad (2.7)$$

Inclusion (2.6) and (2.7) together yield

$$(V_1 + V_2) \cap V_3 = (V_1 \cap V_3) + (V_2 \cap V_3)$$

for $V_1 \subseteq V_3$. □

Using the standard dimension formula for the sum of two subspaces (Theorem 2.43), we obtain

$$\begin{aligned} \dim(V_1 + V_2 + V_3) &= \dim((V_1 + V_2) + V_3) \\ &= \dim(V_1 + V_2) + \dim(V_3) - \dim((V_1 + V_2) \cap V_3) \\ &= \dim(V_1) + \dim(V_2) + \dim(V_3) \\ &\quad - \dim(V_1 \cap V_2) - \dim((V_1 + V_2) \cap V_3). \end{aligned} \quad (2.8)$$

Using the assumption that $V_1 \subseteq V_3$, we calculate:

$$\begin{aligned} \dim((V_1 + V_2) \cap V_3) &= \dim((V_1 \cap V_3) + (V_2 \cap V_3)) \\ &= \dim(V_1 \cap V_3) + \dim(V_2 \cap V_3) - \underbrace{\dim((V_1 \cap V_3) \cap (V_2 \cap V_3))}_{= \dim(V_1 \cap V_2 \cap V_3)}. \end{aligned}$$

Substituting this result into (2.8) yields:

$$\begin{aligned} \dim(V_1 + V_2 + V_3) &= \dim(V_1) + \dim(V_2) + \dim(V_3) - \dim(V_1 \cap V_2) - \dim((V_1 + V_2) \cap V_3) \\ &= \dim(V_1) + \dim(V_2) + \dim(V_3) - \dim(V_1 \cap V_2) \\ &\quad - (\dim(V_1 \cap V_3) + \dim(V_2 \cap V_3) - \dim(V_1 \cap V_2 \cap V_3)) \\ &= \dim(V_1) + \dim(V_2) + \dim(V_3) \\ &\quad - \dim(V_1 \cap V_2) - \dim(V_1 \cap V_3) - \dim(V_2 \cap V_3) \\ &\quad + \dim(V_1 \cap V_2 \cap V_3). \end{aligned}$$

So, under certain conditions (e.g., when $V_1 \subseteq V_3$), the formula is **true**.

- 21** Show that for a finite-dimensional vector space V with subspaces V_1, V_2, V_3 , the following formula holds:

$$\begin{aligned} \dim(V_1 + V_2 + V_3) &= \dim V_1 + \dim V_2 + \dim V_3 \\ &\quad - \frac{\dim(V_1 \cap V_2) + \dim(V_1 \cap V_3) + \dim(V_2 \cap V_3)}{3} \\ &\quad - \frac{\dim((V_1 + V_2) \cap V_3) + \dim((V_1 + V_3) \cap V_2) + \dim((V_2 + V_3) \cap V_1)}{3}. \end{aligned}$$

SOLUTION: Recall that for any two subspaces U and W of a finite-dimensional vector space,

$$\dim(U + W) = \dim U + \dim W - \dim(U \cap W).$$

Applying this formula to the three groupings of $V_1 + V_2 + V_3$ yields:

$$\begin{aligned}\dim((V_1 + V_2) + V_3) &= \dim(V_1 + V_2) + \dim V_3 - \dim((V_1 + V_2) \cap V_3) \\ &= \dim V_1 + \dim V_2 + \dim V_3 - \dim(V_1 \cap V_2) - \dim((V_1 + V_2) \cap V_3),\end{aligned}$$

$$\begin{aligned}\dim(V_1 + (V_2 + V_3)) &= \dim V_1 + \dim(V_2 + V_3) - \dim(V_1 \cap (V_2 + V_3)) \\ &= \dim V_1 + \dim V_2 + \dim V_3 - \dim(V_2 \cap V_3) - \dim((V_2 + V_3) \cap V_1),\end{aligned}$$

$$\begin{aligned}\dim((V_1 + V_3) + V_2) &= \dim(V_1 + V_3) + \dim V_2 - \dim((V_1 + V_3) \cap V_2) \\ &= \dim V_1 + \dim V_2 + \dim V_3 - \dim(V_1 \cap V_3) - \dim((V_1 + V_3) \cap V_2).\end{aligned}$$

Adding these three equations together yields:

$$\begin{aligned}3 \dim(V_1 + V_2 + V_3) &= 3 \dim V_1 + 3 \dim V_2 + 3 \dim V_3 \\ &\quad - (\dim(V_1 \cap V_2) + \dim(V_2 \cap V_3) + \dim(V_1 \cap V_3)) \\ &\quad - (\dim((V_1 + V_2) \cap V_3) + \dim((V_1 + V_3) \cap V_2) + \dim((V_2 + V_3) \cap V_1)).\end{aligned}$$

Dividing both sides by 3 gives the desired formula.

LINEAR MAPS (CHAPTER 3)

VECTOR SPACE OF LINEAR MAPS (SECTION 3A)

DEFINITION AND EXAMPLES OF LINEAR MAPS (SUB-SECTION 3A1)

DEF 3.1 (*linear map*):

A “linear map” from V to W is a function $T : V \rightarrow W$ with the following properties:

- **ADDITIVITY:** $T(u + v) = Tu + Tv \quad \forall u, v \in V$
- **HOMOGENEITY:** $T(\lambda v) = \lambda(Tv) \quad \forall \lambda \in \mathbb{F}, \forall v \in V$

DEF 3.2 (*notation $\mathcal{L}(V, W)$, $\mathcal{L}(V)$*):

The set of all linear maps from V to W is denoted by $\mathcal{L}(V, W)$. For the set of all linear maps from V to itself, we write $\mathcal{L}(V) := \mathcal{L}(V, V)$.

THM 3.4 (*linear map lemma*):

Suppose $v_1, \dots, v_n$ is a basis of V and $w_1, \dots, w_n \in W$. Then there exists a unique linear map $T \in \mathcal{L}(V, W)$ such that

$$Tv_k = w_k \quad \forall k \in \{1, \dots, n\}.$$

PROOF: Since the list $v_1, \dots, v_n$ is a basis of V , every $v \in V$ has a unique expansion

$$v = c_1 v_1 + \dots + c_n v_n,$$

where $c_1, \dots, c_n \in \mathbb{F}$. Define $T : V \rightarrow W$ by

$$T(c_1 v_1 + \dots + c_n v_n) := c_1 w_1 + \dots + c_n w_n.$$

By uniqueness of the basis expansion, this rule defines a well-defined function $T : V \rightarrow W$. In particular, choosing $v = v_k$ (so $c_k = 1$ and $c_i = 0$ for $i \neq k$) gives

$$T(v_k) = w_k.$$

Next, we verify that T is linear. Let $u = a_1 v_1 + \dots + a_n v_n \in V$, $v = b_1 v_1 + \dots + b_n v_n \in V$, and $\lambda \in \mathbb{F}$. Then

$$\begin{aligned} T(u + v) &= T((a_1 + b_1)v_1 + \dots + (a_n + b_n)v_n) \\ &= (a_1 + b_1)w_1 + \dots + (a_n + b_n)w_n \\ &= (a_1 w_1 + \dots + a_n w_n) + (b_1 w_1 + \dots + b_n w_n) \\ &= T(u) + T(v), \end{aligned}$$

and similarly,

$$\begin{aligned} T(\lambda v) &= T(\lambda b_1 v_1 + \dots + \lambda b_n v_n) \\ &= \lambda b_1 w_1 + \dots + \lambda b_n w_n \\ &= \lambda(b_1 w_1 + \dots + b_n w_n) \\ &= \lambda T(v). \end{aligned}$$

Hence, T is linear, so $T \in \mathcal{L}(V, W)$.

Finally, to prove uniqueness, suppose $S \in \mathcal{L}(V, W)$ is any linear map such that $S(v_k) = w_k$ for each $k \in \{1, \dots, n\}$. Then for any $v = c_1 v_1 + \dots + c_n v_n \in V$, we have

$$\begin{aligned} S(v) &= c_1 S(v_1) + \dots + c_n S(v_n) \\ &= c_1 w_1 + \dots + c_n w_n \\ &= T(v), \end{aligned}$$

so $S = T$. Therefore, T is unique. □

DEF 3.5 (addition and scalar multiplication on $\mathcal{L}(V, W)$):

Suppose $S, T \in \mathcal{L}(V, W)$ and $\lambda \in \mathbb{F}$. The sum $S + T \in \mathcal{L}(V, W)$ and the product $\lambda T \in \mathcal{L}(V, W)$ are defined by:

- $(S + T)(v) := Sv + Tv \quad \forall v \in V$
- $(\lambda T)(v) := \lambda(Tv) \quad \forall v \in V$

ALGEBRAIC OPERATIONS ON $\mathcal{L}(V, W)$ (SUB-SECTION 3A2)

THM 3.6 ($\mathcal{L}(V, W)$ is a vector space):

With the operations of addition and scalar multiplication defined above, $\mathcal{L}(V, W)$ is a vector space over $\mathbb{F}$.

DEF 3.7 (product of linear maps):

If $T \in \mathcal{L}(U, V)$ and $S \in \mathcal{L}(V, W)$, then the “product” $ST \in \mathcal{L}(U, W)$ is defined by

$$(ST)(u) := S(Tu) \quad \forall u \in U.$$

THM 3.8 (algebraic properties of products of linear maps):

Let U, V, W, X be vector spaces over $\mathbb{F}$. Then:

- **ASSOCIATIVITY:** $(T_1 T_2) T_3 = T_1 (T_2 T_3)$, whenever $T_3 \in \mathcal{L}(X, U)$, $T_2 \in \mathcal{L}(U, V)$, and $T_1 \in \mathcal{L}(V, W)$.
- **IDENTITY:** $TI_V = I_W T = T$ for all $T \in \mathcal{L}(V, W)$.
- **DISTRIBUTIVE PROPERTIES:** $(S_1 + S_2)T = S_1 T + S_2 T$ and $S(T_1 + T_2) = ST_1 + ST_2$ for all maps with compatible domains and codomains.

THM 3.10 (linear maps take 0 to 0):

If $T \in \mathcal{L}(V, W)$, then $T(0) = 0$.

PROOF: By additivity, $T(0) = T(0 + 0) = T(0) + T(0)$. Adding the additive inverse of $T(0)$ to both sides yields $T(0) = 0$, as desired. $\square$

EXERCISES ABOUT VECTOR SPACE OF LINEAR MAPS 3A

- (7) Show that every linear map from a one-dimensional vector space to itself is multiplication by some scalar. More precisely, prove that if $\dim V = 1$ and $T \in \mathcal{L}(V)$, then there exists $\lambda \in \mathbb{F}$ such that $Tv = \lambda v$ for all $v \in V$.

PROOF: Since $\dim V = 1$, choose a basis v_1 of V . Then every $v \in V$ can be written uniquely as $v = cv_1$ for some $c \in \mathbb{F}$. Because $T(v_1) \in V = \text{span}(v_1)$, there exists a scalar $\lambda \in \mathbb{F}$ such that

$$T(v_1) = \lambda v_1.$$

Then for any $v = cv_1 \in V$, linearity of T implies:

$$T(v) = T(cv_1) = cT(v_1) = c(\lambda v_1) = \lambda(cv_1) = \lambda v.$$

Hence, $Tv = \lambda v$ for all $v \in V$, as desired. $\square$

- (12) Suppose U is a subspace of V with $U \neq V$. Let $S \in \mathcal{L}(U, W)$ with $S \neq 0$ (so $Su \neq 0$ for some $u \in U$). Define $T : V \rightarrow W$ by

$$Tv := \begin{cases} Sv & \text{if } v \in U, \\ 0 & \text{if } v \in V \setminus U. \end{cases}$$

Prove that T is not a linear map on V .

PROOF: Since $S \neq 0$, choose $u \in U$ such that $Su \neq 0$. Because $U \neq V$, there exists $v \in V \setminus U$. We

claim that $u + v \notin U$. Indeed, if $u + v \in U$, then $v = (u + v) - u \in U$, which contradicts $v \notin U$.

Therefore, by definition of T ,

$$T(u + v) = 0.$$

On the other hand,

$$Tu + Tv = Su + 0 = Su \neq 0.$$

Thus, $T(u + v) \neq Tu + Tv$, showing that T is not additive, and therefore, T is not a linear map on V . $\square$

- 13** Suppose V is finite-dimensional. Prove that every linear map on a subspace of V can be extended to a linear map on V . In other words, show that if U is a subspace of V and $S \in \mathcal{L}(U, W)$, then there exists $T \in \mathcal{L}(V, W)$ such that

$$Tu = Su \quad \forall u \in U.$$

PROOF: Let U be a subspace of a finite-dimensional vector space V , and let $S \in \mathcal{L}(U, W)$. By Theorem 2.33, there is a subspace X of V such that $V = U \oplus X$. Let $u_1, \dots, u_m$ be a basis of U , and let $x_{m+1}, \dots, x_n$ be a basis of X . Then

$$u_1, \dots, u_m, x_{m+1}, \dots, x_n$$

is a basis of V . By Theorem 3.4, there exists a unique linear map $T \in \mathcal{L}(V, W)$ satisfying:

$$T(u_i) = S(u_i) \quad \text{for } 1 \leq i \leq m, \quad \text{and} \quad T(x_j) = 0 \quad \text{for } m+1 \leq j \leq n.$$

For any $u \in U$, writing $u = c_1 u_1 + \dots + c_m u_m$ gives:

$$T(u) = \sum_{i=1}^m c_i T(u_i) = \sum_{i=1}^m c_i S(u_i) = S\left(\sum_{i=1}^m c_i u_i\right) = S(u).$$

Thus, $T \in \mathcal{L}(V, W)$ extends S , as required. $\square$

- 15** Suppose $v_1, \dots, v_m$ is a linearly dependent list of vectors in V . Suppose also that $W \neq \{0\}$. Prove that there exist $w_1, \dots, w_m \in W$ such that no $T \in \mathcal{L}(V, W)$ satisfies

$$Tv_k = w_k \quad \text{for all } k \in \{1, \dots, m\}.$$

SOLUTION: Since $v_1, \dots, v_m$ is linearly dependent, there exist scalars $a_1, \dots, a_m \in \mathbb{F}$, not all zero, such that

$$a_1 v_1 + \dots + a_m v_m = 0. \tag{3.1}$$

Choose an index $k \in \{1, \dots, m\}$ such that $a_k \neq 0$. Since $W \neq \{0\}$, choose a non-zero vector $w \in W$. Define the list $w_1, \dots, w_m$ in W by:

$$w_k := w \quad \text{and} \quad w_j := 0 \quad \text{for all } j \neq k.$$

Suppose, for the sake of contradiction, that there exists a linear map $T \in \mathcal{L}(V, W)$ such that $Tv_j = w_j$ for all $j \in \{1, \dots, m\}$. Applying T to both sides of (3.1) gives:

$$0 = T(0) = T(a_1 v_1 + \dots + a_m v_m) = a_1 T(v_1) + \dots + a_m T(v_m) = a_1 w_1 + \dots + a_m w_m = a_k w.$$

Since $a_k \neq 0$, multiplying by $\frac{1}{a_k}$ yields $w = 0$, contradicting the choice of $w \neq 0$. Therefore, no such linear map $T \in \mathcal{L}(V, W)$ can exist.

NULL SPACES AND RANGES (SECTION 3B)

NULL SPACE AND INJECTIVITY (SUB-SECTION 3B1)

DEF 3.11 (*null space, null T*):

For $T \in \mathcal{L}(V, W)$, the “null space” of T , denoted by $\text{null } T$, is the subset of V defined by

$$\text{null } T := \{v \in V \mid Tv = 0\}.$$

THM 3.13 (*null space is a subspace*):

If $T \in \mathcal{L}(V, W)$, then $\text{null } T$ is a subspace of V .

PROOF: We have $0 \in \text{null } T$ because $T(0) = 0$ by Theorem 3.10. Suppose $u, v \in \text{null } T$ and $\lambda \in \mathbb{F}$. Then $T(u + v) = Tu + Tv = 0 + 0 = 0$, so $u + v \in \text{null } T$. Similarly, $T(\lambda u) = \lambda Tu = \lambda 0 = 0$, so $\lambda u \in \text{null } T$. Therefore, $\text{null } T$ is a subspace of V by Theorem 1.34. $\square$

DEF 3.14 (*injectivity*):

A function $T : V \rightarrow W$ is called “injective” if $Tu = Tv \implies u = v$ for all $u, v \in V$.

THM 3.15 (*injectivity criterion via null space*):

Let $T \in \mathcal{L}(V, W)$. Then T is injective if and only if $\text{null } T = \{0\}$.

PROOF: $\implies$ **DIRECTION:** First, assume that T is injective. Since $\text{null } T$ is a subspace of V (by Theorem 3.13), we have $\{0\} \subseteq \text{null } T$. To prove that $\text{null } T \subseteq \{0\}$, suppose $v \in \text{null } T$. Then $T(v) = 0$. Since $T(0) = 0$ by Theorem 3.10 and T is injective, it follows that $v = 0$. Thus, $\text{null } T = \{0\}$.

$\Leftarrow$ **DIRECTION:** Second, assume $\text{null } T = \{0\}$. Let $u, v \in V$ satisfy $Tu = Tv$. Then

$$0 = Tu - Tv = T(u - v).$$

Thus, $u - v \in \text{null } T = \{0\}$, so $u - v = 0$, which gives $u = v$. Therefore, T is injective. $\square$

RANGE AND SURJECTIVITY (SUB-SECTION 3B2)

DEF 3.16 (*range*):

For $T \in \mathcal{L}(V, W)$, the “range” of T , denoted by $\text{range } T$, is the subset of W defined by

$$\text{range } T := \{Tv \mid v \in V\}.$$

THM 3.18 (*the range is a subspace*):

If $T \in \mathcal{L}(V, W)$, then $\text{range } T$ is a subspace of W .

PROOF: Let $T \in \mathcal{L}(V, W)$. By Theorem 3.10, $T(0) = 0$, so $0 \in \text{range } T$. If $w_1, w_2 \in \text{range } T$, then there exist $v_1, v_2 \in V$ such that $Tv_1 = w_1$ and $Tv_2 = w_2$. Then $w_1 + w_2 = Tv_1 + Tv_2 = T(v_1 + v_2) \in \text{range } T$, so $\text{range } T$ is closed under addition. If $w \in \text{range } T$ and $\lambda \in \mathbb{F}$, then there exists $v \in V$ such that $w = Tv$. Then $\lambda w = \lambda Tv = T(\lambda v) \in \text{range } T$, so $\text{range } T$ is closed under scalar multiplication. By Theorem 1.34, $\text{range } T$ is a subspace of W . $\square$

DEF 3.19 (*surjectivity*):

A function $T : V \rightarrow W$ is called “surjective” (or “onto”) if $\text{range } T = W$.

FUNDAMENTAL THEOREM OF LINEAR MAPS (SUB-SECTION 3B3)

THM 3.21 (*Fundamental Theorem of Linear Maps*):

Suppose V is finite-dimensional and $T \in \mathcal{L}(V, W)$. Then $\text{range } T$ is finite-dimensional and $\dim V = \dim \text{null } T + \dim \text{range } T$.

PROOF: Let $u_1, \dots, u_m$ be a basis of $\text{null } T$, so $\dim \text{null } T = m$. By Theorem 2.32, we can extend this to a basis

$$u_1, \dots, u_m, v_1, \dots, v_n \quad \text{of } V.$$

Thus, $\dim V = m + n$. To complete the proof, it suffices to show that $Tv_1, \dots, Tv_n$ is a basis of $\text{range } T$.

Let $v \in V$. Then there exist scalars $a_1, \dots, a_m, b_1, \dots, b_n \in \mathbb{F}$ such that

$$v = a_1 u_1 + \dots + a_m u_m + b_1 v_1 + \dots + b_n v_n.$$

Applying T and using $T(u_i) = 0$ yields:

$$Tv = b_1 Tv_1 + \dots + b_n Tv_n.$$

Since this holds for every $v \in V$, the list $Tv_1, \dots, Tv_n$ spans $\text{range } T$. In particular, $\text{range } T$ is finite-dimensional.

To show that $Tv_1, \dots, Tv_n$ is linearly independent, suppose scalars $c_1, \dots, c_n \in \mathbb{F}$ satisfy:

$$c_1 Tv_1 + \dots + c_n Tv_n = 0.$$

Then $T(c_1 v_1 + \dots + c_n v_n) = 0$, so $c_1 v_1 + \dots + c_n v_n \in \text{null } T$. Because $u_1, \dots, u_m$ is a basis of $\text{null } T$, there exist scalars $d_1, \dots, d_m \in \mathbb{F}$ such that

$$c_1 v_1 + \dots + c_n v_n = d_1 u_1 + \dots + d_m u_m,$$

or equivalently,

$$-d_1 u_1 - \dots - d_m u_m + c_1 v_1 + \dots + c_n v_n = 0.$$

Because $u_1, \dots, u_m, v_1, \dots, v_n$ is a basis of V , it is linearly independent, which forces all $c_i = 0$ (and $d_i = 0$). Thus, $Tv_1, \dots, Tv_n$ is linearly independent and forms a basis of $\text{range } T$. Hence, $\dim \text{range } T = n$, and $\dim V = m + n = \dim \text{null } T + \dim \text{range } T$. $\square$

THM 3.22 (*linear map to a lower-dimensional space is not injective*):

Suppose V and W are finite-dimensional vector spaces such that $\dim V > \dim W$. Then no linear map from V to W is injective.

PROOF: Let $T \in \mathcal{L}(V, W)$ and assume $\dim V > \dim W$. By Theorem 3.21,

$$\begin{aligned} \dim \text{null } T &= \dim V - \dim \text{range } T \\ &\geq \dim V - \dim W \\ &> 0, \end{aligned}$$

where the first inequality holds because $\text{range } T$ is a subspace of W (so $\dim \text{range } T \leq \dim W$ by Theorem 2.37). Since $\dim \text{null } T > 0$, $\text{null } T \neq \{0\}$. Thus, T is not injective by Theorem 3.15. $\square$

THM 3.24 (*linear map to a higher-dimensional space is not surjective*):

Suppose V and W are finite-dimensional vector spaces such that $\dim V < \dim W$. Then no linear map from V to W is surjective.

PROOF: Let $T \in \mathcal{L}(V, W)$ and assume $\dim V < \dim W$. By Theorem 3.21,

$$\begin{aligned} \dim \text{range } T &= \dim V - \dim \text{null } T \\ &\leq \dim V \\ &< \dim W. \end{aligned}$$

Thus, $\dim \text{range } T < \dim W$, which implies $\text{range } T \neq W$. Hence, T is not surjective by Theorem 3.19. $\square$

THM 3.26 (*homogeneous system of linear equations*):

A homogeneous system of linear equations with more variables than equations has non-zero solu-

tions.

PROOF: Let positive integers m and n satisfy $n > m$, and let $A_{j,k} \in \mathbb{F}$ for $j \in \{1, \dots, m\}$ and $k \in \{1, \dots, n\}$.

Consider the system:

$$\sum_{k=1}^n A_{j,k} x_k = 0 \quad \text{for } j = 1, \dots, m.$$

Define $T : \mathbb{F}^n \rightarrow \mathbb{F}^m$ by

$$T(x_1, \dots, x_n) := \left(\sum_{k=1}^n A_{1,k} x_k, \dots, \sum_{k=1}^n A_{m,k} x_k \right).$$

Then T is a linear map. Since $\dim \mathbb{F}^n = n > m = \dim \mathbb{F}^m$, Theorem 3.22 implies that T is not injective. By Theorem 3.15, $\text{null } T \neq \{0\}$, so there exists a non-zero solution $(x_1, \dots, x_n) \in \mathbb{F}^n$. $\square$

THM 3.28 (*system of linear equations with more equations than variables*):

An inhomogeneous system of linear equations with more equations than variables has no solutions for some choice of constant terms.

PROOF: Let positive integers m and n satisfy $m > n$, and let $A_{j,k} \in \mathbb{F}$ for $j \in \{1, \dots, m\}$ and $k \in \{1, \dots, n\}$.

Define $T : \mathbb{F}^n \rightarrow \mathbb{F}^m$ by

$$T(x_1, \dots, x_n) := \left(\sum_{k=1}^n A_{1,k} x_k, \dots, \sum_{k=1}^n A_{m,k} x_k \right).$$

Since $\dim \mathbb{F}^n = n < m = \dim \mathbb{F}^m$, Theorem 3.24 implies that T is not surjective. Therefore, $\text{range } T \neq \mathbb{F}^m$, so there exists $(c_1, \dots, c_m) \in \mathbb{F}^m \setminus \text{range } T$ for which the system has no solution. $\square$

EXERCISES ABOUT NULL SPACES AND RANGES 3B

- ① Give an example of a linear map $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ such that $\text{null } T = \text{range } T$.

SOLUTION: Define $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ by

$$T(x, y) := (y, 0).$$

Then T is linear. We determine its null space and range:

- $T(x, y) = (0, 0) \iff (y, 0) = (0, 0) \iff y = 0$. Thus, $\text{null } T = \{(x, 0) \mid x \in \mathbb{R}\}$.
- $\text{range } T = \{T(x, y) \mid x, y \in \mathbb{R}\} = \{(y, 0) \mid y \in \mathbb{R}\}$.

Hence, $\text{null } T = \text{range } T = \{(x, 0) \mid x \in \mathbb{R}\}$.

- ② Suppose $S, T \in \mathcal{L}(V)$ are such that $\text{range } S \subseteq \text{null } T$. Prove that $(ST)^2 = 0$.

PROOF: For any $v \in V$,

$$(ST)^2(v) = (ST)(ST)(v) = S(T(S(Tv))).$$

Since $Tv \in V$, we have $S(Tv) \in \text{range } S \subseteq \text{null } T$, which implies $T(S(Tv)) = 0$. Applying S gives:

$$(ST)^2(v) = S(0) = 0.$$

Therefore, $(ST)^2 = 0$. $\square$

- ⑧ Suppose $T \in \mathcal{L}(V, W)$ is injective and $v_1, \dots, v_n$ is linearly independent in V . Prove that $Tv_1, \dots, Tv_n$ is linearly independent in W .

PROOF: Suppose scalars $a_1, \dots, a_n \in \mathbb{F}$ satisfy:

$$a_1 Tv_1 + \dots + a_n Tv_n = 0.$$

By linearity of T , this equation can be rewritten as:

$$T(a_1 v_1 + \dots + a_n v_n) = 0.$$

Since T is injective, $\text{null } T = \{0\}$ by Theorem 3.15, so

$$a_1 v_1 + \cdots + a_n v_n = 0.$$

Because $v_1, \dots, v_n$ is linearly independent in V , this forces $a_1 = \cdots = a_n = 0$. Hence, $Tv_1, \dots, Tv_n$ is linearly independent in W . $\square$

- ⑨ Suppose $v_1, \dots, v_n$ spans V and $T \in \mathcal{L}(V, W)$. Show that $Tv_1, \dots, Tv_n$ spans $\text{range } T$.

PROOF: Let $w \in \text{range } T$. Then $w = Tv$ for some $v \in V$. Since $v_1, \dots, v_n$ spans V , there exist scalars $a_1, \dots, a_n \in \mathbb{F}$ such that

$$v = a_1 v_1 + \cdots + a_n v_n.$$

Applying T and using linearity gives:

$$w = Tv = T(a_1 v_1 + \cdots + a_n v_n) = a_1 Tv_1 + \cdots + a_n Tv_n \in \text{span}(Tv_1, \dots, Tv_n).$$

Thus, $\text{range } T \subseteq \text{span}(Tv_1, \dots, Tv_n)$.

Conversely, each $Tv_k \in \text{range } T$, and since $\text{range } T$ is a subspace of W , we have $\text{span}(Tv_1, \dots, Tv_n) \subseteq \text{range } T$. Therefore, $\text{range } T = \text{span}(Tv_1, \dots, Tv_n)$. $\square$

- ⑫ Suppose T is a linear map from $\mathbb{F}^4$ to $\mathbb{F}^2$ such that

$$\text{null } T = \{(x_1, x_2, x_3, x_4) \in \mathbb{F}^4 \mid x_1 = 5x_2 \text{ and } x_3 = 7x_4\}.$$

Prove that T is surjective.

PROOF: Every element of $\text{null } T$ can be written as:

$$(5x_2, x_2, 7x_4, x_4) = x_2(5, 1, 0, 0) + x_4(0, 0, 7, 1) \quad \text{for } x_2, x_4 \in \mathbb{F}.$$

Since $(5, 1, 0, 0)$ and $(0, 0, 7, 1)$ are linearly independent, they form a basis of $\text{null } T$, so $\dim \text{null } T = 2$. By Theorem 3.21,

$$\dim \text{range } T = \dim \mathbb{F}^4 - \dim \text{null } T = 4 - 2 = 2.$$

Since $\text{range } T$ is a subspace of $\mathbb{F}^2$ and $\dim \text{range } T = \dim \mathbb{F}^2 = 2$, Theorem 2.39 implies that $\text{range } T = \mathbb{F}^2$. Therefore, T is surjective. $\square$

- ⑬ Suppose U is a three-dimensional subspace of $\mathbb{R}^8$ and $T \in \mathcal{L}(\mathbb{R}^8, \mathbb{R}^5)$ satisfies $\text{null } T = U$. Prove that T is surjective.

PROOF: By Theorem 3.21,

$$\dim \text{range } T = \dim \mathbb{R}^8 - \dim \text{null } T = 8 - 3 = 5.$$

Since $\text{range } T \subseteq \mathbb{R}^5$ and $\dim \text{range } T = \dim \mathbb{R}^5 = 5$, Theorem 2.39 implies $\text{range } T = \mathbb{R}^5$. Hence, T is surjective. $\square$

- ⑭ Prove that there does not exist a linear map from $\mathbb{F}^5$ to $\mathbb{F}^2$ whose null space equals

$$S := \{(x_1, x_2, x_3, x_4, x_5) \in \mathbb{F}^5 \mid x_1 = 3x_2 \text{ and } x_3 = x_4 = x_5\}.$$

PROOF: Every element of S has the form:

$$(3x_2, x_2, x_3, x_3, x_3) = x_2(3, 1, 0, 0, 0) + x_3(0, 0, 1, 1, 1) \quad \text{for } x_2, x_3 \in \mathbb{F}.$$

The two vectors $(3, 1, 0, 0, 0)$ and $(0, 0, 1, 1, 1)$ are linearly independent, so $\dim S = 2$. For any linear map $T \in \mathcal{L}(\mathbb{F}^5, \mathbb{F}^2)$, Theorem 3.21 gives:

$$\dim \text{null } T = \dim \mathbb{F}^5 - \dim \text{range } T = 5 - \dim \text{range } T \geq 5 - \dim \mathbb{F}^2 = 5 - 2 = 3.$$

Since $\dim \text{null } T \geq 3 \neq 2 = \dim S$, no such linear map T can have $\text{null } T = S$. $\square$

- ⑮ Suppose there exists a linear map $T \in \mathcal{L}(V, W)$ such that its null space $\text{null } T$ and range $\text{range } T$ are both finite-dimensional. Prove that V is finite-dimensional.

PROOF:

Let $u_1, \dots, u_m$ be a basis of $\text{null } T$, and let $w_1, \dots, w_n$ be a basis of $\text{range } T$. Since $w_1, \dots, w_n \in \text{range } T$, there exist vectors $v_1, \dots, v_n \in V$ such that $Tv_j = w_j$ for each $j \in \{1, \dots, n\}$.

We claim that the list

$$u_1, \dots, u_m, v_1, \dots, v_n$$

spans V . To see this, let $v \in V$. Then $Tv \in \text{range } T$, so there exist scalars $c_1, \dots, c_n \in \mathbb{F}$ such that

$$Tv = c_1 w_1 + \dots + c_n w_n = c_1 T v_1 + \dots + c_n T v_n = T(c_1 v_1 + \dots + c_n v_n).$$

By linearity, this implies $T(v - (c_1 v_1 + \dots + c_n v_n)) = 0$, so

$$v - (c_1 v_1 + \dots + c_n v_n) \in \text{null } T.$$

Since $u_1, \dots, u_m$ is a basis of $\text{null } T$, there exist scalars $a_1, \dots, a_m \in \mathbb{F}$ such that

$$v - (c_1 v_1 + \dots + c_n v_n) = a_1 u_1 + \dots + a_m u_m,$$

which gives:

$$v = a_1 u_1 + \dots + a_m u_m + c_1 v_1 + \dots + c_n v_n.$$

Thus, $V = \text{span}(u_1, \dots, u_m, v_1, \dots, v_n)$. Since V is spanned by a finite list, V is finite-dimensional. $\square$

- 16** Suppose V and W are both finite-dimensional. Prove that there exists an injective linear map $T \in \mathcal{L}(V, W)$ if and only if $\dim V \leq \dim W$.

PROOF: $\Rightarrow$ **DIRECTION:** If there exists an injective linear map $T \in \mathcal{L}(V, W)$, then $\text{null } T = \{0\}$ by Theorem 3.15. By Theorem 3.21 and Theorem 2.37,

$$\dim V = \dim \text{null } T + \dim \text{range } T = \dim \text{range } T \leq \dim W.$$

$\Leftarrow$ **DIRECTION:** Suppose $\dim V \leq \dim W$. Let $v_1, \dots, v_n$ be a basis of V , where $n = \dim V$. Since $\dim W \geq n$, there exists a linearly independent list $w_1, \dots, w_n$ in W . By Theorem 3.4, there exists a unique linear map $T \in \mathcal{L}(V, W)$ such that $Tv_k = w_k$ for each $k \in \{1, \dots, n\}$. If $v = \sum_{k=1}^n c_k v_k \in \text{null } T$, then $0 = Tv = \sum_{k=1}^n c_k w_k$, which forces all $c_k = 0$ by linear independence of the w_k . Thus, $\text{null } T = \{0\}$, so T is injective. $\square$

- 17** Suppose V and W are both finite-dimensional. Prove that there exists a surjective linear map from V onto W if and only if $\dim V \geq \dim W$.

PROOF: $\Rightarrow$ **DIRECTION:** Suppose there exists a surjective linear map $T \in \mathcal{L}(V, W)$. Then $\text{range } T = W$, so by Theorem 3.21,

$$\dim V = \dim \text{null } T + \dim \text{range } T = \dim \text{null } T + \dim W \geq \dim W.$$

$\Leftarrow$ **DIRECTION:** Suppose $\dim V \geq \dim W$. Let $m = \dim W$ and $n = \dim V$, so $n \geq m$. Let $v_1, \dots, v_n$ be a basis of V and $w_1, \dots, w_m$ be a basis of W . By Theorem 3.4, define $T \in \mathcal{L}(V, W)$ by:

$$Tv_k := \begin{cases} w_k & \text{if } 1 \leq k \leq m, \\ 0 & \text{if } m < k \leq n. \end{cases}$$

Then for every $w = \sum_{k=1}^m a_k w_k \in W$, setting $v = \sum_{k=1}^m a_k v_k \in V$ yields $Tv = w$. Hence, T is surjective. $\square$

- 18** Suppose V and W are finite-dimensional, and U is a subspace of V . Prove that there exists $T \in \mathcal{L}(V, W)$ such that $\text{null } T = U$ if and only if $\dim U \geq \dim V - \dim W$.

PROOF: $\Rightarrow$ **DIRECTION:** Suppose $T \in \mathcal{L}(V, W)$ satisfies $\text{null } T = U$. By Theorem 3.21 and Theo-

rem 2.37,

$$\dim V = \dim U + \dim \text{range } T \leq \dim U + \dim W \iff \dim U \geq \dim V - \dim W.$$

$\Leftarrow$ **DIRECTION:** Suppose $\dim U \geq \dim V - \dim W$. Then $\dim V - \dim U \leq \dim W$. Let $u_1, \dots, u_k$ be a basis of U , where $k = \dim U$. Extend this to a basis $u_1, \dots, u_k, v_1, \dots, v_m$ of V , where $m = \dim V - \dim U$. Since $m \leq \dim W$, we can choose a linearly independent list $w_1, \dots, w_m$ in W . By Theorem 3.4, define $T \in \mathcal{L}(V, W)$ by:

$$T(u_i) := 0 \quad \text{for } 1 \leq i \leq k, \quad \text{and} \quad T(v_j) := w_j \quad \text{for } 1 \leq j \leq m.$$

Clearly, $U \subseteq \text{null } T$. Conversely, if $v = \sum_{i=1}^k a_i u_i + \sum_{j=1}^m b_j v_j \in \text{null } T$, then $0 = Tv = \sum_{j=1}^m b_j w_j$, so all $b_j = 0$ by linear independence of the w_j . Thus, $v = \sum_{i=1}^k a_i u_i \in U$. Therefore, $\text{null } T = U$. $\square$

- 19** Suppose W is finite-dimensional and $T \in \mathcal{L}(V, W)$. Prove that T is injective if and only if there exists $S \in \mathcal{L}(W, V)$ such that $ST = I_V$.

PROOF: $\Rightarrow$ **DIRECTION:** Suppose T is injective. Then $\dim V \leq \dim W < \infty$, so V is finite-dimensional. Let $v_1, \dots, v_m$ be a basis of V . By Theorem 3.15 and exercise 7, $Tv_1, \dots, Tv_m$ is linearly independent in W . Extend $Tv_1, \dots, Tv_m$ to a basis $Tv_1, \dots, Tv_m, w_{m+1}, \dots, w_n$ of W . By Theorem 3.4, define $S \in \mathcal{L}(W, V)$ by:

$$S(Tv_k) := v_k \quad \text{for } 1 \leq k \leq m, \quad \text{and} \quad S(w_j) := 0 \quad \text{for } m+1 \leq j \leq n.$$

Then for any $v = \sum_{k=1}^m c_k v_k \in V$,

$$ST(v) = S\left(\sum_{k=1}^m c_k Tv_k\right) = \sum_{k=1}^m c_k S(Tv_k) = \sum_{k=1}^m c_k v_k = v.$$

Thus, $ST = I_V$.

$\Leftarrow$ **DIRECTION:** Suppose $S \in \mathcal{L}(W, V)$ satisfies $ST = I_V$. If $v \in \text{null } T$, then $Tv = 0$, so $v = I_V(v) = (ST)(v) = S(Tv) = S(0) = 0$. Hence, $\text{null } T = \{0\}$, so T is injective. $\square$

- 20** Suppose W is finite-dimensional and $T \in \mathcal{L}(V, W)$. Prove that T is surjective if and only if there exists $S \in \mathcal{L}(W, V)$ such that $TS = I_W$.

PROOF: $\Rightarrow$ **DIRECTION:** Suppose T is surjective. Let $w_1, \dots, w_m$ be a basis of W . Since T is surjective, for each $k \in \{1, \dots, m\}$ there exists $v_k \in V$ such that $Tv_k = w_k$. By Theorem 3.4, define $S \in \mathcal{L}(W, V)$ by $S(w_k) := v_k$ for $1 \leq k \leq m$. Then for any $w = \sum_{k=1}^m a_k w_k \in W$,

$$TS(w) = T\left(\sum_{k=1}^m a_k v_k\right) = \sum_{k=1}^m a_k Tv_k = \sum_{k=1}^m a_k w_k = w.$$

Thus, $TS = I_W$.

$\Leftarrow$ **DIRECTION:** Suppose $S \in \mathcal{L}(W, V)$ satisfies $TS = I_W$. For any $w \in W$, setting $v = S(w) \in V$ gives $Tv = T(S(w)) = (TS)(w) = w$. Therefore, T is surjective. $\square$

- 21** Suppose V is finite-dimensional, U is a subspace of V , and $T \in \mathcal{L}(V, W)$. Prove that

$$X := \{v \in V \mid Tv \in U\}$$

is a subspace of V and

$$\dim X = \dim \text{null } T + \dim(U \cap \text{range } T).$$

PROOF: First, we show that X is a subspace of V :

- Since $T(0) = 0 \in U$, we have $0 \in X$.
- If $v_1, v_2 \in X$, then $T(v_1), T(v_2) \in U$, so $T(v_1 + v_2) = T(v_1) + T(v_2) \in U$, hence $v_1 + v_2 \in X$.
- If $v \in X$ and $\lambda \in \mathbb{F}$, then $T(\lambda v) = \lambda T(v) \in U$, hence $\lambda v \in X$.

Thus, X is a subspace of V .

Now consider the restriction $T|_X: X \rightarrow W$. By Theorem 3.21,

$$\dim X = \dim \text{null}(T|_X) + \dim \text{range}(T|_X).$$

Since $0 \in U$, $\text{null } T \subseteq X$, so $\text{null}(T|_X) = \text{null } T$. Furthermore, $\text{range}(T|_X) = \{Tv \mid v \in X\} = \{Tv \mid v \in V \text{ and } Tv \in U\} = U \cap \text{range } T$. Substituting these identities yields:

$$\dim X = \dim \text{null } T + \dim(U \cap \text{range } T).$$

□

- (22) (a) Suppose U, V, W are finite-dimensional vector spaces, $S \in \mathcal{L}(V, W)$, and $T \in \mathcal{L}(U, V)$.

Prove that

$$\dim \text{null } ST \leq \dim \text{null } S + \dim \text{null } T.$$

PROOF: Applying Theorem 3.21 to T and to ST gives:

$$\dim U = \dim \text{null } T + \dim \text{range } T,$$

$$\dim U = \dim \text{null } ST + \dim \text{range } ST.$$

Equating these expressions yields:

$$\dim \text{null } ST = \dim \text{null } T + \dim \text{range } T - \dim \text{range } ST. \quad (3.2)$$

Now consider the restriction $S|_{\text{range } T}: \text{range } T \rightarrow W$. Its range is $\text{range}(S|_{\text{range } T}) = \text{range } ST$.

Applying Theorem 3.21 to $S|_{\text{range } T}$ gives:

$$\dim \text{range } T = \dim \text{null}(S|_{\text{range } T}) + \dim \text{range } ST.$$

Since $\text{null}(S|_{\text{range } T}) = \text{null } S \cap \text{range } T \subseteq \text{null } S$, we have $\dim \text{null}(S|_{\text{range } T}) \leq \dim \text{null } S$.

Therefore,

$$\dim \text{range } T - \dim \text{range } ST = \dim \text{null}(S|_{\text{range } T}) \leq \dim \text{null } S.$$

Substituting this into (3.2) yields:

$$\dim \text{null } ST \leq \dim \text{null } S + \dim \text{null } T.$$

□

- (b) Give an example of $S, T \in \mathcal{L}(\mathbb{F}^5)$ with $ST = 0$ and $\dim \text{range } TS = 2$.

SOLUTION: Define $S, T \in \mathcal{L}(\mathbb{F}^5)$ by:

$$S(x_1, x_2, x_3, x_4, x_5) := (0, 0, 0, x_4, x_5),$$

$$T(x_1, x_2, x_3, x_4, x_5) := (x_4, x_5, 0, 0, 0).$$

Then for any $x \in \mathbb{F}^5$,

$$ST(x) = S(x_4, x_5, 0, 0, 0) = (0, 0, 0, 0, 0) = 0,$$

so $ST = 0$. On the other hand,

$$TS(x) = T(0, 0, 0, x_4, x_5) = (x_4, x_5, 0, 0, 0).$$

Thus, $\text{range } TS = \{(x_4, x_5, 0, 0, 0) \mid x_4, x_5 \in \mathbb{F}\}$, which has dimension 2.

- (24) Suppose $\dim V = 5$ and $S, T \in \mathcal{L}(V)$ are such that $ST = 0$. Prove that $\dim \text{range } TS \leq 2$.

PROOF:

Since $ST = 0$, we have $S(Tv) = 0$ for all $v \in V$, so $\text{range } T \subseteq \text{null } S$. Therefore,

$$\dim \text{range } T \leq \dim \text{null } S.$$

By Theorem 3.21, $\dim \text{null } S = 5 - \dim \text{range } S$, so

$$\dim \text{range } T \leq 5 - \dim \text{range } S \iff \dim \text{range } S + \dim \text{range } T \leq 5.$$

Since $\text{range } TS = T(\text{range } S) \subseteq \text{range } T$, we have $\dim \text{range } TS \leq \dim \text{range } T$. Also, $\dim \text{range } TS \leq$

$\dim \text{range } S$. Summing these two inequalities gives:

$$2 \dim \text{range } TS \leq \dim \text{range } S + \dim \text{range } T \leq 5 \implies \dim \text{range } TS \leq \frac{5}{2}.$$

Since $\dim \text{range } TS$ must be an integer, $\dim \text{range } TS \leq 2$. $\square$

- (25) Suppose W is finite-dimensional and $S, T \in \mathcal{L}(V, W)$. Prove that $\text{null } S \subseteq \text{null } T$ if and only if there exists $E \in \mathcal{L}(W)$ such that $T = ES$.

PROOF: $\implies$ **DIRECTION:** Suppose $\text{null } S \subseteq \text{null } T$. Let $w_1, \dots, w_m$ be a basis of $\text{range } S \subseteq W$. Since each $w_j \in \text{range } S$, there exist $v_1, \dots, v_m \in V$ such that $Sv_j = w_j$ for $j \in \{1, \dots, m\}$. Extend $w_1, \dots, w_m$ to a basis $w_1, \dots, w_m, w_{m+1}, \dots, w_n$ of W . By Theorem 3.4, define $E \in \mathcal{L}(W)$ by:

$$E(w_j) := T(v_j) \quad \text{for } 1 \leq j \leq m, \quad \text{and} \quad E(w_k) := 0 \quad \text{for } m+1 \leq k \leq n.$$

Now let $v \in V$. Then $Sv \in \text{range } S$, so $Sv = \sum_{j=1}^m c_j w_j = \sum_{j=1}^m c_j Sv_j = S\left(\sum_{j=1}^m c_j v_j\right)$. This implies $S\left(v - \sum_{j=1}^m c_j v_j\right) = 0$, so $v - \sum_{j=1}^m c_j v_j \in \text{null } S \subseteq \text{null } T$. Therefore,

$$T\left(v - \sum_{j=1}^m c_j v_j\right) = 0 \implies Tv = \sum_{j=1}^m c_j T(v_j).$$

On the other hand,

$$ES(v) = E\left(\sum_{j=1}^m c_j w_j\right) = \sum_{j=1}^m c_j E(w_j) = \sum_{j=1}^m c_j T(v_j) = Tv.$$

Thus, $T = ES$.

$\Leftarrow$ **DIRECTION:** Suppose there exists $E \in \mathcal{L}(W)$ such that $T = ES$. If $v \in \text{null } S$, then $Sv = 0$, so $Tv = E(Sv) = E(0) = 0$ by Theorem 3.10. Thus, $v \in \text{null } T$, proving $\text{null } S \subseteq \text{null } T$. $\square$

- (26) Suppose V is finite-dimensional and $S, T \in \mathcal{L}(V, W)$. Prove that $\text{range } S \subseteq \text{range } T$ if and only if there exists $E \in \mathcal{L}(V)$ such that $S = TE$.

PROOF: $\implies$ **DIRECTION:** Suppose $\text{range } S \subseteq \text{range } T$. Let $v_1, \dots, v_n$ be a basis of V . For each $k \in \{1, \dots, n\}$, $Sv_k \in \text{range } S \subseteq \text{range } T$, so there exists $u_k \in V$ such that $T(u_k) = S(v_k)$. By Theorem 3.4, there exists a unique linear map $E \in \mathcal{L}(V)$ such that $E(v_k) = u_k$ for all $k \in \{1, \dots, n\}$.

Then for any $v = \sum_{k=1}^n c_k v_k \in V$,

$$TE(v) = T\left(\sum_{k=1}^n c_k E(v_k)\right) = \sum_{k=1}^n c_k T(u_k) = \sum_{k=1}^n c_k S(v_k) = S(v).$$

Thus, $S = TE$.

$\Leftarrow$ **DIRECTION:** Suppose there exists $E \in \mathcal{L}(V)$ such that $S = TE$. Then for any $w \in \text{range } S$, $w = Sv = T(Ev) \in \text{range } T$. Hence, $\text{range } S \subseteq \text{range } T$. $\square$

- (27) Suppose $P \in \mathcal{L}(V)$ and $P^2 = P$. Prove that $V = \text{null } P \oplus \text{range } P$.

PROOF: For any $v \in V$, we can write:

$$v = (v - Pv) + Pv.$$

Applying P to the first term gives:

$$P(v - Pv) = Pv - P^2v = Pv - Pv = 0,$$

which shows that $v - Pv \in \text{null } P$. Since $Pv \in \text{range } P$, we have $V = \text{null } P + \text{range } P$.

To show that the sum is direct, suppose $u \in \text{null } P \cap \text{range } P$. Since $u \in \text{range } P$, there exists $w \in V$ such that $u = Pw$. Since $u \in \text{null } P$, we also have $Pu = 0$. Therefore,

$$u = Pw = P^2w = P(Pw) = Pu = 0.$$

Thus, $\text{null } P \cap \text{range } P = \{0\}$. By ??, $V = \text{null } P \oplus \text{range } P$.

□

MATRICES (SECTION 3C)

REPRESENTING A LINEAR MAP BY A MATRIX (SUB-SECTION 3C1)

DEF 3.31 (*matrix of a linear map, $\mathcal{M}(T)$*):

Let $T \in \mathcal{L}(V, W)$ and let $v_1, \dots, v_n$ be a basis of V and let $w_1, \dots, w_m$ be a basis of W . The “matrix of T ” with respect to these bases is the m -by- n matrix $\mathcal{M}(T)$ whose entries $A_{j,k}$ are defined by

$$Tv_k = A_{1,k}w_1 + \dots + A_{m,k}w_m = \sum_{j=1}^m A_{j,k}w_j.$$

If the bases are not clear from the context, one writes $\mathcal{M}(T, (v_1, \dots, v_n), (w_1, \dots, w_m))$. One remembers how to construct $\mathcal{M}(T)$, if one writes the bases on left hand side and on top of the matrix. It looks as follows:

$$\mathcal{M}(T) := \begin{matrix} & v_1 & \cdots & v_k & \cdots & v_n \\ \begin{matrix} w_1 \\ \vdots \\ w_m \end{matrix} & \left(\begin{array}{cccc} & & & \\ & & A_{1,k} & \\ & & \vdots & \\ & & & A_{m,k} \end{array} \right) \end{matrix}$$

DEF (*matrix notation for operators (see later)*):

For $T \in \mathcal{L}(V)$ we use the notation

$$\mathcal{M}(T, (v_1, \dots, v_n)) := \mathcal{M}(T, (v_1, \dots, v_n), (v_1, \dots, v_n))$$

ADDITION AND SCALAR MULTIPLICATION OF MATRICES (SUB-SECTION 3C2)

THM 3.35 (*matrix of the sum of linear maps*):

$\mathcal{M}(S + T) = \mathcal{M}(S) + \mathcal{M}(T)$ for $S, T \in \mathcal{L}(V, W)$

THM 3.38 (*matrix of a scalar times a linear map*):

$\mathcal{M}(\lambda T) = \lambda \mathcal{M}(T)$ for $\lambda \in \mathbb{F}$ and $T \in \mathcal{L}(V, W)$

DEF 3.39 (*notation $\mathbb{F}^{m,n}$*):

$\mathbb{F}^{m,n}$ denotes the set of all m -by- n matrices with entries in $\mathbb{F}$, where $m, n \in \mathbb{N}$

THM 3.40 (*dimension of $\mathbb{F}^{m,n}$*):

$\mathbb{F}^{m,n}$ is a vector space with $\dim \mathbb{F}^{m,n} = mn$.

PROOF: It is easy to verify that $\mathbb{F}^{m,n}$ is a vector space. The list of length mn that consists of distinct m -by- n matrices that have 0 in all entries except a 1 in one entry is a basis of $\mathbb{F}^{m,n}$. Hence, $\dim \mathbb{F}^{m,n} = mn$. $\square$

MATRIX MULTIPLICATION (SUB-SECTION 3C3)

Let $v_1, \dots, v_n$ be a basis of V , $w_1, \dots, w_m$ be a basis of W and $u_1, \dots, u_p$ be a basis of U . Consider the linear maps $T : U \rightarrow V$ and $S : V \rightarrow W$. The composition ST is a linear map from U to W . Is it the case that

$$\mathcal{M}(ST) = \mathcal{M}(S)\mathcal{M}(T) ?$$

Later on, matrix multiplication will be defined accordingly such that this question has a positive answer. For now, let $A := \mathcal{M}(S) \in \mathbb{F}^{m,n}$ and $B := \mathcal{M}(T) \in \mathbb{F}^{n,p}$. The matrices of S and T look as

follows,

$$A := \mathcal{M}(S) = \begin{matrix} & v_1 & \cdots & v_r & \cdots & v_n \\ \begin{matrix} w_1 \\ \vdots \\ w_j \\ \vdots \\ w_m \end{matrix} & \begin{pmatrix} & & A_{1,r} & & \\ & & \vdots & & \\ & & A_{j,r} & & \\ & & \vdots & & \\ & & A_{m,r} & & \end{pmatrix} \end{matrix}, \quad B := \mathcal{M}(T) = \begin{matrix} & u_1 & \cdots & u_k & \cdots & u_p \\ \begin{matrix} v_1 \\ \vdots \\ v_r \\ \vdots \\ v_n \end{matrix} & \begin{pmatrix} & & B_{1,k} & & \\ & & \vdots & & \\ & & B_{r,k} & & \\ & & \vdots & & \\ & & B_{n,k} & & \end{pmatrix} \end{matrix}$$

For $1 \leq k \leq p$, we have

$$(ST)u_k = S(Tu_k) = S\left(\sum_{r=1}^n B_{r,k}v_r\right) = \sum_{r=1}^n B_{r,k}Sv_r = \sum_{r=1}^n B_{r,k} \sum_{j=1}^m A_{j,r}w_j = \sum_{j=1}^m \left(\sum_{r=1}^n A_{j,r}B_{r,k}\right)w_j$$

Thus, $\mathcal{M}(ST)$ is the m -by- p matrix whose entry in row j , column k equals $\sum_{r=1}^n A_{j,r}B_{r,k}$.

$$\mathcal{M}(ST) = \begin{matrix} & u_1 & \cdots & u_k & \cdots & u_p \\ \begin{matrix} w_1 \\ \vdots \\ w_j \\ \vdots \\ w_m \end{matrix} & \begin{pmatrix} & & \sum_{r=1}^n A_{1,r}B_{r,k} & & \\ & & \vdots & & \\ & & \sum_{r=1}^n A_{j,r}B_{r,k} & & \\ & & \vdots & & \\ & & \sum_{r=1}^n A_{m,r}B_{r,k} & & \end{pmatrix} \end{matrix}$$

DEF 3.41 (*matrix multiplication*):

Let $A \in \mathbb{F}^{m,n}$ and $B \in \mathbb{F}^{n,p}$. Then $AB \in \mathbb{F}^{m,p}$ whose entry in row j , column k , is given by the equation

$$(AB)_{j,k} := \sum_{r=1}^n A_{j,r}B_{r,k}.$$

The octave code for calculating the entry j,k of the result of A,B with the “*sum-scheme*” like above looks as follows

```

1  function retval = standard_matrix_mult (A,B)
2      m = rows(A);
3      n = columns(A);
4      p = columns(B);
5      retval=zeros(m,p)
6      if (columns(A) != rows(B))
7          error('columns of A and rows of B dont match');
8      end
9      for (j=1:m)
10         for (k=1:p)
11             for (r=1:n)
12                 retval(j,k)=retval(j,k)+A(j,r)*B(r,k);
13             end
14         end
15     end
16 end
    
```

THM 3.43 (*matrix of product of linear maps*):

If $T \in \mathcal{L}(U, V)$ and $S \in \mathcal{L}(V, W)$, then

$$\mathcal{M}(ST) = \mathcal{M}(S)\mathcal{M}(T).$$

PROOF: See at the beginning of the sub-subsection. □

DEF 3.44 (*notation: $A_{j,\bullet}, A_{\bullet,k}$*):

Let $A \in \mathbb{F}^{m,n}$. Then

- If $1 \leq j \leq m$, then $A_{j,\bullet}$ denotes the 1-by- n matrix consisting of row j of A .
- If $1 \leq k \leq n$, then $A_{\bullet,k}$ denotes the m -by-1 matrix consisting of the column k of A .

THM 3.46 (*entry of matrix product equals row times column*):

If $A \in \mathbb{F}^{m,n}$ and $B \in \mathbb{F}^{n,p}$, then we have for $1 \leq j \leq m$ and $1 \leq k \leq p$

$$(AB)_{j,k} = A_{j,\bullet} B_{\bullet,k}$$

The octave code for multiplying rows and columns directly at entry j, k looks as follows

```

1  function retval = row_wise_matrix_mult (A,B)
2      m = rows(A);
3      n = columns(A);
4      p = columns(B);
5      retval=zeros(m,p)
6      if (n != rows(B))
7          error('columns of A and rows of B dont match');
8      end
9      for (j=1:m)
10         for (k=1:p)
11             retval(j,k)=A(j,:)*B(:,k);
12         end
13     end
14 end
    
```

THM 3.48 (*column of matrix product equals matrix times column*):

If $A \in \mathbb{F}^{m,n}$ and $B \in \mathbb{F}^{n,p}$, then we have for $1 \leq k \leq p$

$$(AB)_{\bullet,k} = AB_{\bullet,k}$$

In other words, column k of AB equals A times column k of B . The octave code looks as follows

```

1  function retval = matrix_times_columns_matrix_mult (A,B)
2      m = rows(A);
3      p = columns(B);
4      retval=zeros(m,p)
5      if (columns(A) != rows(B))
6          error('columns of A and rows of B dont match');
7      end
8      for (k=1:p)
9          retval(:,k)=A*B(:,k);
10     end
11 end
    
```

THM 3.50 (*linear combination of columns*):

Suppose $A \in \mathbb{F}^{m,n}$ and $b = (b_1, \dots, b_n)^t \in \mathbb{F}^{n,1}$. Then

$$Ab = b_1 A_{\bullet,1} + \dots + b_n A_{\bullet,n}.$$

See 3.54 for the definition of $(^t)$.

THM 3.51 (*matrix multiplication as linear comb. of columns or rows*):

Let $C \in \mathbb{F}^{m,c}$ and $R \in \mathbb{F}^{c,n}$.

COLUMN-WISE: If $k \in \{1, \dots, n\}$, then column k of $CR \in \mathbb{F}^{m,n}$ is a linear combination of the columns of

C , with the coefficients of this linear combination coming from column k of R [$R_{\bullet,k}$].

$CR =$

$$\left(\underbrace{\begin{pmatrix} C_{1,1} \\ \vdots \\ C_{m,1} \end{pmatrix} R_{1,1} + \cdots + \begin{pmatrix} C_{1,c} \\ \vdots \\ C_{m,c} \end{pmatrix} R_{c,1}}_{\text{1st column}} \cdots \underbrace{\begin{pmatrix} C_{1,1} \\ \vdots \\ C_{m,1} \end{pmatrix} R_{1,k} + \cdots + \begin{pmatrix} C_{1,c} \\ \vdots \\ C_{m,c} \end{pmatrix} R_{c,k}}_{\text{kth column}} \cdots \underbrace{\begin{pmatrix} C_{1,1} \\ \vdots \\ C_{m,1} \end{pmatrix} R_{1,n} + \cdots + \begin{pmatrix} C_{1,c} \\ \vdots \\ C_{m,c} \end{pmatrix} R_{c,n}}_{\text{last column}} \right)$$

The octave-code for column-wise multiplication is as follows.

```

1  function retval = column_wise_matrix_mult (C,R)
2      m = rows(C);
3      c = columns(C);
4      n = columns (R);
5      retval=zeros(m,n);
6      if (c != rows(R))
7          error('columns of C and rows of R dont match');
8      end
9      for (k=1:n)
10         for (i=1:c)
11             retval(:,k)=retval(:,k)+C(:,i)*R(i,k);
12         end
13     end
14 end
    
```

Row-wise: If $j \in \{1, \dots, m\}$, then row j of CR is a linear combination of the rows of R , with the coefficients of this linear combination coming from row j of C [$C_{j,\bullet}$]

$$CR = \begin{pmatrix} C_{1,1}(R_{1,1}, \dots, R_{1,n}) + C_{1,2}(R_{2,1}, \dots, R_{2,n}) + \cdots + C_{1,c}(R_{c,1}, \dots, R_{c,n}) \\ C_{2,1}(R_{1,1}, \dots, R_{1,n}) + C_{2,2}(R_{2,1}, \dots, R_{2,n}) + \cdots + C_{2,c}(R_{c,1}, \dots, R_{c,n}) \\ \vdots \\ C_{j,1}(R_{1,1}, \dots, R_{1,n}) + C_{j,2}(R_{2,1}, \dots, R_{2,n}) + \cdots + C_{j,c}(R_{c,1}, \dots, R_{c,n}) \\ \vdots \\ C_{m,1}(R_{1,1}, \dots, R_{1,n}) + C_{m,2}(R_{2,1}, \dots, R_{2,n}) + \cdots + C_{m,c}(R_{c,1}, \dots, R_{c,n}) \end{pmatrix}$$

The octave-code for row-wise multiplication is as follows.

```

1  function retval = row_wise_matrix_mult (C,R)
2      m = rows(C);
3      c = columns(C);
4      n = columns (R);
5      retval=zeros(m,n);
6      if (c != rows(R))
7          error('columns of C and rows of R dont match');
8      end
9      for (j=1:m)
10         for (i=1:c)
11             retval(j,:) = retval(j,:) + C(j,i) * R(i,:);
12         end
13     end
14 end
    
```

COLUMN-ROW FACTORIZATION AND RANK OF A MATRIX (SUB-SECTION 3C4)

DEF 3.52 (*column rank, row rank*):

For $A \in \mathbb{F}^{m,n}$, the “column rank” of A is the dimension of the span of the columns of A in $\mathbb{F}^{m,1}$. The

“row rank” of A is the dimension of the span of the rows of A in $\mathbb{F}^{1,n}$.

DEF 3.54 (*transpose, A^t*):

The “transpose” of a matrix $A \in \mathbb{F}^{m,n}$, denoted by $A^t \in \mathbb{F}^{n,m}$, is obtained by interchanging rows and columns. Its entries are given by the equation

$$(A^t)_{k,j} = A_{j,k} \quad \forall k \in \{1, \dots, m\}, j \in \{1, \dots, n\}.$$

THM 3.56 (*column-row factorization*):

Suppose $A \in \mathbb{F}^{m,n}$ with column rank $c \geq 1$. Then there exists $C \in \mathbb{F}^{m,c}$ and $R \in \mathbb{F}^{c,n}$ such that

$$A = CR.$$

THM 3.57 (*column rank and row rank*):

If $A \in \mathbb{F}^{m,n}$, then column rank of A = row rank of A . This leads to the next definition.

DEF 3.58 (*rank*):

The “rank” of a matrix $A \in \mathbb{F}^{m,n}$ is the column rank of A .

EXERCISES ABOUT MATRICES 3C

INVERTIBILITY AND ISOMORPHISMS (SECTION 3D)

INVERTIBLE LINEAR MAPS (SUB-SECTION 3D1)

DEF 3.59 (*invertible, inverse*):

$S \in \mathcal{L}(W, V)$ with $ST = I_V$ and $TS = I_W$ is called the “inverse” of the “invertible” linear map $T \in \mathcal{L}(V, W)$.

THM 3.60 (*inverse is unique*):

An invertible linear map has a unique inverse.

PROOF: Suppose $T \in \mathcal{L}(V, W)$ is invertible and S_1 and S_2 are inverses of T . Hence,

$$S_1 = S_1 I = S_1 (TS_2) = (S_1 T) S_2 = I S_2 = S_2.$$

Therefore, $S_1 = S_2$. □

DEF 3.61 (*notation T^{-1}*):

Since the inverse is unique, the inverse is denoted by $T^{-1} \in \mathcal{L}(W, V)$ for $T \in \mathcal{L}(V, W)$. We have

$$T^{-1}T = I_V \text{ and } TT^{-1} = I_W.$$

THM 3.62 (*invertibility, injectivity and surjectivity*):

A linear map is invertible $\iff$ it is injective and surjective.

THM 3.65 (*injectivity is equivalent to surjectivity*):

Suppose V and W are finite-dimensional vector spaces with $\dim V = \dim W$, and $T \in \mathcal{L}(V, W)$. Then

$$T \text{ is invertible } \iff T \text{ is injective } \iff T \text{ is surjective.}$$

Equivalently:

$$T \text{ is not invertible } \iff T \text{ is not injective } \iff T \text{ is not surjective.}$$

PROOF: The rank-nullity theorem (3.21) or fundamental theorem of linear maps states that

$$\dim V = \dim \text{null } T + \dim \text{range } T \tag{3.3}$$

If T is injective, $\dim \text{null } T = 0$ and therefore

$$\dim \text{range } T = \dim V = \dim W,$$

which means T is surjective ($\text{range } T = \dim W$). This is because every linearly independent list of vectors of length $\dim W$ is a basis by 2.38. And that's how dimension is defined. (TODO)

If T is surjective, we have $\dim \text{range } T = \dim W$ to begin with and therefore (3.3) becomes

$$\dim \text{null } T = \dim V - \dim \text{range } T = \dim V - \dim W = 0.$$

which makes T also injective. Since injectivity and surjectivity together imply invertibility, this ends the proof. □

THM 3.68 ($ST = I \iff TS = I$ on vector spaces of the same dimension):

Suppose V and W are finite-dimensional vector spaces of the same dimension. Let $S \in \mathcal{L}(W, V)$, and $T \in \mathcal{L}(V, W)$. Then $ST = I \iff TS = I$.

PROOF: $\Rightarrow$ **DIRECTION:** Let $S \in \mathcal{L}(W, V)$, and $T \in \mathcal{L}(V, W)$ such that $ST = I$. If $v \in V$ and $Tv = 0$, then

$$v = Iv = (ST)v = S(Tv) = S(0) = 0.$$

Thus, $\text{null } T = \{0\}$ and therefore, T is injective by 3.15. Because V and W have the same dimension, T is invertible by 3.65. Now, multiply both sides of the equation $ST = I$ by T^{-1} to conclude:

$$S = T^{-1}.$$

Hence, $TS = TT^{-1} = I$, as desired.

⇐ **DIRECTION:** Right above, we have shown that $ST = I$ implies that $TS = I$. To prove the implication in the other direction, simply reverse the roles of S and T and V and W , showing that $TS = I$ implies that $ST = I$. $\square$

ISOMORPHIC VECTOR SPACES (SUB-SECTION 3D2)

DEF 3.69:

An isomorphism is an invertible linear map. Two vector spaces are called isomorphic if there is an isomorphism from one vector space to the other. $T : V \rightarrow W$

THM 3.70 (dimension shows whether vector spaces are isomorphic):

Two finite-dimensional vector spaces U, V over $\mathbb{F}$ are isomorphic $\iff$ they have the same dimension (i.e., $\dim U = \dim V$).

PROOF: exercise

⇒ **DIRECTION:** rank-nullity theorem

⇐ **DIRECTION:** surjectivity and injectivity. $\square$

THM 3.71 ($\mathcal{L}(V, W)$ and $\mathbb{F}^{m,n}$ are isomorphic):

Suppose $v_1, \dots, v_n$ is a basis of V and $w_1, \dots, w_m$ is a basis of W . Then $\mathcal{M}$ is an isomorphism between $\mathcal{L}(V, W)$ and $\mathbb{F}^{m,n}$.

PROOF: exercise (injectivity and surjectivity) $\square$

THM 3.72 ($\dim \mathcal{L}(V, W) = (\dim V) \cdot (\dim W)$):

Suppose V, W are finite-dimensional. Then $\mathcal{L}(V, W)$ is finite-dimensional and

$$\dim \mathcal{L}(V, W) = (\dim V) \cdot (\dim W).$$

PROOF: Suppose $v_1, \dots, v_n$ is a basis of V and $w_1, \dots, w_m$ is a basis of W . By 3.71, there is an isomorphism between $\mathcal{L}(V, W)$ and $\mathbb{F}^{m,n}$, and thus, they have the same dimension. Now, 3.40 tells us that $\dim \mathbb{F}^{m,n} = mn$, therefore

$$\dim \mathcal{L}(V, W) = \dim \mathbb{F}^{m,n} = m \cdot n = (\dim V) \cdot (\dim W),$$

as desired. $\square$

LINEAR MAPS THOUGHT OF AS MATRIX MULTIPLICATION (SUB-SECTION 3D3)

DEF 3.73 (matrix of a vector, $\mathcal{M}(u)$):

Let $u \in V$ and $v_1, \dots, v_m$ be a basis of V such that

$$u = b_1 v_1 + \dots + b_m v_m. \text{ We define } \mathcal{M}(u) := \begin{pmatrix} b_1 \\ \vdots \\ b_m \end{pmatrix}.$$

The vector depends on the basis, but it is not included in the notation.

THM 3.75 ($\mathcal{M}(T)_{\bullet,k} = \mathcal{M}(Tv_k)$):

Suppose $T \in \mathcal{L}(V, W)$ and $v_1, \dots, v_n$ is a basis of V and $w_1, \dots, w_m$ is a basis of W . Let $1 \leq k \leq n$. Then the k^{th} column of $\mathcal{M}(T)$, which is denoted by $\mathcal{M}(T)_{\bullet,k}$, equals $\mathcal{M}(Tv_k)$.

$$\mathcal{M}(T)_{\bullet,k} = \mathcal{M}(Tv_k) \quad \forall k \in \{1, \dots, n\}.$$

THM 3.76 (linear maps act like matrix multiplication):

Suppose $T \in \mathcal{L}(V, W)$ and $v \in V$. Suppose $v_1, \dots, v_n$ is a basis of V and $w_1, \dots, w_m$ is a basis of W . Then

$$\mathcal{M}(Tv) = \mathcal{M}(T) \cdot \mathcal{M}(v)$$

Or using different notation:

$$\mathcal{M}(Tv, (v_1, \dots, v_n), (w_1, \dots, w_m)) = \mathcal{M}(T, (v_1, \dots, v_n), (w_1, \dots, w_m)) \cdot \mathcal{M}(v)$$

THM 3.78 (*dimension of range T and column rank of $\mathcal{M}(T)$*):

Suppose $\dim V < \infty$ and $\dim W < \infty$. Then for $T \in \mathcal{L}(V, W)$:

$$\dim \text{range } T = \text{column rank of } \mathcal{M}(T)$$

CHANGE OF BASIS (SUB-SECTION 3D4)

DEF 3.80 (*invertible, inverse, A^{-1}*):

A square matrix A is called invertible, if there is some square matrix B of the same size such that

$$AB = BA = I.$$

We call B the inverse of A denoted by $A^{-1} \equiv B$. Rules:

- $(A^{-1})^{-1} = A$
- $(AC)^{-1} = C^{-1}A^{-1}$ (Because $(AC)(C^{-1}A^{-1}) = I$ and $(C^{-1}A^{-1})(AC) = I$)

THM 3.81 (*matrix of product of linear maps*):

Suppose $T \in \mathcal{L}(U, V)$ and $S \in \mathcal{L}(V, W)$. If $u_1, \dots, u_m$ is a basis of U , $v_1, \dots, v_n$ is a basis of V and $w_1, \dots, w_p$ is a basis of W . Note that $\dim U = m$, $\dim V = n$, $\dim W = p$. Then we have:

$$\begin{aligned} \mathcal{M}(ST, (u_1, \dots, u_m), (w_1, \dots, w_p)) &= \\ \mathcal{M}(S, (v_1, \dots, v_n), (w_1, \dots, w_p)) \cdot \mathcal{M}(T, (u_1, \dots, u_m), (v_1, \dots, v_n)) \end{aligned}$$

Or using different notation:

$$\mathcal{M}(ST) = \mathcal{M}(S) \cdot \mathcal{M}(T)$$

PROOF: This is just another way of writing 3.43. □

THM 3.82 (*matrix of identity operator with respect to two bases*):

Suppose that $u_1, \dots, u_m$ and $v_1, \dots, v_m$ are bases of V . Then the matrices

$$\mathcal{M}(I, (u_1, \dots, u_m), (v_1, \dots, v_m)) \quad \text{and} \quad \mathcal{M}(I, (v_1, \dots, v_m), (u_1, \dots, u_m))$$

are invertible, and each is the inverse of the other.

PROOF: Careful application of 3.81 yields

$$\mathcal{M}(I_V) = I_n = \mathcal{M}(I_V I_V) = \mathcal{M}(I_V, (v_1, \dots, v_m), (u_1, \dots, u_m)) \cdot \mathcal{M}(I_V, (u_1, \dots, u_m), (v_1, \dots, v_m)).$$

Now, interchange the roles of the u 's and the v 's, getting

$$\mathcal{M}(I_V) = I_n = \mathcal{M}(I_V I_V) = \mathcal{M}(I_V, (u_1, \dots, u_m), (v_1, \dots, v_m)) \cdot \mathcal{M}(I_V, (v_1, \dots, v_m), (u_1, \dots, u_m)).$$

So, the product of both matrices equal to I_n , which is the identity in $\mathbb{F}^{n,n}$. □

THM 3.84 (*change-of-basis formula*):

Let $T \in \mathcal{L}(V)$. Suppose $u_1, \dots, u_m$ and $v_1, \dots, v_m$ are bases of V . Let

$$A := \mathcal{M}(T, (u_1, \dots, u_m)),$$

$$B := \mathcal{M}(T, (v_1, \dots, v_m)), \text{ and}$$

$$C := \mathcal{M}(I, (u_1, \dots, u_m), (v_1, \dots, v_m)).$$

Then

$$A = C^{-1}BC.$$

THM 3.86:

If $v_1, \dots, v_m$ is a basis V and $T \in \mathcal{L}V$ is invertible, then

$$\mathcal{M}(T^{-1}) = (\mathcal{M}(T))^{-1},$$

where both matrices are with respect to the basis $v_1, \dots, v_m$, that is

$$\mathcal{M}(T^{-1}, (v_1, \dots, v_m)) = \mathcal{M}(T, (v_1, \dots, v_m))^{-1}.$$

EXERCISES ABOUT INVERTIBILITY AND ISOMORPHISMS 3D

- (1) Suppose $T \in \mathcal{L}(V, W)$ is invertible. Show that T^{-1} is invertible and $(T^{-1})^{-1} = T$.

PROOF: T^{-1} is the unique element in $\mathcal{L}(W, V)$ such that

$$T^{-1}T = I_V \text{ and } TT^{-1} = I_W.$$

But this makes T the unique element in $\mathcal{L}(V, W)$ such that

$$T(T^{-1}) = I_W \text{ and } T^{-1}T = I_V.$$

This makes T the inverse of T^{-1} . Since the inverse is unique, we conclude that $(T^{-1})^{-1} = T$. $\square$

- (2) Suppose $T \in \mathcal{L}(U, V)$ and $S \in \mathcal{L}(V, W)$ are both invertible linear maps. Prove that $ST \in \mathcal{L}(U, W)$ is invertible and that $(ST)^{-1} = T^{-1}S^{-1}$.

PROOF: Let $T \in \mathcal{L}(U, V)$ and $S \in \mathcal{L}(V, W)$ be both invertible linear maps. We compute:

$$(ST)(T^{-1}S^{-1}) = S(TT^{-1})S^{-1} = SI_VS^{-1} = SS^{-1} = I_W$$

$$(T^{-1}S^{-1})(ST) = T^{-1}(S^{-1}S)T = T^{-1}I_VT = T^{-1}T = I_U$$

Therefore, (ST) is invertible and its inverse is $(ST)^{-1} = T^{-1}S^{-1}$. $\square$

- (3) Suppose V is finite-dimensional and $T \in \mathcal{L}(V)$. Prove that the following are equivalent:

- (a) T is invertible.
- (b) $Tv_1, \dots, Tv_n$ is a basis of V for every basis $v_1, \dots, v_n$ of V .
- (c) $Tv_1, \dots, Tv_n$ is a basis of V for some basis $v_1, \dots, v_n$ of V .

PROOF: For all proof-steps, let $T \in \mathcal{L}(V)$ and V be a finite-dimensional vector space. In each step of the proof, we show one equivalence listed above.

STEP 1: Clearly, (b) implies (c).

STEP 2: We start by showing that (c) implies (a). Actually, the proof that (b) implies (a) would look the same. Suppose $Tv_1, \dots, Tv_n$ is a basis of V for some basis $v_1, \dots, v_n$. Since $\dim V = n$, we have $\text{range } T = V$. It follows that $Tv_1, \dots, Tv_n$ is a basis of $\text{range } T$. So T is a surjective map on V by the definition 3.19 of surjectivity. Using theorem 3.65, we conclude that T is invertible. It states that for a linear map from a vector space to another one of the same dimension, invertibility, injectivity and surjectivity are equivalent.

STEP 3: Now we want to show that (c) implies (b). So let $Tv_1, \dots, Tv_n$ be a basis of V for some basis of $v_1, \dots, v_n$ of V . Let $u_1, \dots, u_n$ be another basis of V . We already know that (c) $\implies$ (a), so T is injective. This means, that $Tu_1, \dots, Tu_n$ is a basis of V as well. Since $u_1, \dots, u_n$ was arbitrary, we have shown that (c) $\implies$ (b).

STEP 4: Lastly, we want to show that (a) implies (c). Suppose $T \in \mathcal{L}(V)$ such that T is invertible. Let $v_1, \dots, v_n$ be a basis of V . Let $a_1, \dots, a_n \in \mathbb{F}$ such that

$$0 = a_1Tv_1 + \dots + a_nTv_n$$

Hence,

$$\begin{aligned} T^{-1}(0) &= T^{-1}(a_1 T v_1 + \cdots a_n T v_n) \iff \\ 0 &= a_1 T^{-1} T v_1 + \cdots + a_n T^{-1} T v_n \iff \\ 0 &= a_1 v_1 + \cdots + a_n v_n. \end{aligned}$$

Therefore, $0 = a_1 = \cdots = a_n$, because all the v 's are linearly independent. We conclude that $T v_1, \dots, T v_n$ is a linearly independent list of length $n = \dim V$ and hence a basis of V . Thus, (a) implies (c).

Overall, we have shown that (b) $\implies$ (c), (c) $\implies$ (b), (c) $\implies$ (a), and (a) $\implies$ (c). So all three statements are equivalent. $\square$

- ⑤ Suppose V is finite-dimensional, U is a subspace of V , and $S \in \mathcal{L}(U, V)$. Prove that there exists an invertible linear map $T \in \mathcal{L}(U, V)$ such that $Tu = Su \ \forall u \in U \iff S$ is injective.

PROOF: Suppose V is finite-dimensional, U is a subspace of V and $S \in \mathcal{L}(U, V)$.

$\implies$ **DIRECTION:** Suppose for every $u \in U$ we have that $Tu = Su$. Let $u_1, u_2 \in U$ such that $S(u_1) = S(u_2)$. This implies that

$$\begin{aligned} T(u_1) &= T(u_2) \\ T^{-1}T(u_1) &= T^{-1}T(u_2) \\ u_1 &= u_2 \end{aligned}$$

Hence, S is injective.

$\Leftarrow$ **DIRECTION:** Suppose S is injective. Let $u_1, \dots, u_m$ be a basis of U and $Su_1, \dots, Su_m$ be a basis of range S . We extend

$$u_1, \dots, u_m, v_{m+1}, \dots, v_n \tag{3.4}$$

to a basis of V , and

$$Su_1, \dots, Su_m, r_{m+1}, \dots, r_n \tag{3.5}$$

to a basis of V as well. We define $T \in \mathcal{L}(V)$ for each $v \in V$ as follows:

$$\begin{aligned} T(v) &= T(c_1 u_1 + \cdots + c_m u_m + c_{m+1} v_{m+1} + \cdots + c_n v_n) \\ &:= c_1 Su_1 + \cdots + c_m Su_m + c_{m+1} r_{m+1} + \cdots + c_n r_n. \end{aligned}$$

Clearly, $\forall u \in U : T(u) = S(u)$. Now we have to show that T is injective and surjective. If $v_1, v_2 \in V$ such that

$$y_1 = T(v_1) = T(v_2) = y_2$$

All the c coefficients of y_1 and y_2 , which are defined over the basis (3.5) match the c coefficients of v_1 and v_2 , which are defined over the basis (3.4). So these representations are unique and therefore $v_1 = v_2$. We conclude that T is injective. To prove that T is surjective, we have to show that for any given $y \in V$ there is a $v \in V$ such that $T(v) = y$. So let

$$y := c_1 Su_1 + \cdots + c_m Su_m + c_{m+1} r_{m+1} + \cdots + c_n r_n.$$

Then there is exactly one $v \in V$ such that $T(v) = y$, namely

$$v := c_1 u_1 + \cdots + c_m u_m + c_{m+1} v_{m+1} + \cdots + c_n v_n.$$

For this choice of y and v , we have $T(v) = y$. Hence, T is injective and surjective. This makes T invertible. $\square$

PRODUCTS AND QUOTIENTS OF VECTOR SPACES (SECTION 3E)

PRODUCTS OF VECTOR SPACES (SUB-SECTION 3E1)

DEF 3.87 (*product of vector spaces*):

The product, addition and scalar multiplication of a list of vector spaces $V_1, \dots, V_m$ is defined as follows:

$$\begin{aligned} V_1 \times \cdots \times V_m &\equiv \{(v_1, \dots, v_m) \mid v_1 \in V_1, \dots, v_m \in V_m\} \\ (u_1, \dots, u_m) + (v_1, \dots, v_m) &\equiv (u_1 + v_1, \dots, u_m + v_m) \\ \lambda(v_1, \dots, v_m) &\equiv (\lambda v_1, \dots, \lambda v_m) \end{aligned}$$

THM 3.89 (*product of vector space is a vector space*):

$V_1 \times \cdots \times V_m$ together with addition and scalar multiplication is a vector space over $\mathbb{F}$.

THM 3.92 (*dimension of a product is the sum of dimensions*):

Suppose $V_1, \dots, V_m$ are finite-dimensional vector spaces. Then $V_1 \times \cdots \times V_m$ is finite-dimensional and $\dim(V_1 \times \cdots \times V_m) = \dim V_1 + \cdots + \dim V_m$.

PROOF: Choose a basis $v_{k_1}, \dots, v_{k_{\dim V_k}}$ for each V_k . For each basis vector of each V_k , consider the element of $V_1 \times \cdots \times V_m$ that equals the basis vector in the k -th slot and 0 in the other slots.

$$\underbrace{\begin{pmatrix} v_{1_1} \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \begin{pmatrix} v_{1_2} \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \dots, \begin{pmatrix} v_{1_{\dim V_1}} \\ 0 \\ \vdots \\ 0 \end{pmatrix}}_{\text{constructed from basis vectors of } V_1}, \dots, \underbrace{\begin{pmatrix} 0 \\ \vdots \\ v_{k_1} \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ \vdots \\ v_{k_2} \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \dots, \begin{pmatrix} 0 \\ \vdots \\ v_{k_{\dim V_k}} \\ 0 \\ \vdots \\ 0 \end{pmatrix}}_{\text{constructed from basis vectors of } V_k}, \dots, \underbrace{\begin{pmatrix} 0 \\ \vdots \\ 0 \\ v_{m_1} \end{pmatrix}, \begin{pmatrix} 0 \\ \vdots \\ 0 \\ v_{m_2} \end{pmatrix}, \dots, \begin{pmatrix} 0 \\ \vdots \\ 0 \\ v_{m_{\dim V_m}} \end{pmatrix}}_{\text{constructed from basis vectors of } V_m}$$

The list of all such vector is linearly independent and spans $V_1 \times \cdots \times V_m$. Thus it is a basis of $V_1 \times \cdots \times V_m$ and the length of this basis is $(\dim V_1 + \cdots + \dim V_m)$. $\square$

In the next result, the map Γ is surjective by the definition of $V_1 + \cdots + V_m$ because of closedness. Thus the last word in the result below could be changed from “*injective*” to “*invertible*”.

THM 3.93 (*products and direct sums*):

Let $\Gamma : V_1 \times \cdots \times V_m \rightarrow V_1 + \cdots + V_m$ such that

$$\Gamma(v_1, \dots, v_m) = v_1 + \cdots + v_m \quad \forall v_1 \in V_1, \dots, v_m \in V_m.$$

Then

$$V_1 + \cdots + V_m \text{ is a direct sum} \iff \Gamma \text{ is injective/invertible.}$$

PROOF: By 3.15, Γ is injective $\iff v_1 = \cdots = v_m = 0$ [Stated correctly, Γ is injective $\iff \text{null } \Gamma = \{\vec{0}\}$].

This means, the only way to write 0 as a sum $v_1 + \cdots + v_m$, where each $v_k \in V_k$, is by taking each $v_k = 0$. Thus, 1.45 shows that Γ is injective $\iff V_1 + \cdots + V_m$ is a direct sum, as desired. $\square$

THM 3.94 (*a sum is a direct sum if and only if the dimensions add up*):

Suppose V is finite-dimensional and $V_1, \dots, V_m$ are subspaces of V . Then

$$V_1 + \cdots + V_m \text{ is a direct sum (i.e. } \sum_{i=1}^m V_i = \bigoplus_{i=1}^m V_i) \iff \dim(V_1 + \cdots + V_m) = \dim V_1 + \cdots + \dim V_m$$

PROOF: The map $\Gamma : V_1 \times \cdots \times V_m \rightarrow V_1 + \cdots + V_m$ in 3.93 is surjective. Thus, by the rank-nullity theorem

3.21, Γ is injective $\iff \dim \text{null} \Gamma = 0$ and therefore

$$\underbrace{\dim(V_1 \times \cdots \times V_m)}_{\text{domain}} = \underbrace{\dim \text{null} \Gamma}_{=0} + \underbrace{\dim(V_1 + \cdots + V_m)}_{\text{codomain} \subseteq V}$$

$$\dim(V_1 + \cdots + V_m) = \dim(V_1 \times \cdots \times V_m)$$

We can use 3.92 to conclude that

$$\dim(V_1 \times \cdots \times V_m) = \dim V_1 + \cdots + \dim V_m \text{ and therefore}$$

$$\dim(V_1 + \cdots + V_m) = \dim V_1 + \cdots + \dim V_m$$

Since Γ is injective, 3.93 tells us that $V_1 + \cdots + V_m$ is a direct sum. So, we have that

$$\dim(V_1 + \cdots + V_m) = \dim V_1 + \cdots + \dim V_m$$

if and only if $V_1 + \cdots + V_m$ is a direct sum.

SPECIAL CASE FOR $m=2$: For the special case where $m = 2$ we can use 2.37 where

$$\dim(V_1 + V_2) = \dim V_1 + \dim V_2 - \dim(V_1 \cap V_2)$$

and we get $\dim(V_1 \cap V_2) = 0 \iff V_1 + V_2 = V_1 \oplus V_2$ by 2.43. □

QUOTIENT SPACES (SUB-SECTION 3E2)

DEF 3.95 (notation $v + U$):

$v + U \equiv \{v + u \mid u \in U\}$ for $v \in V$ and $U \subseteq V$.

DEF 3.97 (translate):

The set $v + U$ is said to be a “translate” of U , where $v \in V$ and $U \subseteq V$.

DEF 3.99 (quotient space, V/U):

Suppose U is a subspace of V . Then the “quotient space” V/U is the set of all translates of U . Thus

$$V/U \equiv \{v + U \mid v \in V\}.$$

EXAMPLE 3.100:

If $U = \{(x, 2x) \in \mathbb{R}^2 \mid x \in \mathbb{R}\} \implies \mathbb{R}^2/U$ is the set of all lines with slope 2.

If U is a plane in $\mathbb{R}^3 \implies \mathbb{R}^3/U$ is the set of all planes parallel to U .

THM 3.101 (two translates of a subspace are equal or disjoint):

If U is a subspace of V and $v, w \in V$ (not a typo!). Then

$$v - w \in U \iff v + U = w + U \iff (v + U) \cap (w + U) \neq \emptyset$$

PROOF: STEP 1: First suppose $v - w \in U$. If $u \in U$, then

$$v + u = (w - w + v + u) = w + \underbrace{(v - w) + u}_{\in U}, \text{ thus } v + u \in w + U.$$

Similarly, because $v - w \in U$ we also have $w - v \in U$, and thus,

$$w + u = (v - v + w + u) = v + \underbrace{(w - v) + u}_{\in U}, \text{ hence } w + u \in v + U.$$

We have shown that $\forall u \in U : v + u \in w + U$; which is equivalent to

$$v + U \subseteq w + U$$

and $\forall u \in U : w + u \in v + U$; which is equivalent to

$$w + U \subseteq v + U.$$

This implies $v + U = w + U$, given that $v - w \in U$, which proves the first equivalence.

STEP 2: The equation $v + U = w + U$ implies that $(v + U) \cap (w + U) \neq \emptyset$.

STEP 3: Now, suppose $(v + U) \cap (w + U) \neq \emptyset$. Thus, there exist $u_1, u_2 \in U$ such that

$$v + u_1 = w + u_1,$$

Therefore, $v - w = u_2 - u_1$. Hence, $v - w \in U$. □

DEF 3.102 (*addition and scalar multiplication on V/U*):

Suppose U is a subspace of V . Then “addition” and “scalar multiplication” are defined on V/U by

$$(v + U) + (w + U) \equiv (v + w) + U$$

$$\lambda(v + U) \equiv (\lambda v) + U$$

$\forall v, w \in V$ and $\forall \lambda \in \mathbb{F}$.

THM 3.103 (*quotient space is a vector space*):

V/U is a vector space with additive identity $0 + U$ which is equal to U and the additive inverse $(-v) + U$.

DEF 3.104 (*quotient map, π*):

Suppose $U \subseteq V$. the “quotient map” $\pi : V \rightarrow V/U$ is the linear map defined by

$$\pi(v) = v + U \quad \forall v \in V.$$

THM 3.105:

Suppose $\dim V < \infty$ and $U \subseteq V$.

$$\implies \dim V/U = \dim V - \dim U.$$

DEF 3.106 (*notation $\tilde{T}$*):

Suppose $T \in \mathcal{L}(V, W)$. Define $\tilde{T} : V/(\text{null } T) \rightarrow W$ by

$$\tilde{T}(v + \text{null } T) = Tv$$

THM 3.107 (*null space and range of $\tilde{T}$*):

Suppose $T \in \mathcal{L}(V, W)$ Then

- (a) $\tilde{T} \circ \pi = T$, where π is the quotient map V onto $V/(\text{null } T)$;
- (b) $\tilde{T}$ is injective;
- (c) $\text{range } \tilde{T} = \text{range } T$;
- (d) $V/(\text{null } T)$ and $\text{range } T$ are isomorphic vector spaces.

EXERCISES ABOUT PRODUCTS AND QUOTIENTS OF VECTOR SPACES 3E

- (6) Suppose that $v, x \in V$ and that U, W are subspaces of V such that $v + U = x + W$. Prove that $U = W$. **PROOF:** Since $v \in v + U$ and $v + U = x + W$, it follows that $v \in x + W$. Hence, there is $w_0 \in W$ with $v = x + w_0$, so $(x - v) = -w_0 \in W$. Now,

$$U = (-v) + (v + U) = (-v) + (x + W) = (x - v) + W = W,$$

where the last equality uses $x - v \in W$ and the fact that translating a subspace by one of its own elements leaves it unchanged (i.e., $(x - v) + W = W$). Hence, $U = W$. □

- (10) Suppose $A_1 = v + U_1$ and $A_2 = w + U_2$ for some $v, w \in V$ and some subspaces U_1, U_2 of V . Prove that the intersection $A_1 \cap A_2$ is either a translate of some subspace of V or is the empty set.

PROOF: By the definition of intersection, we get

$$A_1 \cap A_2 = \{x \in V : x = v + u_1 \text{ and } x = w + u_2, u_1 \in U_1, u_2 \in U_2\}.$$

The condition $v + u_1 = w + u_2$ is equivalent to

$$v - w = u_2 - u_1, u_1 \in U_1, u_2 \in U_2 \iff v - w \in U_1 + U_2.$$

If $v - w \notin U_1 + U_2$, there is no solution for u_1 and u_2 and $A_1 \cap A_2 = \emptyset$. If a solution exists, put

$$\begin{aligned} r^0 &:= v + u_1^0 \\ &:= w + u_2^0, \end{aligned}$$

where $r^0 \in A_1 \cap A_2$ and for appropriate $u_1^0 \in U_1, u_2^0 \in U_2$ (i.e., $v - w = u_2^0 - u_1^0$). Define

$$S := \{r^0 + u : u \in U_1 \cap U_2\} = r^0 + (U_1 \cap U_2).$$

We want to show that $A_1 \cap A_2 = S$. Therefore, take any $x \in A_1 \cap A_2$, so that $x = v + u_1$ and $x = w + u_2$ with $u_1 \in U_1, u_2 \in U_2$. Now, it follows that

$$\begin{aligned} x - r^0 &= (v + u_1) - (v + u_1^0) = u_1 - u_1^0 \in U_1, \text{ and} \\ x - r^0 &= (w + u_2) - (w + u_2^0) = u_2 - u_2^0 \in U_2, \end{aligned}$$

and therefore, $x - r^0 \in U_1 \cap U_2$. Since $x = r^0 + (x - r^0)$, it is immediate that

$$x \in r^0 + (U_1 \cap U_2).$$

Hence, $A_1 \cap A_2 \subseteq S$. Conversely, pick any $u \in U_1 \cap U_2$, so that $r^0 + u \in S$. Then

$$\begin{aligned} r^0 + u &= (v + u_1^0) + u = v + (u_1^0 + u) \in v + U_1 = A_1, \text{ and} \\ r^0 + u &= (w + u_2^0) + u = w + (u_2^0 + u) \in w + U_2 = A_2. \end{aligned}$$

Thus, $S \subseteq A_1 \cap A_2$. Therefore, we have shown that

$$A_1 \cap A_2 = \begin{cases} S, & \text{if } v - w \in U_1 + U_2, \\ \emptyset, & \text{otherwise.} \end{cases}$$

□

DUALITY (SECTION 3F)

DUAL SPACE AND DUAL MAP (SUB-SECTION 3F1)

DEF 3.108:

A “linear functional” ϕ is an element of $\mathcal{L}(V, \mathbb{F})$. So $\phi \in \mathcal{L}(V, \mathbb{F})$

Examples:

$$\begin{aligned}\phi : \mathbb{R}^3 &\rightarrow \mathbb{R}, & \phi(x, y, z) &\mapsto 4x - 5y - 2z \\ \phi : \mathbb{F}^n &\rightarrow \mathbb{F}, & \phi(x_1, \dots, x_n) &\mapsto c_1x_1 + \dots + c_nx_n \\ \phi : \mathcal{P}(\mathbb{R}) &\rightarrow \mathbb{R}, & \phi(p) &\mapsto 3p''(5) + 7p(4) \\ \phi : \mathcal{P}(\mathbb{R}) &\rightarrow \mathbb{R}, & \phi(p) &\mapsto \int_0^1 p(x)dx\end{aligned}$$

THM 3.110:

The dual space of V , denoted by V' or V^* , is the vector space of all linear functionals on V , i.e.

$$V' := \mathcal{L}(V, \mathbb{F}).$$

THM 3.111 ($\dim V' = \dim V$):

Suppose V is finite-dimensional. Then V' is also finite-dimensional and

$$\dim V' = \dim V.$$

PROOF: By ????, we have $\dim V' = \dim \mathcal{L}(V, \mathbb{F}) = (\dim V) \cdot (\dim \mathbb{F}) = \dim V$. □

DEF 3.112:

If $v_1, \dots, v_m$ is a basis of V , then the “dual basis” of $v_1, \dots, v_m$ is the list $\varphi_1, \dots, \varphi_m \in V'$, where each φ_j is the linear functional such that

$$\varphi_j(v_k) = \delta_{j,k} = \begin{cases} 1, & \text{if } k = j \\ 0, & \text{if } k \neq j \end{cases}$$

Note that this is not the definition of each φ_j but a property of each φ_j .

THM 3.114:

Let $v_1, \dots, v_m$ be a basis of V . $\varphi_1, \dots, \varphi_m$ is called “a dual basis” of $v_1, \dots, v_m$ because $\forall v \in V$:

$$v = \varphi_1(v) \cdot v_1 + \dots + \varphi_m(v) \cdot v_m$$

PROOF: Let $v \in V$ such that $v = c_1v_1 + \dots + c_mv_m$. Applying φ_j on both sides gives $\varphi_j(v) = c_j$ for $j = 1, \dots, m$. Hence, $v = \varphi_1(v)v_1 + \dots + \varphi_m(v)v_m$. □

THM 3.116 (basis of the dual space):

The dual basis of a basis of V is a basis of the dual space V' .

PROOF: Suppose $v_1, \dots, v_n$ is a basis of V . Let $\varphi_1, \dots, \varphi_n$ denote its dual basis. Suppose

$$a_1\varphi_1 + \dots + a_n\varphi_n = 0 \text{ for some } a_1, \dots, a_n \in \mathbb{F}.$$

Which means that we get a function in $\mathcal{L}(V, \mathbb{F})$ that always returns 0. In other words, $\forall v \in V$:

$$(a_1\varphi_1 + \dots + a_n\varphi_n)(v) = 0. \text{ This implies}$$

$$(a_1\varphi_1 + \dots + a_n\varphi_n)(v_k) = a_k = 0 \quad \forall k \in \{1, \dots, n\}.$$

Thus $a_1 = \dots = a_n = 0$. Hence the list $\varphi_1, \dots, \varphi_n$ is linearly independent. Since the list is a linearly independent list in V' whose length equals $\dim V'$ by Theorem 3.111, we conclude that $\varphi_1, \dots, \varphi_n$ is a basis of V' by Theorem 2.38. □

DEF 3.118 (dual map, T'):

The “dual map” T' of $T \in \mathcal{L}(V, W)$ is the linear map $T' \in \mathcal{L}(W', V') = \mathcal{L}(\mathcal{L}(W, \mathbb{F}), \mathcal{L}(V, \mathbb{F}))$ defined as

follows.

$$T'(\varphi) := \varphi \circ T \quad \text{for every } \varphi \in W' = \mathcal{L}(W, \mathbb{F}).$$

Therefore, we have that $T'(\varphi) \in V' = \mathcal{L}(V, \mathbb{F})$. Hence, $T' \in \mathcal{L}(W', V')$ is indeed a linear map from W' to V' , since

- $\varphi, \psi \in W' \implies T'(\varphi + \psi) = (\varphi + \psi) \circ T = \varphi \circ T + \psi \circ T = T'(\varphi) + T'(\psi)$
- $\lambda \in \mathbb{F}, \varphi \in W' \implies T'(\lambda\varphi) = (\lambda\varphi) \circ T = \lambda(\varphi \circ T) = \lambda T'(\varphi)$

THM 3.120 (*algebraic properties of dual maps*):

$$T \in \mathcal{L}(V, W) \implies$$

- (a) $(S + T)' = S' + T' \quad \forall S \in \mathcal{L}(V, W)$
- (b) $(\lambda T)' = \lambda T' \quad \forall \lambda \in \mathbb{F}$
- (c) $(ST)' = T'S' \quad \forall S \in \mathcal{L}(W, U)$

DUAL SPACE AND RANGE OF DUAL OF LINEAR MAP (SUB-SECTION 3F2)

DEF 3.121 (*annihilator, U^0*):

For $U \subseteq V$, the “annihilator” of U , denoted by U^0 , is defined by

$$U^0 = \{\varphi \in V' \mid \varphi(u) = 0 \quad \forall u \in U\}.$$

THM 3.124 (*the annihilator is a subspace*):

If $U \subseteq V$, then U^0 is a subspace of V'

THM 3.125 (*dimension of the annihilator*):

If $U \subseteq V$ and $\dim V < \infty$, then

$$\dim U^0 = \dim V - \dim U.$$

THM 3.127 (*condition for the annihilator to equal $\{0\}$ or the whole space*):

If $\dim V < \infty$ and $U \subseteq V$, then

- (a) $U^0 = \{0\} \iff U = V$
- (b) $U^0 = V' \iff U = \{0\}$

THM 3.128 (*the null space of T'*):

If $\dim V < \infty$, $\dim W < \infty$ and $T \in \mathcal{L}(V, W)$, then

- (a) $\text{null } T' = (\text{range } T)^0$
- (b) $\dim \text{null } T' = \dim \text{null } T + \dim W - \dim V$

THM 3.129 (*T surjective, T' injective*):

If $\dim V < \infty$, $\dim W < \infty$ and $T \in \mathcal{L}(V, W)$, then

$$T \text{ is surjective} \iff T' \text{ is injective}$$

THM 3.130 (*the range of T'*):

If $\dim V < \infty$, $\dim W < \infty$ and $T \in \mathcal{L}(V, W)$, then

- (a) $\dim \text{range } T' = \dim \text{range } T$
- (b) $\text{range } T' = (\text{null } T)^0$

THM 3.131 (*T injective, T' surjective*):

If $\dim V < \infty$, $\dim W < \infty$ and $T \in \mathcal{L}(V, W)$, then

$$T \text{ is injective} \iff T' \text{ is surjective}.$$

MATRIX OF DUAL OF LINEAR MAP (SUB-SECTION 3F3)

THM 3.132:

Suppose V, W are finite-dimensional and $T \in \mathcal{L}(V, W)$.

$$\implies \mathcal{M}(T') = (\mathcal{M}(T))^T$$

THM 3.133:

If $A \in \mathbb{F}^{m,n}$, then the column rank of A equals the row rank of A .

EXERCISES ABOUT DUALITY 3F

- ① Explain why each linear functional is surjective or the zero map.

SOLUTION: $\varphi \in V' = \mathcal{L}(V, \mathbb{F})$ implies that for every $\lambda \in \mathbb{F}$, and every $v \in V$,

$$\varphi(\lambda v) = \lambda \varphi(v).$$

If φ is not the zero map, then φ evaluated at $v \in V$ or $\lambda v \in V$ is a scalar in $\mathbb{F}$ for some $v \in V$ for which it holds that $\varphi(v) \neq 0$. In this case, we have $\varphi(v) \in \mathbb{F}$ and $\varphi(\lambda v) = \lambda \varphi(v) \in \mathbb{F}$. Hence, $\text{range } \varphi = \mathbb{F}$, since $\lambda \in \mathbb{F}$ was arbitrary.

- ③ Suppose V is finite-dimensional and $v \in V$ is nonzero. Prove that there exists $\varphi \in V'$ such that $\varphi(v) = 1$.

SOLUTION: Let $v_1, \dots, v_n$ be a basis of V . Let $v \in V$ be nonzero such that $v = a_1 v_1 + \dots + a_n v_n$, where $a_i \in \mathbb{F}$. Since v is nonzero, there exists a k such that $a_k \neq 0$. Define $\varphi : V \rightarrow \mathbb{R}$ by

$$\varphi(u) = \frac{c_k}{a_k},$$

whenever $u = c_1 v_1 + \dots + c_n v_n \in V$. Note that this definition corresponds to the use of the linear map lemma (3.4), where we define φ on each basis vector as follows:

$$\varphi(v_1) = 0, \dots, \varphi(v_{k-1}) = 0, \varphi(v_k) = \frac{1}{a_k}, \varphi(v_{k+1}) = 0, \dots, \varphi(v_n) = 0.$$

Hence, $\varphi(a_1 v_1 + \dots + a_n v_n) = \frac{a_k}{a_k} = 1$. It is easy to check that φ is a linear map.

- ⑤ Suppose V is finite dimensional and U is a subspace to V , where $U \neq V$. Prove that there exists a $\varphi \in V'$ such that $\varphi(u) = 0$ for every $u \in U$ but $\varphi \neq 0$.

SOLUTION: Choose $W \subset V$ such that $V = U \oplus W$. Hence, there is a basis

$$u_1, \dots, u_n, w_1, \dots, w_m$$

of V such that the u 's and w 's form a basis of U and W respectively. We define $\varphi \in V'$ such that for $v \in V$, written as

$$v = a_1 u_1 + \dots + a_n u_n + b_1 w_1 + \dots + b_m w_m,$$

we have

$$\varphi(v) = b_1 + \dots + b_m.$$

It is easy to see that $\varphi(u) = 0$ for every $u \in U$ (since $U \cap W = \{0\}$, by 1.46) and that φ is linear.

- ⑩ Suppose m is positive integer.

(a) Show that $1, x - 5, \dots, (x - 5)^m$ is a basis of $\mathcal{P}_m(\mathbb{R})$.

(b) What is the dual basis of the basis in (a)?

SOLUTION: We first start the exercise by solving it for $\mathcal{P}_4(\mathbb{R})$ straight forward before solving it more abstract and generic for $\mathcal{P}_m(\mathbb{R})$. Recall that $e_0 = x^0, e_1 = x^1, \dots, e_4 = x^4$ is the standard basis of $\mathcal{P}_4(\mathbb{R})$. We want to show that $v_0 = 1, v_1 = x - 5, \dots, v_4 = (x - 5)^4$ is a basis as well. If we do

some rewriting and expansion of the formulas, we get

$$\begin{aligned} v_0 &= (x^1 - 5x^0)^0 = 1x^0, \\ v_1 &= (x^1 - 5x^0)^1 = -5x^0 + x^1, \\ v_2 &= (x^1 - 5x^0)^2 = 25x^0 - 10x^1 + x^2, \\ v_3 &= (x^1 - 5x^0)^3 = -125x^0 + 75x^1 - 15x^2 + x^3, \\ v_4 &= (x^1 - 5x^0)^4 = 625x^0 - 500x^1 + 150x^2 - 20x^3 + x^4. \end{aligned}$$

Now, for a given $p \in \mathcal{P}_4(\mathbb{R})$ so that $p(x) = b_0x^0 + b_1x^1 + \dots + b_4x^4$ ($b_j \in \mathbb{F}$) we want to show that there exists $a_i \in \mathbb{F}$ such that $\sum_{j=0}^4 b_j e_j = \sum_{j=0}^4 a_j v_j$. So let's expand

$$\begin{aligned} \sum_{j=0}^4 a_j v_j &= a_0(1x^0) + a_1(-5x^0 + x^1) + a_2(25x^0 - 10x^1 + x^2) \\ &\quad + a_3(-125x^0 + 75x^1 - 15x^2 + x^3) + a_4(625x^0 - 500x^1 + 150x^2 + x^4) \\ &= (a_0 - 5a_1 + 25a_2 - 125a_3 + 625a_4) + (a_1 - 10a_2 + 75a_3 - 500a_4)x \\ &\quad + (a_2 - 15a_3 + 150a_4)x^2 + (a_3 - 20a_4)x^3 + a_4x^4 \\ &= b_0x^0 + b_1x^1 + \dots + b_4x^4 = \sum_{j=0}^4 b_j e_j. \end{aligned}$$

We can solve this for a_j by back substitution:

$$\begin{aligned} a_4 &= b_4, \\ a_3 &= b_3 + 20a_4, \\ a_2 &= b_2 + 15a_3 - 150a_4, \\ a_1 &= b_1 + 10a_2 + 75a_3 - 500a_4, \text{ and finally} \\ a_0 &= b_0 + 5a_1 - 25a_2 + 125a_3 - 625a_4. \end{aligned}$$

Hence, $v_1, \dots, v_4$ is indeed a basis of $\mathcal{P}_4(\mathbb{R})$. Now, we want to find the dual basis $\psi_0, \psi_1, \dots, \psi_4$ of $v_1, \dots, v_4$. Recall that the condition

$$\phi_j(v_k) = \delta_{j,k} = \begin{cases} 1, & j = k \\ 0, & j \neq k \end{cases}$$

must hold, and we thus want to define each ϕ_j on the standard basis $e_0, \dots, e_4$. So let's start with ψ_0 , where $\psi_0(v_0) = 1$ and zero on all other v 's.

$$\begin{aligned} \psi_0(v_0) &= 1\psi_0(x^0) = 1. \implies \psi_0(x^0) = 1. \\ \psi_0(v_1) &= -5\psi_0(x^0) + \psi_0(x^1) = 0. \implies \psi_0(x^1) = 5. \\ \psi_0(v_2) &= 25\psi_0(x^0) - 10\psi_0(x^1) + \psi_0(x^2) = 0. \implies \psi_0(x^2) = 25. \\ \psi_0(v_3) &= -125\psi_0(x^0) + 75\psi_0(x^1) - 15\psi_0(x^2) + \psi_0(x^3) = 0. \implies \psi_0(x^3) = 125. \\ \psi_0(v_4) &= 625\psi_0(x^0) - 500\psi_0(x^1) + 150\psi_0(x^2) - 20\psi_0(x^3) + \psi_0(x^4) = 0. \implies \psi_0(x^4) = 625. \end{aligned}$$

Now, the same for ψ_1 :

$$\begin{aligned} \psi_1(v_0) &= 1\psi_1(x^0) = 0. \implies \psi_1(x^0) = 0. \\ \psi_1(v_1) &= -5\psi_1(x^0) + \psi_1(x^1) = 1. \implies \psi_1(x^1) = 1. \\ \psi_1(v_2) &= 25\psi_1(x^0) - 10\psi_1(x^1) + \psi_1(x^2) = 0. \implies \psi_1(x^2) = 10. \\ \psi_1(v_3) &= -125\psi_1(x^0) + 75\psi_1(x^1) - 15\psi_1(x^2) + \psi_1(x^3) = 0. \implies \psi_1(x^3) = 75. \\ \psi_1(v_4) &= 625\psi_1(x^0) - 500\psi_1(x^1) + 150\psi_1(x^2) - 20\psi_1(x^3) + \psi_1(x^4) = 0. \implies \psi_1(x^4) = 500. \end{aligned}$$

Now, ψ_2 :

$$\begin{aligned} \psi_2(v_0) &= 1\psi_2(x^0) = 0. \implies \psi_2(x^0) = 0. \\ \psi_2(v_1) &= -5\psi_2(x^0) + \psi_2(x^1) = 0. \implies \psi_2(x^1) = 0. \\ \psi_2(v_2) &= 25\psi_2(x^0) - 10\psi_2(x^1) + \psi_2(x^2) = 1. \implies \psi_2(x^2) = 1. \\ \psi_2(v_3) &= -125\psi_2(x^0) + 75\psi_2(x^1) - 15\psi_2(x^2) + \psi_2(x^3) = 0. \implies \psi_2(x^3) = 15. \\ \psi_2(v_4) &= 625\psi_2(x^0) - 500\psi_2(x^1) + 150\psi_2(x^2) - 20\psi_2(x^3) + \psi_2(x^4) = 0. \implies \psi_2(x^4) = 150. \end{aligned}$$

For ψ_3 we get $\psi_3(x^0) = \psi_3(x_1) = \psi_3(x^2) = 0$ and $\psi_3(x^3) = 1$. Now what is left to calculate is

$$\psi_3(v_4) = 625\psi_2(x^0) - 500\psi_2(x^1) + 150\psi_2(x^2) - 20\psi_2(x^3) + \psi_2(x^4) = 0.$$

Hence, $\psi_4(v_4) = 20$. For ψ_4 we get $\psi_4(x^0) = \psi_4(x_1) = \dots = \psi_4(x^3) = 0$, and $\psi_4(v_4) = 1$, and thus, $\psi_4(x^4) = 1$. Now, let's take a more generic approach making extensive use of the binomic formula

$$(a+b)^n = \sum_{i=0}^n \binom{n}{i} a^i b^{n-i}, \quad (a, b \in \mathbb{R}, n \in \mathbb{N}),$$

and Pascal's identity

$$\binom{n}{k} + \binom{n}{k+1} = \binom{n+1}{k+1}, \quad (n, k \in \mathbb{N}).$$

We start by rewriting once again

$$v_0 = (x^1 - 5x^0)^0 = \binom{0}{0}(-5)^0 e_0 = e^0,$$

$$v_1 = (x^1 - 5x^0)^1 = \binom{1}{0}(-5)^1 e_0 + \binom{1}{1}(-5)^0 e_1,$$

$$v_2 = (x^1 - 5x^0)^2 = \binom{2}{0}(-5)^2 e_0 + \binom{2}{1}(-5)^1 e_1 + \binom{2}{2}(-5)^0 e_2,$$

$\vdots$

$$v_m = (x^1 - 5x^0)^m = \binom{m}{0}(-5)^m e_0 + \binom{m}{1}(-5)^{m-1} e_1 + \dots + \binom{m}{m}(-5)^0 e_m = \sum_{i=0}^m \binom{m}{i}(-5)^{m-i} e_i.$$

Now, $\psi_j(e_k) = \binom{k}{j} 5^{k-j}$, taylor polynomials. (why?).

ChatGPT: Now, why is $\psi_j(e_k) = \binom{k}{j} \cdot 5^{k-j}$? We want to define the dual basis $\{\psi_j\}_{j=0}^m$ of $\{v_k\}_{k=0}^m$, meaning we require

$$\psi_j((x-5)^k) = \delta_{j,k}.$$

Using the binomial theorem, we can expand each $(x-5)^k$ as

$$(x-5)^k = \sum_{i=0}^k \binom{k}{i} (-5)^{k-i} e_i.$$

To satisfy the condition $\psi_j((x-5)^k) = \delta_{j,k}$, it is enough to define ψ_j on the standard basis $\{e_k\}$ by

$$\psi_j(e_k) = \binom{k}{j} \cdot 5^{k-j}.$$

This works because ψ_j corresponds to applying the j -th derivative at $x = 5$, divided by $j!$, to polynomials:

$$\psi_j(p(x)) = \frac{1}{j!} p^{(j)}(5),$$

and

$$\frac{1}{j!} \cdot \frac{d^j}{dx^j} x^k \Big|_{x=5} = \binom{k}{j} \cdot 5^{k-j}.$$

Thus, $\{\psi_j\}_{j=0}^m$ is indeed the dual basis of $\{v_k\}_{k=0}^m$.

11 Suppose $v_1, \dots, v_n$ is a basis of V and $\varphi_1, \dots, \varphi_n$ is the corresponding dual basis of V' . Suppose $\psi \in V'$. Prove that

$$\psi = \psi(v_1)\varphi_1 + \dots + \psi(v_n)\varphi_n. \quad (3.6)$$

SOLUTION: For a given $\psi \in V'$, let $\Gamma \in V'$ defined by

$$\Gamma := \psi(v_1)\varphi_1 + \dots + \psi(v_n)\varphi_n$$

just like in (3.6). We want to show that $\Gamma = \psi$. Observe that Γ is a linear functional, since it is

the linear combination of some linear functionals $\varphi_1, \dots, \varphi_n$ (which are the dual basis of V). In this case, $\psi(v_1), \dots, \psi(v_n)$ are the scalars of this sum, which do not change for a given ψ and a given basis of V . If we now apply Γ to each v_k (for $k = 1, \dots, n$), we get

$$\begin{aligned}\Gamma(v_k) &= (\psi(v_1)\varphi_1 + \dots + \psi(v_n)\varphi_n)(v_k) \\ &= \psi(v_1)\varphi_1(v_k) + \dots + \psi(v_k)\varphi_n(v_k) + \dots + \psi(v_n)\varphi_n(v_k) \\ &= \psi(v_1) \cdot 0 + \dots + \psi(v_k) \cdot 1 + \dots + \psi(v_k) \cdot 0 \\ &= \psi(v_k).\end{aligned}$$

Hence, Γ and ψ agree on every basis vector v_k . Therefore, $\Gamma = \psi$.

14 Define $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ by

$$T(x, y, z) := (4x + 5y + 6z, 7x + 8y + 9z).$$

Suppose φ_1, φ_2 denotes the dual basis of the standard basis of $\mathbb{R}^2$ and ψ_1, ψ_2, ψ_3 denotes the dual basis of the standard basis of $\mathbb{R}^3$.

(a) Describe the linear functionals $T'(\varphi_1)$ and $T'(\varphi_2)$.

(b) Write $T'(\varphi_1)$ and $T'(\varphi_2)$ as a linear combination of ψ_1, ψ_2, ψ_3 .

SOLUTION: Let's start by getting comfortable about the notation. Let e_1, e_2, e_3 denote the standard basis of $\mathbb{R}^3$ and e_1, e_2 denote the standard basis of $\mathbb{R}^2$. In this case, $T(x, y, z) = (4x + 5y + 6z, 7x + 8y + 9z)$ actually means

$$T(xe_1 + ye_2 + ze_3) = (4x + 5y + 6z)e_1 + (7x + 8y + 9z)e_2$$

Thus, we have $Te_1 = 4e_1 + 7e_2$, $Te_2 = 5e_1 + 8e_2$, and $Te_3 = 6e_1 + 9e_2$. The Matrix $\mathcal{M}(T)$ of T would look like this:

$$\mathcal{M}(T) = \begin{matrix} & \begin{matrix} e_1 & e_2 & e_3 \end{matrix} \\ \begin{matrix} e_1 \\ e_2 \end{matrix} & \begin{pmatrix} 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} \end{matrix}$$

Now, $T'(\varphi_1) = \varphi_1 \circ T$ and $T'(\varphi_2) = \varphi_2 \circ T$. Hence,

$$T'(\varphi_1)(xe_1 + ye_2 + ze_3) = \varphi_1((4x + 5y + 6z)e_1 + (7x + 8y + 9z)e_2) = 4x + 5y + 6z, \text{ and}$$

$$T'(\varphi_2)(xe_1 + ye_2 + ze_3) = \varphi_2((4x + 5y + 6z)e_1 + (7x + 8y + 9z)e_2) = 7x + 8y + 9z,$$

which answers question (a). For question (b), we use the result (3.6) from question 11 and write

$$T'(\varphi_1) = \sum_{k=1}^3 T'(\varphi_1)(e_k)\psi_k = 4\psi_1 + 5\psi_2 + 6\psi_3, \text{ and}$$

$$T'(\varphi_2) = \sum_{k=1}^3 T'(\varphi_2)(e_k)\psi_k = 7\psi_1 + 8\psi_2 + 9\psi_3$$

VARIATION OF QUESTION 14: The question becomes more interesting if we use another basis for $\mathbb{R}^2$ and $\mathbb{R}^3$, say $v_1 = (1, 1), v_2 = (2, 1)$ for $\mathbb{R}^2$ and $w_1 = (2, 2, 0), w_2 = (4, 2, 0), w_3 = (0, 0, 1)$ for $\mathbb{R}^3$. In this case, let ϕ_1, ϕ_2 denote the dual basis of v_1, v_2 and ψ_1, ψ_2, ψ_3 denote the dual basis of w_1, w_2, w_3 . Now, we have to calculate ϕ and ψ first. Per definition, we have

$$\phi(v) = \phi(a_1v_1 + a_2v_2) = a_1\phi_1(v_1) + a_2\phi_2(v_2).$$

And we must assure that $\phi_1(v_1) = 1, \phi_1(v_2) = 0$ and $\phi_2(v_1) = 0$ and $\phi_2(v_2) = 1$. Using the facts that $v_1 = e_1 + e_2$ and $v_2 = 2e_1 + e_2$, we get

$$\begin{vmatrix} 1\phi_1(e_1) + 1\phi_1(e_2) = 1 \\ 2\phi_1(e_1) + 1\phi_1(e_2) = 0 \end{vmatrix} \implies \begin{vmatrix} 1\phi_1(e_1) + 1\phi_1(e_2) = 1 \\ -2\phi_1(e_2) = -2 \end{vmatrix},$$

and hence, $\phi_1(e_1) = -1$ and $\phi_1(e_2) = 2$. Likewise,

$$\begin{cases} 1\phi_2(e_1) + 1\phi_2(e_2) = 0 \\ 2\phi_2(e_1) + 1\phi_2(e_2) = 1 \end{cases} \implies \begin{cases} 1\phi_2(e_1) + 1\phi_2(e_2) = 0 \\ -2\phi_2(e_2) = 1 \end{cases}.$$

Therefore, $\phi_2(e_1) = 1$ and $\phi_2(e_2) = -1$. To write ϕ_1 and ϕ_2 as members of $\mathbb{R}^2 \rightarrow \mathbb{R}$, we have $\phi_1(x, y) = -x + 2y$ and $\phi_2(x, y) = x - y$. Now,

$$T'(\phi_1) = \phi_1 \circ T = -(4x + 5y + 6z) + 2(7x + 8y + 9z) = 10x + 11y + 12z, \text{ and}$$

$$T'(\phi_2) = \phi_2 \circ T = 4x + 5y + 6z - (7x + 8y + 9z) = -3x - 3y - 3z = -3(x + y + z).$$

With the same approach, we can calculate the dual basis for w_1, w_2, w_3 . We get $\psi_1(x, y, z) = -\frac{1}{2}x + y$, $\psi_2(x, y, z) = \frac{1}{2}x - \frac{1}{2}y$ and $\psi_3(x, y, z) = z$. To answer question (b), we use the theorem (3.6) from question 11 and write

$$\begin{aligned} T'(\varphi_1) &= \sum_{k=1}^3 T'(\varphi_1)(w_k)\psi_k = (20 + 22 + 0)\psi_1 + (40 + 22 + 0)\psi_2 + (0 + 0 + 12)\psi_3 \\ &= 42\psi_1 + 62\psi_2 + 12\psi_3, \text{ and} \end{aligned}$$

$$T'(\varphi_2) = \sum_{k=1}^3 T'(\varphi_2)(w_k)\psi_k = -12\psi_1 - 18\psi_2 - 3\psi_3.$$

16 Suppose W is finite-dimensional and $T \in \mathcal{L}(V, W)$. Prove that

$$T' = 0 \iff T = 0.$$

PROOF: $\Rightarrow$ **DIRECTION:** Assume $T' = 0$. By definition, $T' = \varphi \circ T = 0$ for every $\varphi \in W'$. If $W = \{0\}$, then necessarily $T = 0$, so $T' = 0$ holds trivially. For this reason, assume for now that $W \neq 0$ and let $w_1, \dots, w_n$ be a basis of W . Suppose for the sake of contradiction that $T \neq 0$. Hence, there exists a nonzero $v \in V$ and a nonzero $w \in W$ such that $w = T(v)$. Express w as $w = a_1w_1 + \dots + a_nw_n$. Since $w \neq 0$, pick k such that $a_k \neq 0$. Define $\varphi \in W'$ by

$$\varphi(b_1w_1 + \dots + b_nw_n) = \frac{b_k}{a_k},$$

where b_k is the k -th coordinate of φ 's input. Hence, $\varphi(w) = 1$. But then

$$T'(\varphi)(v) = \varphi(T(v)) = \varphi(w) = 1 \neq 0.$$

Thus, $T'(\varphi) \neq 0$, contradicting $T' = 0$. Hence, $T = 0$.

$\Leftarrow$ **DIRECTION:** If $T = 0$, then for each $\varphi \in W'$ and $T' \in \mathcal{L}(W', V')$ we have

$$T'(\varphi)(v) = \varphi(T(v)) = \varphi(0) = 0.$$

Hence, $T'(\varphi)$ is always sent to a map in W' that is the zero map. $\square$

16 Suppose V and W are finite-dimensional and $T \in \mathcal{L}(V, W)$. Prove that

$$T \text{ is invertible} \iff T' \in \mathcal{L}(W', V') \text{ is invertible.}$$

PROOF: $\Rightarrow$ **DIRECTION:** Let $T \in \mathcal{L}(V, W)$ be invertible and $T^{-1} \in \mathcal{L}(W, V)$ be the inverse of T . We want first want to show that $(T^{-1})'$ is the inverse of T' , that is

$$(T^{-1})' = (T')^{-1}.$$

For $(T^{-1})' \in \mathcal{L}(V', W')$ to be an inverse of $T' \in \mathcal{L}(W', V')$, it must hold that $(T^{-1})'T' = I_{W'}$ and $T'(T^{-1})' = I_{V'}$. Note that $I_W \neq I_{W'}$ and $I_V \neq I_{V'}$. Therefore, let's compute for $\psi \in V'$:

$$((T^{-1})' \circ T')(\psi) = (T^{-1})'(\psi \circ T^{-1}) = (\psi \circ T) \circ T^{-1} = \psi \circ (TT^{-1}) = \psi.$$

Thus, $T'(T^{-1})' = I_{V'}$. The other composition works by the same argument. Let $\varphi \in W'$. So,

$$(T' \circ (T^{-1})')(\varphi) = T'(\varphi \circ T^{-1}) = (\varphi \circ T^{-1}) \circ T = \varphi \circ (T^{-1}T) = \varphi,$$

for that $(T^{-1})'T' = I_{W'}$. Hence, T' is invertible with $(T^{-1})' = (T')^{-1}$.

$\Leftarrow$ **DIRECTION:** If T' is invertible, then it is surjective (and also injective) by 3.65. Take any nonzero $v \in V$ and $\varphi \in V'$ with $\varphi(v) \neq 0$. Surjectivity of T' yields a $\psi \in W'$ with $T'(\psi) = \varphi$, or $\varphi = \psi \circ T$. Therefore,

$$0 \neq \varphi(v) = \psi(Tv) = (\psi \circ T)(v).$$

This implies that $Tv \neq 0$, and thus, $\text{null } T = \{0\}$ (since v was arbitrary and nonzero). This forces T to be injective. Because T' is invertible, we have $\dim W' = \dim V'$, and hence, $\dim W = \dim V$ (see 3.111). An injective linear map between finite-dimensional vector spaces of the same dimension is surjective, so T is bijective, and therefore, invertible. $\square$

POLYNOMIALS (CHAPTER 4)

$\mathbb{F}$ denotes $\mathbb{R}$ or $\mathbb{C}$.

ZEROS OF POLYNOMIALS (SECTION 4A)

$$p : \mathbb{F} \rightarrow \mathbb{F}, p(z) = a_0 + a_1 z + \cdots + a_m z^m \quad \forall z \in \mathbb{F}.$$

DEF 4.5 (zero of a polynomial):

$\lambda \in \mathbb{F}$ is called a “zero” (or “root”) of a polynomial p if $p(\lambda) = 0$.

THM 4.6 (each zero of a polynomial corresponds to a degree-one factor):

Let $p \in \mathcal{P}_m(\mathbb{F})$ and $\lambda \in \mathbb{F}$. Then

$$p(\lambda) = 0 \iff \exists q \in \mathcal{P}_{m-1}(\mathbb{F}) : p(z) = (z - \lambda)q(z) \quad \forall z \in \mathbb{F}.$$

PROOF: Let $p \in \mathcal{P}(\mathbb{F})_m$ for $m \in \mathbb{N}$.

$\Leftarrow$ **DIRECTION:** Suppose there is $q \in \mathcal{P}(\mathbb{F})$ such that $p(z) = (z - \lambda)q(z) \quad \forall z \in \mathbb{F}$. Then

$$p(\lambda) = (\lambda - \lambda)q(\lambda) = 0, \text{ as desired.}$$

$\Rightarrow$ **DIRECTION:** Now suppose $p(\lambda) = 0$. Let $a_0, a_1, \dots, a_m \in \mathbb{F}$ such that

$$p(z) \equiv a_0 + a_1 z + \cdots + a_m z^m.$$

Thus

$$p(z) = p(z) - p(\lambda) = a_1(z - \lambda) + a_2(z^2 - \lambda^2) + \cdots + a_m(z^m - \lambda^m) \quad \forall z \in \mathbb{F}.$$

Using the binomic formula $z^k - \lambda^k = (z - \lambda) \sum_{j=1}^k \lambda^{j-1} z^{k-j} \quad \forall k \in \{1, \dots, m\}$, we get

$$p(z) = (z - \lambda) \underbrace{\left(a_1 + a_2(z + \lambda) + a_3(z^2 + \lambda z + \lambda^2) + a_4(z^3 + \lambda z^2 + \lambda^2 z + \lambda^3) + \cdots + a_m \sum_{j=1}^m \lambda^{j-1} z^{m-j} \right)}_{\text{a polynomial with degree } m-1}$$

Thus p equals $z - \lambda$ times some polynomial of degree $m - 1$, as desired. $\square$

In case you are wondering where the a_0 went by subtracting $p(\lambda)$ from $p(z)$, it is hidden in the roots!

For example $p \in \mathcal{P}_1(\mathbb{F}) : p = a_0 + a_1 z = \left(z - \left(-\frac{a_0}{a_1} \right) \right) (a_1)$ where $\left(-\frac{a_0}{a_1} \right)$ is the only root!

THM 4.8 (degree m implies at most m roots / zeros):

A polynomial $p \in \mathcal{P}_m(\mathbb{F})$ of degree m has at most m zeros in $\mathbb{F}$.

PROOF: For $m = 1$, $a_0 + a_1 z$ has only one zero, which is $-\frac{a_0}{a_1}$.

Now for $m > 1$ let us assume the desired result holds for $m - 1$.

If $p \in \mathcal{P}_m(\mathbb{F})$ has no zeros, we are done.

If p has a zero $\lambda \in \mathbb{F}$, by 4.6, there is a $q \in \mathcal{P}(\mathbb{F})_{m-1}$ such that

$$p(z) = (z - \lambda)q(z).$$

Since q has at most $m - 1$ zeros, p has at most m zeros. $\square$

DIVISION ALGORITHM FOR POLYNOMIALS (SECTION 4B)

THM 4.9:

Suppose $p, s \in \mathcal{P}(\mathbb{F})$, $s \neq 0$. Then $\exists q, r \in \mathcal{P}(\mathbb{F})$ such that

$$p = sq + r \quad \text{and} \quad \deg r < \deg s.$$

PROOF: Let $n \equiv \deg p$ and $m \equiv \deg s$.

If $n < m$, then take $q = 0$ and $r = p$.

Thus now we assume $n \geq m$. The list

$$1, z, \dots, z^{m-1}, s, zs, \dots, z^{n-m}s \quad (4.1)$$

is linearly independent because each polynomial in this list has a different degree.

The list has length $1 + (m-1) + 1 + (n-m) = n+1$, which equals $\dim \mathcal{P}_n(\mathbb{F})$. Hence (4.1) is a basis of $\mathcal{P}_n(\mathbb{F})$ by 2.38. Therefore, we have $a_0, a_1, \dots, a_{m-1} \in \mathbb{F}$ and $b_0, b_1, \dots, b_{n-m} \in \mathbb{F}$ such that

$$\begin{aligned} p &= a_0 + a_1 z + a_2 z^2 + \dots + a_{m-1} z^{m-1} + b_0 s + b_1 zs + b_2 z^2 s + \dots + b_{n-m} z^{n-m} s \\ &= \underbrace{(a_0 + a_1 z + a_2 z^2 + \dots + a_{m-1} z^{m-1})}_r + \underbrace{(s)}_s \underbrace{(b_0 + b_1 z + b_2 z^2 + \dots + b_{n-m} z^{n-m})}_q \\ &= r + sq = sq + r, \text{ with } \deg r = m-1 \leq \deg s = m, \end{aligned}$$

as desired.

The uniqueness of $q, r \in \mathcal{P}(\mathbb{F})$ follows from the uniqueness of the a 's and b 's, because (4.1) is a basis of $\mathcal{P}_n(\mathbb{F})$. $\square$

FACTORIZATION OF POLYNOMIALS OVER $\mathbb{C}$ (SECTION 4C)

THM 4.12 (FUNDAMENTAL THEOREM OF ALGEBRA, first version):

Every non-constant polynomial with complex coefficients has a zero in $\mathbb{C}$.

THM 4.13 (FUNDAMENTAL THEOREM OF ALGEBRA, second version):

If $p \in \mathcal{P}(\mathbb{C})$ is nonconstant polynomial (i.e. $\deg p \geq 1$), then p has a unique factorization

$$p(z) = c \cdot (z - \lambda_1) \cdots (z - \lambda_m)$$

where $c, \lambda_1, \dots, \lambda_m \in \mathbb{C}$. This factorization is unique except for the order of the factors.

PROOF: Let $p \in \mathcal{P}(\mathbb{C})$ and put $m := \deg p$. If $m = 1$, then

$$p = a_0 + a_1 z = a_1 \cdot \left(z - \frac{a_0}{a_1} \right),$$

so the the desired factorization exists and is unique.

Hence, assume that $m > 1$ and that the desired factorization exists and is unique for all polynomials of degree $m-1$. By the first version of the fundamental theorem of algebra 4.12, p has a zero $\lambda \in \mathbb{C}$. By 4.6, there is polynomial q of degree $m-1$ such that

$$p(z) = (z - \lambda)q(z) \quad \forall z \in \mathbb{C}.$$

Our induction hypothesis implies that q has the desired factorization, which when plugged into the equation above, gives the desired factorization of p .

UNIQUENESS: The number c is uniquely determined as the coefficient a_m of z^m in p . So we only need to show that except for the order, there is only one way to choose $\lambda_1, \dots, \lambda_m$. If we have some $\tau_1, \dots, \tau_m \in \mathbb{C}$ such that

$$\cancel{c} \cdot (z - \lambda_1) \cdots (z - \lambda_m) = \cancel{c} \cdot (z - \tau_1) \cdots (z - \tau_m) \quad \forall z \in \mathbb{C},$$

then because the left side of the equation above equals 0 when $z = \lambda_1$, one of the τ 's on the right hand side equals λ_1 , because every zero of a polynomial corresponds to a degree-one factor by 4.6. Without loss of generality, we can assume by relabeling, that $\tau_1 = \lambda_1$. Now, if $z \neq \lambda_1$, we can use division to get

$$\begin{aligned} \cancel{(z - \lambda_1)}(z - \lambda_2) \cdots (z - \lambda_m) &= \cancel{(z - \tau_1)}(z - \tau_2) \cdots (z - \tau_m) & \forall z \in \mathbb{C} \\ (z - \lambda_2) \cdots (z - \lambda_m) &= (z - \tau_2) \cdots (z - \tau_m) & \forall z \in \mathbb{C} \setminus \{\lambda_1\} \end{aligned} \quad (4.2)$$

$$0 = \left((z - \lambda_2) \cdots (z - \lambda_m) \right) - \left((z - \tau_2) \cdots (z - \tau_m) \right) \quad \forall z \in \mathbb{C} \setminus \{\lambda_1\} \quad (4.3)$$

Note that the last 2 equations hold for all $z \in \mathbb{C}$ except possibly $z = \lambda_1$. Remember that our goal is to use induction on the degree of the polynomial. So it would come in handy if the remaining polynomials (4.2) would be equal after cancelation and behave like any other polynomial. But we have derived those equations by the assumption that $z \neq \lambda_1$ using division. Here comes the trick. Let us assume that $(z - \lambda_2) \cdots (z - \lambda_m)$ and $(z - \tau_2) \cdots (z - \tau_m)$ are not always equal. Thus, with the standard laws of polynomial arithmetic, we can form a new polynomial $s \in \mathcal{P}_{m-1}(\mathbb{C})$ with

$$s(z) := ((z - \lambda_2) \cdots (z - \lambda_m)) - ((z - \tau_2) \cdots (z - \tau_m)),$$

where we assume that s is not the zero polynomial. Using the last equation in (4.3), we know that $s(z) = 0 \quad \forall z \in \mathbb{C} \setminus \{\lambda_1\}$. But we know that s can have at most $m - 1$ roots by 4.8. A contradiction, so it must be the case that s is the zero polynomial. Hence, $\forall z \in \mathbb{C} : s(z) = 0$. The zero polynomial is allowed to have infinitely many roots. This is only the case if $\forall z \in \mathbb{C}$:

$$(z - \lambda_2) \cdots (z - \lambda_m) = (z - \tau_2) \cdots (z - \tau_m).$$

So, we get two identical polynomials after canceling out $(z - \lambda_1)$. By our induction hypothesis, the equation above implies that except for the order, the λ 's are the same as the τ 's, completing the proof of uniqueness. $\square$

FACTORIZATION OF POLYNOMIALS OVER $\mathbb{R}$ (SECTION 4D)

THM 4.14 (*polynomials with real coefficients have nonreal zeros in pairs*):

Suppose $p \in \mathcal{P}(\mathbb{C})$ is a polynomial with real coefficients. If $\lambda \in \mathbb{C}$ is a zero/root of p , then so is $\bar{\lambda}$.

PROOF: Let $p \in \mathcal{P}(\mathbb{C})$ be defined by

$$p(z) := a_0 + a_1 z + \cdots + a_m z^m \quad (\forall z \in \mathbb{C}),$$

where $a_0, \dots, a_m \in \mathbb{R}$. Suppose $\lambda \in \mathbb{C}$ is a zero of p . Then

$$a_0 + a_1 \lambda + \cdots + a_m \lambda^m = 0$$

$$\overline{a_0 + a_1 \lambda + \cdots + a_m \lambda^m} = \bar{0}$$

$$\overline{a_0} + \overline{a_1 \lambda} + \cdots + \overline{a_m \lambda^m} = 0$$

$$a_0 + a_1 \bar{\lambda} + \cdots + a_m \bar{\lambda}^m = 0$$

The equation above shows that $\bar{\lambda}$ is a zero of p as well. $\square$

THM 4.15 (*factorization of a quadratic polynomial*):

Let $p(x) = x^2 + bx + c$ ($\forall x \in \mathbb{R}$) such that $b, c \in \mathbb{R}$. Then p can be written as

$$p(x) = (x - \lambda_1)(x - \lambda_2) \quad (\forall x \in \mathbb{R})$$

with $\lambda_1, \lambda_2 \in \mathbb{R} \iff b^2 \geq 4c$.

PROOF: Write

$$\begin{aligned} x^2 + bx + c &= \left(x^2 + 2 \left(\frac{b}{2} \right) x + \left(\frac{b}{2} \right)^2 \right) + \left(c - \frac{b^2}{4} \right) \\ &= \left(x + \frac{b}{2} \right)^2 + \left(c - \frac{b^2}{4} \right). \end{aligned}$$

Therefore, if $b^2 < 4c$, the first and the second term are always positive and by 4.6, no factoriza-

tion is possible, because p has no zeros. If $b^2 \geq 4c$, set $d^2 := (b^2/4 - c)$ and rewrite p as

$$\begin{aligned} p(x) &= x^2 + bx + c \\ &= \left(x + \frac{b}{2}\right)^2 - d^2 \\ &= \left(\left(x + \frac{b}{2}\right) + d\right) \cdot \left(\left(x + \frac{b}{2}\right) - d\right) \\ &= \left(x - \left(-d - \frac{b}{2}\right)\right) \cdot \left(x - \left(d - \frac{b}{2}\right)\right) \\ &= (x - \lambda_1) \cdot (x - \lambda_2) \quad (\forall x \in \mathbb{R}) \end{aligned}$$

for $\lambda_1 := -d - b/2$ and $\lambda_2 := d - b/2$, which is the same as the well known midnight formula:

$$\lambda_{1,2} = \frac{-b \pm \sqrt{b^2 - 4c}}{2}$$

The formula above also shows that $b^2 \geq 4c$ is a necessary condition for real solutions (i.e., for $\lambda_{1,2}$). $\square$

REMARK: The next result gives a factorization of a polynomial over $\mathbb{R}$. The second version of the fundamental theorem of algebra 4.13 gives a factorization of p with complex coefficients. Complex but nonreal zeros of p come in pairs (see 4.14). Multiplying terms of the form $(x - \lambda)$ and $(x - \bar{\lambda})$ together yields

$$(x^2 - 2(\operatorname{Re} \lambda)x + |\lambda|^2),$$

which is a quadratic term of the required form. But 4.14 does not state that these two factors appear the same number of times, as needed to make the idea work. However, the next proof works around this point.

In the next result, either m or M may equal to 0. The numbers $\lambda_1, \dots, \lambda_m$ are precisely the real zeros of p , for these are the only real values of x for which the right side of the equation in the next result equals 0.

THM 4.16 (*factorization of a polynomial over $\mathbb{R}$*):

Suppose $p \in \mathcal{P}(\mathbb{R})$ is non-constant. Then p has a unique factorization:

$$p(x) = c(x - \lambda_1) \cdots (x - \lambda_m)(x^2 + b_1x + c_1) \cdots (x^2 + b_Mx + c_M)$$

where $c, \lambda_1, \dots, \lambda_m, b_1, \dots, b_M, c_1, \dots, c_M \in \mathbb{R}$, with $b_k^2 < 4c_k \quad \forall k \in \{1, \dots, M\}$.

PROOF: First we will prove that the desired factorization exists, and after that, we will prove uniqueness.

Think of p as an element of $\mathcal{P}(\mathbb{C})$, not $\mathcal{P}(\mathbb{R})$, but with real coefficients like stated in the theorem. If all (complex) zeros of p are real, then we have the desired factorization by the second version of the fundamental theorem of algebra 4.13. Thus suppose p has a complex zero $\lambda \in \mathbb{C}$ with $\lambda \notin \mathbb{R}$. By 4.14, $\bar{\lambda}$ is a zero of p as well. Thus we write $\forall x \in \mathbb{R}$

$$\begin{aligned} p(x) &= (x - \lambda)(x - \bar{\lambda})q(x) \\ &= (x^2 - 2(\operatorname{Re} \lambda)x + |\lambda|^2)q(x), \end{aligned}$$

for some $q \in \mathcal{P}(\mathbb{C})$, with $\deg q = \deg p - 2$. If we can prove that q has real coefficients, then using induction on the degree of such polynomial completes the proof of the existence part of this result. If we solve for $q(x)$, we get

$$q(x) = \frac{p(x)}{x^2 - 2(\operatorname{Re} \lambda)x + |\lambda|^2} \quad \forall x \in \mathbb{R}.$$

The equation above implies that $q(x) \in \mathbb{R} \quad \forall x \in \mathbb{R}$. Let $n := \deg p$. Then we can rewrite q as

$$q(x) = a_0 + a_1x + \cdots + a_{n-2}x^{n-2} \quad \text{where } a_0, \dots, a_{n-2} \in \mathbb{C},$$

because we don't know if the a 's are all real yet. But we know that $\forall x \in \mathbb{R} : q(x) \in \mathbb{R}$. Thus we have $\forall x \in \mathbb{R}$:

$$0 = \operatorname{Im} q(x) = (\operatorname{Im} a_0) + (\operatorname{Im} a_1)x + \cdots + (\operatorname{Im} a_{n-2})x^{n-2}$$

Since according to 4.8 a complex polynomial with degree $n-2$ can have at most $n-2$ zeros, this equation can not hold for all $x \in \mathbb{R}$. This implies that $\operatorname{Im} a_0, \dots, \operatorname{Im} a_{n-2}$ all equal to 0, because only the zero polynomial is allowed to have infinitely many zeros. Thus all the coefficients of q are real, as desired. Hence the wanted factorization exists by induction.

UNIQUENESS: A factor of p of the form $x^2 + b_kx + c_k$ with $b_k^2 < 4c_k$ can be uniquely written as $(x - \lambda_k)(x - \overline{\lambda_k})$ with $\lambda_k \in \mathbb{C}$. Two different factorizations of p as an element of $\mathcal{P}(\mathbb{R})$ would lead to two different factorizations of p as an element of $\mathcal{P}(\mathbb{C})$, contradicting using again the second version of the fundamental theorem of algebra 4.13. $\square$

EXERCISES ABOUT POLYNOMIALS 4D

- ④ Suppose m is a positive integer. Is the set

$$\{0\} \cup \{p \in \mathcal{P}(\mathbb{F}) : \deg p = m\} = S$$

a subspace of $\mathcal{P}(\mathbb{F})$?

SOLUTION: The set in question is not a subspace of $\mathcal{P}(\mathbb{F})$, because it is not closed under addition. Take $m = 4$, then $p = 1 + 2x^2 + 3x^3 + 4x^4$ and $s = 4x^4$ are both in S , but $p - s$ is not.

- ⑤ Is the set

$$\{0\} \cup \{p \in \mathcal{P}(\mathbb{F}) : \deg p \text{ is even}\} = S$$

a subspace of $\mathcal{P}(\mathbb{F})$?

SOLUTION: The same counter-example as in the previous question applies to this one.

- ⑥ Suppose that m and n are positive integers with $m \leq n$, and suppose $\lambda_1, \dots, \lambda_m \in \mathbb{F}$ are distinct. Prove that there exists a polynomial $p \in \mathcal{P}(\mathbb{F})$ with $\deg p = n$ such that $0 = p(\lambda_1) = \dots = p(\lambda_m)$ and such that p has no other zeros.

PROOF: Define $p \in \mathcal{P}_n(\mathbb{F})$ by

$$p(z) = (z - \lambda_1)(z - \lambda_2) \cdots (z - \lambda_{m-1}) \underbrace{(z - \lambda_m)(z - \lambda_m) \cdots (z - \lambda_m)}_{n-m+1 \text{ times}}$$

$$= \left(\prod_{k=1}^{m-1} (z - \lambda_k) \right) (z - \lambda_m)^{n-m+1} \quad \forall z \in \mathbb{F}.$$

Clearly, $\deg p = (m - 1) + (n - m + 1) = n$, and $p(\lambda_k) = 0$ for each $k \in \{1, \dots, m\}$. Furthermore, if $\lambda \in \mathbb{F}$ satisfies $p(\lambda) = 0$, then since $\mathbb{F}$ is a field (with no zero divisors), at least one factor $(\lambda - \lambda_k)$ must be zero. Hence $\lambda \in \{\lambda_1, \dots, \lambda_m\}$, meaning p has no other zeros. $\square$

EIGENVALUES AND EIGENVECTORS (CHAPTER 5)

INVARIANT SUBSPACES (SECTION 5A)

EIGENVALUES (SUB-SECTION 5A1)

DEF 5.1 (operator):

A linear map from a vector space to itself is called an “operator”.

(Suppose $T \in \mathcal{L}(V)$, then may be $T|_{V_k}$ is not an operator on a subspace V_k)

DEF 5.2 (invariant subspace):

Let $T \in \mathcal{L}(V)$. A subspace U of V is called “invariant under T ” if $\forall u \in U : Tu \in U$.

Thus U is invariant under T if $T|_U$ is an operator on U .

EXAMPLE 5.3:

Let $T \in \mathcal{L}(\mathcal{P}(\mathbb{R}))$ such that $Tp = p'$. Let $U = \mathcal{P}_4(\mathbb{R}) \subseteq \mathcal{P}(\mathbb{R})$. Then U is invariant under T because if $p \in U$, $\deg p = 4$ and $\deg(p') = 3$.

EXAMPLE 5.4:

Let $T \in \mathcal{L}(V)$. Then $\{0\}, V, \text{null } T, \text{range } T$ are all invariant.

(Sometimes, $\text{null } T = \{0\}$ and $\text{range } T = V$ if T is invertible.)

INVARIANT SUBSPACES OF DIMENSION ONE:

Take any $v \in V, v \neq 0$ and let

$$U := \{\lambda v \mid \lambda \in \mathbb{F}\} = \text{span}(v),$$

then U is a one-dimensional subspace of V .

If U is invariant under an operator $T \in \mathcal{L}(V)$, then $Tv \in U \implies \exists \lambda \in \mathbb{F} : Tv = \lambda v$.

Conversely if $Tv = \lambda v, \lambda \in \mathbb{F}$, then $\text{span}(v)$ is a one-dimensional subspace of V invariant under T .

DEF 5.5 (eigenvalue):

Suppose $T \in \mathcal{L}(V)$. A scalar $\lambda \in \mathbb{F}$ is called an “eigenvalue of T ” if there exists a nonzero (i.e., $v \neq 0$) $v \in V$ such that

$$Tv = \lambda v.$$

THM 5.7 (equivalent conditions to be an eigenvalue):

The following are equivalent for $T \in \mathcal{L}(V)$ and $\lambda \in \mathbb{F}$.

- (a) λ is an eigenvalue of T .
- (b) $T - \lambda I$ is not injective.
- (c) $T - \lambda I$ is not surjective.
- (d) $T - \lambda I$ is not invertible.

PROOF: Conditions (a) and (b) are equivalent because the eigenvector v is a solution to $Tv = \lambda v$, which is equivalent to $(T - \lambda I)v = 0$. Hence, $\dim \text{null}(T - \lambda I) > 0$, which implies that $T - \lambda I$ is not injective. (b), (c) and (d) are therefore equivalent by 3.65. $\square$

DEF 5.8 (eigenvector):

Let $T \in \mathcal{L}(V)$. A vector $v \in V$ is called an “eigenvector” of T corresponding to λ if $v \neq 0$ and $Tv = \lambda v$. In other words: A vector $v \in V, v \neq 0$ is an eigenvector corresponding to $\lambda \iff v \in \text{null}(T - \lambda I_V)$

THM 5.11 (*linearly independent eigenvectors*):

Every list of eigenvectors of T corresponding to distinct eigenvalues of T is linearly independent.

PROOF: Suppose the desired result is false. Then there exists a smallest list of length m of linearly dependent eigenvectors $v_1, \dots, v_m$ with distinct eigenvalues $\lambda_1, \dots, \lambda_m$. Since an eigenvector is unequal to the zero vector, m must be ≥ 2 .

Because of the minimality of m and because our list is linearly dependent:

$$\exists a_1, \dots, a_m, \text{ where not all } a\text{'s are equal to } 0, \text{ such that } a_1 v_1 + \dots + a_m v_m = 0.$$

Now we apply $(T - \lambda_m I)$ on both sides of the equation and get

$$\begin{aligned} & (a_1 \lambda_1 v_1 - a_1 \lambda_m v_1) + \dots + \\ & (a_{m-1} \lambda_{m-1} v_{m-1} - a_{m-1} \lambda_m v_{m-1}) + \underbrace{(a_m \lambda_m v_m - a_m \lambda_m v_m)}_{=0} = 0 \end{aligned}$$

Which is the same as:

$$a_1 \underbrace{(\lambda_1 - \lambda_m)}_{\neq 0} v_1 + \dots + a_{m-1} \underbrace{(\lambda_{m-1} - \lambda_m)}_{\neq 0} v_{m-1} = 0.$$

By the minimality of m , the list $v_1, \dots, v_{m-1}$ is linearly independent. Hence $a_1(\lambda_1 - \lambda_m) = \dots = a_{m-1}(\lambda_{m-1} - \lambda_m) = 0$. Since all eigenvalues are distinct, this forces $a_1 = \dots = a_{m-1} = 0$. Returning to the original equation yields $a_m v_m = 0$, which implies $a_m = 0$ (since $v_m \neq 0$), contradicting that not all coefficients were zero. Therefore, no such linearly dependent list of eigenvectors can exist. $\square$

THM 5.12 (*number of eigenvalues*):

Each operator $T \in \mathcal{L}(V)$ on V has at most $\dim V$ distinct eigenvalues.

PROOF: Let $T \in \mathcal{L}(V)$. Suppose $\lambda_1, \dots, \lambda_m$ are distinct eigenvalues with corresponding eigenvectors $v_1, \dots, v_m \in V$. Hence, Theorem 5.11 implies that the list $v_1, \dots, v_m$ is linearly independent. Thus, ?? implies that $m \leq \dim V$. $\square$

POLYNOMIALS APPLIED TO OPERATORS (SUB-SECTION 5A2)

DEF 5.13 (*notation T^m*):

Let $T \in \mathcal{L}(V)$ and $m \in \mathbb{N}^+$

- $T^m := \underbrace{T \cdots T}_{m \text{ times}}$ or $T^m := T^{m-1} \cdot T$ such that $T^m \in \mathcal{L}(V)$
- $T^0 := I_V$
- If T is invertible with inverse T^{-1} then $T^{-m} \in \mathcal{L}(V)$ is defined by $T^{-m} := (T^{-1})^m$

$T^m T^n = T^{m+n}$ and $(T^m)^n = T^{mn}$ for $m, n \in \mathbb{Z}$ if T is invertible. And for $m, n \in \mathbb{N}$ if T is not invertible.

DEF 5.14 (*notation $p(T)$*):

For $\mathcal{P}(\mathbb{F}) \ni p(z) = a_0 + a_1 z + a_2 z^2 + \dots + a_m z^m$ and $T \in \mathcal{L}(V)$ we define:

$$p(T) := a_0 I + a_1 T + a_2 T^2 + \dots + a_m T^m.$$

Note that $p(T) \in \mathcal{L}(V)$

THM 5.17 (*multipliative properties*):

Suppose $p, q \in \mathcal{P}(\mathbb{F})$ and $T \in \mathcal{L}(V)$. Then

$$(pq)(T) = p(T)q(T) = q(T)p(T).$$

PROOF: When a product of polynomials is expanded using the distributive property, it does not matter if the symbol is z or T . Suppose $p(z) = \sum_{j=0}^m a_j z^j$ and $q(z) = \sum_{k=0}^n b_k z^k$ for all $z \in \mathbb{F}$. Then

$$(pq)(z) = \sum_{j=0}^m \sum_{k=0}^n a_j b_k z^{j+k} \quad \text{and} \quad (pq)(T) = \sum_{j=0}^m \sum_{k=0}^n a_j b_k T^{j+k} = \sum_{j=0}^m a_j T^j \sum_{k=0}^n b_k T^k$$

which is the same as $(pq)(T) = p(T)q(T)$ as well as $(pq)(T) = q(T)p(T)$ $\square$

THM 5.18 (*null space and range of $p(T)$ are invariant under T*):

For every $T \in \mathcal{L}(V)$, and every $p \in \mathcal{P}(\mathbb{F})$,

$\text{null } p(T)$ and $\text{range } p(T)$ are invariant under T .

PROOF: **STEP 1:** First, let $u \in \text{null } p(T)$. Then,

$$p(T)u = 0.$$

Since $p(T)$ commutes with T (by associativity and distributivity of polynomial evaluation),

$$\begin{aligned} (p(T))(Tu) &= (p(T)T)(u) \\ &= (Tp(T))(u) \\ &= T(p(T)u) \\ &= T(0) = 0. \end{aligned}$$

Hence, $Tu \in \text{null } p(T)$.

STEP 2: Second, let $u \in \text{range } p(T)$. Then, $u = p(T)v$ for some $v \in V$. Thus,

$$Tu = T(p(T)v) = p(T)(Tv),$$

so, $Tu \in \text{range } p(T)$. $\square$

EXERCISES ABOUT INVARIANT SUBSPACES 5A

① Suppose $T \in \mathcal{L}(V)$ and U is a subspace of V . Then

(a) $U \subseteq \text{null } T \implies U$ is invariant under T .

(b) $\text{range } T \subseteq U \implies U$ is invariant under T .

PROOF: (a). If $v \in U$, then $Tv = 0 \in U$, since U is a subspace and therefore $0 \in U$.

(b). Let $w \in U \subseteq V$. $\implies Tw \in \text{range } T$, and since $\text{range } T \subseteq U$, $Tw \in U$. $\square$

② Suppose that $T \in \mathcal{L}(V)$ and $V_1, \dots, V_m$ are subspaces of V invariant under T . Prove that $V_1 + \dots + V_m$ is invariant under T .

PROOF: Let $w \in V_1 + \dots + V_m$ such that $w = v_1 + \dots + v_m$ where $v_1 \in V_1, \dots, v_m \in V_m$. Then $Tw = Tv_1 + \dots + Tv_m$ such that $Tv_1 \in V_1, \dots, Tv_m \in V_m$ because $V_1, \dots, V_m$ are invariant under T . Therefore we have $Tw \in V_1 + \dots + V_m$. So $V_1 + \dots + V_m$ is invariant under T as well. $\square$

③ Suppose $T \in \mathcal{L}(V)$. Prove that the intersection of every collection of subspaces of V invariant under T is invariant under T .

PROOF: Let $w \in V_1 \cap \dots \cap V_m \subseteq V$ where each V_i is invariant under T . Then $w \in V_1, \dots, w \in V_m$, and thus, $Tw \in V_1, \dots, Tw \in V_m$. Hence, $Tw \in V_1 \cap \dots \cap V_m$. $\square$

⑫ Suppose $V = U \oplus W$, where U and W are nonzero subspaces of V . Define $P \in \mathcal{L}(V)$ by $P(u+w) = u$ for each $u \in U$ and each $w \in W$. Find all eigenvalues and eigenvectors of P .

SOLUTION: We have an eigenvalue $\lambda_1 = 1$ for all vectors $u \in U$, because P acts as the identity on U . We also have an eigenvalue $\lambda_0 = 0$ for all $w \in W$, because P acts as the zero operator there.

⑮ Suppose V is finite-dimensional, $T \in \mathcal{L}(V)$, and $\lambda \in \mathbb{F}$. Show that λ is an eigenvalue of $T \iff \lambda$ is an eigenvalue of the dual operator $T' \in \mathcal{L}(V')$.

SOLUTION: The definition of T' states that

$$T'(\varphi) = \varphi \circ T \quad \forall \varphi \in V'.$$

Hence, being an eigenvalue λ and an eigenvector $\varphi \in V' = \mathcal{L}(V, \mathbb{F})$ of T' means

$$T'(\varphi) = \lambda\varphi.$$

$\Rightarrow$ **DIRECTION:** If $u \in V$ is an eigenvector of T with an eigenvalue λ such that $T(u) = \lambda u$, we can extend u to a basis $u, v_2, \dots, v_n$ of V ($n = \dim V$). Define $\varphi \in V'$ by

$$\varphi(u) = 1, \quad \varphi(v_i) = 0 \quad (i = 2, \dots, n),$$

on the basis vectors by the linear map lemma 3.4.

then we have for every $\varphi \in V'$:

$$\begin{aligned} T'(\varphi)(u) &= (\varphi \circ T)(u) \\ &= \varphi(T(u)) \\ &= \varphi(\lambda u) \\ &= \lambda\varphi(u) \end{aligned}$$

So, if we construct ψ such that for every $v \in V, v = au + \sum_{i \geq 2} b_i v_i$ we have

$$(T'\varphi)(v) = (\varphi \circ T)(u) = \varphi(aTu + \sum_{i \geq 2} b_i T v_i) = \varphi(a\lambda u) + \sum_{i \geq 2} b_i T v_i$$

$$\psi(v) = \begin{cases} \varphi(v), & v \in E(\lambda, T) \text{ and } v \notin \text{null } T \\ 0, & \text{otherwise} \end{cases}$$

we have found an eigenvector ψ with the same eigenvalue λ .

Or: λ is eigenvalue of $T \iff T - \lambda I$ is not injective $\iff (T - \lambda I)'$ is not surjective $\iff T' - \lambda I$ is not invertible $\iff \lambda$ is an eigenvalue of T' .

21 Suppose $T \in \mathcal{L}(V)$ is invertible

(a) Suppose $\lambda \in \mathbb{F}$ with $\lambda \neq 0$. Prove that λ is an eigenvalue of $T \iff \frac{1}{\lambda}$ is an eigenvalue of T^{-1} .

(b) Prove that T and T^{-1} have the same eigenvectors.

SOLUTION: Let $v \in V$ be such that $T(v) = \lambda v$. This is the case $\iff T^{-1}T(v) = T^{-1}(\lambda v) \iff T^{-1}(v) = \frac{1}{\lambda}v$. Note that in a field $\mathbb{F}$, we can “replace” $\frac{1}{\lambda} \in \mathbb{F}$ with its multiplicative inverse $\alpha \in \mathbb{F}$. Since $(T^{-1})^{-1} = T$, we have proven (a) and (b).

22 Suppose $T \in \mathcal{L}(V)$ and there exist nonzero vectors $u, w \in V$ such that

$$Tu = 3w \quad \text{and} \quad Tw = 3u$$

Prove that 3 or -3 is an eigenvalue of T

SOLUTION: If T is defined as above, then $T(u+w) = 3w+3u = 3(u+w)$. Hence, $u+w$ gets sent to $3(u+w)$, which makes 3 an eigenvalue of T . If we change the definition of T to

$$T(u) = 3(-w) \quad \text{and} \quad T(-w) = 3u,$$

then we have

$$T(u+w) = T(u+(-1)(-1)w) = T(u) - T(-w) = -3w - 3u = -3(u+w).$$

Thus, $u+w$ gets sent to $-3(u+w)$ and -3 is an eigenvalue of T as well. Note that we didn't prove that those are the only eigenvalues.

23 Suppose V is finite-dimensional and $T \in \mathcal{L}(V)$. Prove that T has an eigenvalue $\iff$ there exists a subspace of V of dimension $\dim V - 1$ that is invariant under T .

KI Bemerkung: Hier standen zwei fehlerhafte bzw. unvollständige Beweisversuche. Ich habe diese durch einen korrekten und eleganten Beweis über den dualen Operator ersetzt, der auch das Ergebnis von Aufgabe 15 direkt ausnutzt.

PROOF: $\Rightarrow$ **DIRECTION:** Suppose T has an eigenvalue $\lambda \in \mathbb{F}$. By Exercise 15, λ is also an eigenvalue of the dual operator $T' \in \mathcal{L}(V')$. Thus, there exists a nonzero linear functional $\varphi \in V'$ such that $T'(\varphi) = \lambda\varphi$. This means that for every $u \in V$, we have $\varphi(Tu) = (T'\varphi)(u) = (\lambda\varphi)(u) = \lambda\varphi(u)$. Let $U = \text{null } \varphi$. Because $\varphi \neq 0$, the Fundamental Theorem of Linear Maps tells us that $\dim U = \dim V - \dim(\text{range } \varphi) = \dim V - 1$. If $u \in U$, then $\varphi(u) = 0$, so $\varphi(Tu) = \lambda\varphi(u) = 0$, which implies $Tu \in \text{null } \varphi = U$. Hence, U is a subspace of dimension $\dim V - 1$ that is invariant under T .

$\Leftarrow$ **DIRECTION:** Suppose there exists a subspace $U \subseteq V$ of dimension $\dim V - 1$ that is invariant under T . Let $U^0 \subseteq V'$ be the annihilator of U . Since $\dim U = \dim V - 1$, we know that $\dim U^0 = \dim V - \dim U = 1$. Because U is invariant under T , its annihilator U^0 is invariant under the dual operator T' . Since U^0 is a one-dimensional subspace invariant under T' , any nonzero functional $\varphi \in U^0$ is an eigenvector of T' . Hence, T' has an eigenvalue. By Exercise 15, T must also have an eigenvalue. $\square$

- (25) Suppose $T \in \mathcal{L}(V)$ and u, w are eigenvectors of T such that $u + w$ is also an eigenvector of T . Prove that u and w are eigenvectors of T corresponding to the same eigenvalue.

PROOF: Let $u, w \in V \setminus \{0\}$ such that $Tu = \lambda_u u$, $Tw = \lambda_w w$, and $T(u + w) = \lambda_{u+w}(u + w)$. Since

$$T(u + w) = Tu + Tw = \lambda_u u + \lambda_w w = \lambda_{u+w}(u + w) = \lambda_{u+w}u + \lambda_{u+w}w,$$

it follows that

$$(\lambda_u - \lambda_{u+w})u + (\lambda_w - \lambda_{u+w})w = 0.$$

If u and w are linearly independent, the coefficients must vanish, so $\lambda_u = \lambda_{u+w} = \lambda_w$. Otherwise, if u and w are linearly dependent, we can write $u = \alpha w$ for some $\alpha \neq 0$. Applying T to u in two ways gives

$$Tu = T(\alpha w) = \alpha Tw = \alpha \lambda_w w \text{ and}$$

$$Tu = \lambda_u u = \lambda_u \alpha w.$$

Hence, $\lambda_u = \lambda_w$. $\square$

- (26) Let u, v

- (33) Suppose $T \in \mathcal{L}(V)$ and m is a positive integer.

(a) Prove that T is injective $\iff T^m$ is injective.

(b) Prove that T is surjective $\iff T^m$ is surjective.

PROOF: We show (a) and (b) both with induction, first in the $\Rightarrow$ direction, and then in the $\Leftarrow$ direction.

(a) $\Rightarrow$ **DIRECTION:** Suppose T is injective. For $m = 1$ the statement is trivial ($T^1 = T$). Fix $m \geq 1$ and suppose as induction hypothesis that T^m is injective. If $T^{m+1}(v_1) = T^{m+1}(v_2)$, then

$$T(T^m(v_1)) = T(T^m(v_2)).$$

Since T is injective, $T^m(v_1) = T^m(v_2)$. The induction hypothesis forces $v_1 = v_2$. Therefore, T^{m+1} is injective as well. (In general, function composition is injective. If $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ are injective, so is $g \circ f : X \rightarrow Z$. So, a finite composition of injective maps is injective.)

$\Leftarrow$ **DIRECTION:** Assume T^m is injective for some fixed $m \geq 1$. If $m = 1$ we are done. If

$m > 1$, take $v_1, v_2 \in V$ with $T(v_1) = T(v_2)$. Apply T^{m-1} to both sides to get $T^{m-1}(T(v_1)) = T^{m-1}(T(v_2))$. Injectivity of T^m forces $v_1 = v_2$. Thus, T is injective.

(b) $\Rightarrow$ **DIRECTION:** Suppose T is surjective. For $m = 1$ the statement is trivial. Fix $m \geq 1$ and assume that T^m is surjective. Let $y \in V$ be arbitrary. Since T is surjective, there exists $z \in V$ with $T(z) = y$. By the induction hypothesis, there exists $v \in V$ with $T^m(v) = z$. Thus,

$$T^{m+1}(v) = T(T^m(v)) = T(z) = y.$$

Hence, T^{m+1} is surjective.

(Or: A finite composition of surjective maps (not necessarily linear) is surjective.)

$\Leftarrow$ **DIRECTION:** Assume T^m is surjective for some fixed $m \geq 1$. Let $y \in V$ be arbitrary. Choose $v \in V$ with $T^m(v) = y$ (always possible, since T^m is surjective). Put $x := T^{m-1}(v)$. Then

$$y = T(x) = T(T^{m-1}(v)) = T^m(v).$$

Hence, T is surjective.

□

40 Suppose $S, T \in \mathcal{L}(V)$, and S is invertible. Suppose $p \in \mathcal{P}(\mathbb{F})$ is a polynomial. Prove that

$$p(STS^{-1}) = Sp(T)S^{-1}.$$

PROOF: We first want to prove that $(STS^{-1})^m = ST^mS^{-1}$ for every $m \in \mathbb{N}_{\geq 0}$. To this end, we use induction over m . For $m = 0$, the statement holds because

$$(STS^{-1})^0 = I = SIS^{-1} = ST^0S^{-1}.$$

If $m = 1$ the claim is immediate. Hence, assume $m \geq 1$. Then

$$\begin{aligned} (STS^{-1})^{m+1} &= STS^{-1}(STS^{-1})^m \\ &= ST \underbrace{S^{-1}S}_{I} T^m S^{-1} \\ &= ST^{m+1}S^{-1}, \end{aligned}$$

This completes the induction. Let $p(z) = a_0z^0 + a_1z^1 + \cdots + a_mz^m$, $a_j \in \mathbb{F}$. Then

$$\begin{aligned} p(STS^{-1}) &= a_0I + a_1(STS^{-1}) + a_2(STS^{-1})^2 + \cdots + a_m(STS^{-1})^m = \sum_{j=0}^m a_j (STS^{-1})^j \\ &= a_0I \underbrace{SS^{-1}}_{=I=(SIS^{-1})^0} + a_1(STS^{-1}) + a_2(ST^2S^{-1}) + \cdots + a_m(ST^mS^{-1}) = \sum_{j=0}^m a_j (ST^jS^{-1}) \\ &= S(a_0I)S^{-1} + S(a_1T)S^{-1} + S(a_2T^2)S^{-1} + \cdots + S(a_mT^m)S^{-1} = \sum_{j=0}^m S(a_jT^j)S^{-1} \\ &= S\left(a_0I + a_1T + a_2T^2 + \cdots + a_mT^m\right)S^{-1} = S\left(\sum_{j=0}^m a_jT^j\right)S^{-1} \\ &= Sp(T)S^{-1}. \end{aligned}$$

Here we have been making use of the distributive property of linear maps, see 3.8. And we used that scalar multiples of the identity (i.e. a_0I) commute with every operator. □

THE MINIMAL POLYNOMIAL (SECTION 5B)

EXISTENCE OF EIGENVALUES ON COMPLEX VECTOR SPACES (SUB-SECTION 5B1)

THM 5.19 (*existence of eigenvalues*):

Every operator on a finite-dimensional nonzero, complex vector space has an eigenvalue.

PROOF: Suppose V is finite-dimensional with $\dim V \geq 1$ and $T \in \mathcal{L}(V)$. Let $v \in V$ be nonzero. Then the list

$$v, Tv, T^2v, \dots, T^n v$$

of length $n + 1$ cannot be linearly independent. Hence, some non-trivial linear combination of these vectors is zero, so there exists a non-constant polynomial $p \in \mathcal{P}(\mathbb{C})$ of minimal degree such that

$$p(T)v = 0.$$

By the first version of the fundamental theorem of algebra 4.12, there exists a root $\exists \lambda \in \mathbb{C}$ with

$$p(\lambda) = 0.$$

By 4.6, we know that $\exists q \in \mathcal{P}(\mathbb{C})$ such that we can factor out $(z - \lambda)$ and write

$$p(z) = (z - \lambda)q(z) \quad \forall z \in \mathbb{C}.$$

Applying this to T gives

$$0 = p(T)v = (T - \lambda I)(q(T)v).$$

If $q(T)v$ were zero, then q (of strictly smaller degree) would also annihilate v , contradicting the minimality of p . Thus

$$q(T)v \neq 0$$

and so λ is an eigenvalue of T with eigenvector $q(T)v$. □

EIGENVALUES AND THE MINIMAL POLYNOMIAL (SUB-SECTION 5B2)

DEF 5.20 (*monic polynomials*):

A “*monic polynomial*” is a polynomial whose highest-degree coefficient equals 1.

EXAMPLE 5.21:

$$p(z) = 2 + 9z^2 + z^7, \quad \deg p = 7.$$

THM 5.22 (*existence, uniqueness, and degree of minimal polynomial*):

Suppose $T \in \mathcal{L}(V)$. Then there exists a unique monic polynomial $p \in \mathcal{P}(\mathbb{F})$ of smallest degree such that

$$p(T) = 0.$$

Furthermore, $\deg p \leq \dim V$.

PROOF: Before beginning our inductive proof, we examine some base cases. We first prove the existence of a monic polynomial that annihilates T and later establish its uniqueness; these two parts are independent.

BASE CASE: If $\dim V = 0$, then $V = \{\vec{0}\}$, and the identity operator I is the zero-operator on V . We let $p(z) := 1$ so that

$$p(T) = I = 0.$$

ANOTHER BASE CASE: Now suppose that $\dim V = 1$. For any nonzero $v \in V$, the list v, Tv is

linearly dependent (since it has length 2), so there exists a scalar $c_0 \in \mathbb{F}$ such that

$$c_0 v + T v = 0.$$

This follows from the linear dependence lemma Theorem 2.19. If we choose $q(z) := c_0 + z \quad \forall z \in \mathbb{F}$, then

$$q(T)v = (c_0 I + T)(v) = c_0 v + T v = 0.$$

In the case where T is the zero operator, one may instead choose $q(z) = z$. So we have $\dim \text{null } q(T) = 1$ and therefore $\dim \text{range } q(T) = \dim V - \dim \text{null } q(T) = 1 - 1 = 0$. Thus, every vector in this one-dimensional space is a scalar multiple of v and maps to zero by $q(T)$. Thus, $q(z)$ serves as a minimal polynomial for T .

YET ANOTHER BASE CASE: Suppose $\dim V = 2$. For any nonzero $v \in V$, the list $v, T v, T^2 v$ is linearly dependent (since it has length 3). According to the linear dependence lemma Theorem 2.19, we either have

$$c_0 v + T v = 0 \quad \text{or} \quad c_0 v + c_1 T v + T^2 v = 0,$$

for some scalars $c_0, c_1 \in \mathbb{F}$. If we define $q(z) := c_0 + z$ in the first case or $q(z) := c_0 + c_1 z + z^2$ in the second case accordingly, then $q(T)v = 0$.

In the first case where $(-1)c_0 v = T v$ and $q(z) = c_0 + z$, the vector v by itself is linearly independent and thus $\dim \text{null } q(T) \geq 1$. Therefore

$$\dim \text{range } q(T) \leq \dim V - \dim \text{null } q(T) \leq 1.$$

Since $\text{range } q(T)$ is at most one-dimensional and is invariant under T (by Theorem 5.18), we can restrict T to this subspace. In this restricted setting $\text{range } q(T)$, we know how to construct the desired polynomial; denote it by s , defined such that

$$s\left(T|_{\text{range } q(T)}\right) = 0.$$

Just for clarity, the equation above means that s is the zero-operator of $T|_{\text{range } q(T)}$. Therefore,

$$\forall u \in V : ((sq)(T))(u) = s(T)(q(T)u) = 0.$$

Thus, sq is a candidate for our minimal polynomial of T .

A similar reasoning goes for the case where $(-1)c_0 v + (-1)c_1 T v = T^2 v$ and $q(z) = c_0 + c_1 z + z^2$. Here we have v and $T v$ as linearly independent vectors. Note that if $k \in \mathbb{N}$, then

$$q(T)(T^k v) = T^k(q(T)v) = T^k(0) = 0. \quad (5.1)$$

Hence the equation (5.1) above tells us that $v \in \text{null } q(T)$ and $T v \in \text{null } q(T)$. Here in this case we conclude that $\dim \text{null } q(T) \geq 2$ and therefore $\dim \text{range } q(T) = 0$. Since $\dim \text{null } q(T) = 2$, we have that

$$\text{null } q(T) = V.$$

This means that every vector of V gets mapped to 0 if $q(T)$ is applied. So we have found a minimal monic polynomial!

GENERAL CASE: As we can see a pattern now, we can use induction on $\dim V$ and assume $\dim V > 2$ and that the theorem holds for all vector spaces of smaller dimension than $\dim V$. Let $v \in V$ be nonzero. Then the list

$$v, T v, \dots, T^{\dim V} v \quad (5.2)$$

(which has length $1 + \dim V$) is linearly dependent. By the linear dependence lemma (Theorem 2.19), there is a smallest positive integer $m \leq \dim V$ and some $c_0, \dots, c_{m-1} \in \mathbb{F}$ such that

$$c_0 v + c_1 T v + \dots + c_{m-1} T^{m-1} v + T^m v = 0.$$

Let us define $q \in \mathcal{P}_m(\mathbb{F})$ as

$$q(z) := c_0 + c_1 z + \cdots + c_{m-1} z^{m-1} + z^m$$

accordingly. Thus, we have

$$q(T)v = 0.$$

Note that $q(z)$ is a monic polynomial and that $\deg q = m$. Observe that if $k \in \mathbb{N}$, then

$$q(T)(T^k v) = T^k(q(T)v) = T^k(0) = 0. \quad (5.3)$$

By the linear dependence lemma (Theorem 2.19) and the definition of m , the list

$$v, Tv, \dots, T^{m-1}v \quad (5.4)$$

from before is linearly independent (This is not the same list as (5.2) since $m \leq \dim V$). This fact together with equation (5.3) imply that the vectors of the list (5.4) all belong to $\text{null } q(T)$. Hence,

$$\dim \text{null } q(T) \geq m.$$

Using the Rank-Nullity Theorem, this implies:

$$\dim \text{range } q(T) = \dim V - \dim \text{null } q(T) \leq \dim V - m.$$

Because $\text{range } q(T)$ is invariant under T by Theorem 5.18, we can apply our induction hypothesis to the operator $T|_{\text{range } q(T)}$. Hence, there exists a monic polynomial $s \in \mathcal{P}(\mathbb{F})$ such that $\deg s \leq \dim V - m$ and

$$s\left(T|_{\text{range } q(T)}\right) = 0.$$

In other words, $\forall u \in \text{range } q(T) : s(T)(u) = 0$. So we can safely say that

$$s(T)|_{\text{range } q(T)} = s\left(T|_{\text{range } q(T)}\right).$$

Because $\forall u \in V : q(T)u \in \text{range } q(T)$, we conclude:

$$\forall u \in V : (sq)(T)(u) = s(T)(q(T)u) = 0.$$

Therefore, sq is a monic polynomial such that

$$\deg sq \leq (\dim V - m) + m = \dim V$$

and $(sq)(T) = 0$.

PROOF OF UNIQUENESS: Let $p \in \mathcal{P}(\mathbb{F})$ be a monic polynomial of smallest degree such that

$$p(T) = 0.$$

Let $r \in \mathcal{P}(\mathbb{F})$ be another monic polynomial of the same degree such that $r(T) = 0$. This implies

$$(p - r)(T) = 0.$$

Moreover, $\deg(p - r) < \deg p = \deg r$. If $p - r \neq 0$, we could divide $p - r$ by the coefficient of the highest order term in $p - r$ to obtain a monic polynomial that, when applied to T , yields the zero-operator. This polynomial would have a smaller degree than p or r , which would be a contradiction. Therefore, $p - r = 0 \iff p = r$. $\square$

DEF 5.24 (minimal polynomial):

Let $T \in \mathcal{L}(V)$. The “minimal polynomial of T ” is the unique monic polynomial $p \in \mathcal{P}(\mathbb{F})$ of smallest degree such that $p(T) = 0$.

COMPUTATION OF THE MINIMAL POLYNOMIAL: Find the smallest $m \in \mathbb{N}$ such that:

$c_0 I + c_1 T + \cdots + c_{m-1} T^{m-1} = -T^m$ has a solution $c_0, \dots, c_{m-1} \in \mathbb{F}$. Solve for $m = 1, 2, \dots, \dim V$

Even faster (usually), pick $v \in V$ with $v \neq 0$ and consider the equation

$$c_0 v + c_1 Tv + \cdots + c_{\dim V - 1} T^{\dim V - 1} v = -T^{\dim V} v.$$

If this equation has a unique solution, as happens most of the time,

$$c_0, c_1, c_2, \dots, c_{\dim V-1}, 1$$

are the coefficients of the minimal polynomial of T .

THM 5.27 (*eigenvalues are the zeros of the minimal polynomial*):

Let $T \in \mathcal{L}(V)$, which means T is an operator and let V be finite-dimensional ($\dim v \neq \infty$). Then

- (a) The zeros of the minimal polynomial of T are the eigenvalues of T .
- (b) If V is a complex vector space, the minimal polynomial has the form

$$(z - \lambda_1) \cdots (z - \lambda_m),$$

where $\lambda_1, \dots, \lambda_m$ are the eigenvalues of T , possibly with repetitions (may be $\lambda_i = \lambda_j$ for $i \neq j$).

PROOF: Let p be the minimal polynomial of T .

- (a) $\Rightarrow$ **DIRECTION:** Suppose $\lambda \in \mathbb{F}$ is a zero of p . Hence, we can write p as $p(z) = (z - \lambda)q(z)$, where q is a monic polynomial with coefficients in $\mathbb{F}$ (see 4.6). As we can write

$$p(T) = 0,$$

we can also write

$$0 = (T - \lambda I)(q(T)v) \quad \forall v \in V.$$

Because q is of lesser degree than p (i.e. $\deg q = (\deg p) - 1$) and p is the minimal polynomial of T , there exists at least one vector $u \in V$ such that

$$q(T)u \neq 0,$$

which makes $q(T)u$ an eigenvector with eigenvalue λ .

$\Leftarrow$ **DIRECTION:** Conversely, suppose $\lambda \in \mathbb{F}$ is an eigenvalue of T . Thus, there exists a nonzero $u \in V$ such that $Tu = \lambda u$. Repeated applications of T on both sides of this equation show that

$$T^k u = \lambda^k u \quad \forall k \in \mathbb{N}.$$

This implies that

$$p(T)u = p(\lambda)u.$$

Because p is the minimal polynomial of T , we have $p(T)u = p(\lambda)u = 0$. Since u is nonzero, we have $p(\lambda) = 0$. This makes λ a zero of p .

- (b) use (a) and the second version of the fundamental theorem of algebra (4.13). □

THM 5.29 (*every “zero-polynomial” is a multiple of the minimal polynomial*):

Let $T \in \mathcal{L}(V)$ and $q \in \mathcal{P}(\mathbb{F})$. Then $q(T) = 0 \iff q$ is a multiple of the minimal polynomial of T .

Or equivalently: The minimal polynomial of T divides every polynomial q for which $q(T) = 0$ (sometimes written as $m_T \mid q$).

PROOF: Let p denote the minimal polynomial of T .

- $\Rightarrow$ **DIRECTION:** Suppose $q(T) = 0$. By (4.9) there exists $s, r \in \mathcal{P}(\mathbb{F})$ such that

$$q = ps + r, \quad \deg r < \deg p.$$

We have

$$0 = q(T) = p(T)s(T) + r(T) = r(T). \tag{5.5}$$

The equation above implies that $r = 0$. Otherwise, dividing r by its highest-degree coefficient would produce a monic polynomial that when applied to T gives 0. A contradiction because $\deg r < \deg p$ and p is minimal. Thus, (5.5) becomes the equation $q = ps$, as desired

$\Leftarrow$ **DIRECTION:** Suppose $q = ps$ for $q, p, s \in \mathcal{P}(\mathbb{F})$. We have

$$q(T) = p(T)s(T) = 0s(T) = 0,$$

as desired. Which proves both directions. $\square$

THM 5.31 (*minimal polynomial of a restriction operator*):

If U is a subspace of V , then the minimal polynomial of T is a polynomial multiple of the minimal polynomial of $T|_U$.

PROOF: Suppose p is the minimal polynomial of $T \in \mathcal{L}(V)$ and $U \subseteq V$. Then

$$p(T)v = 0 \quad \forall v \in V.$$

In particular,

$$p(T)u = 0 \quad \forall u \in U.$$

Thus $p(T|_U) = 0$. Now, the previous theorem 5.29 tells us that p is a polynomial multiple of the minimal polynomial of $T|_U$. [This means that the degree of p is at least as large]. $\square$

THM 5.32 (*invertibility and the constant term of the minimal polynomial*):

Let $T \in \mathcal{L}(V)$. Then

T is not invertible $\iff$ the constant term of the minimal polynomial of T is 0.

PROOF: T is not invertible $\iff$ (5.7) 0 is an eigenvalue of $T \iff$ (5.27) 0 is a zero of $p \iff$ the constant term of p is 0. (In the first equivalence, we have actually used that 0 is an eigenvalue of T if and only if $T - 0I$ is not invertible, according to 5.7.) $\square$

EIGENVALUES ON ODD-DIMENSIONAL REAL VECTOR SPACES (SUB-SECTION 5B3)

THM 5.33 (*even-dimensional null space*):

Suppose $\mathbb{F} = \mathbb{R}$ and V is finite-dimensional. Suppose also that $T \in \mathcal{L}(V)$ and $b, c \in \mathbb{R}$ with $b^2 < 4c$. Then $\dim \text{null}(T^2 + bT + cI)$ is an even number.

PROOF: Recall that $\text{null}(T^2 + bT + cI)$ is invariant under T , by 5.18. Let

$$N := \text{null}(T^2 + bT + cI) \text{ and}$$

$$S := T|_N.$$

With these definitions above, we can assume that

$$S^2 + bS + cI = 0, \tag{5.6}$$

where 0 means the zero operator. We now need to prove that $\dim N$ is even. Suppose $\lambda \in \mathbb{R}$ and $v \in N$ are such that $Sv = \lambda v$. Then

$$0 = (S^2 + bS + cI)v = (\lambda^2 + b\lambda + c)v = \underbrace{\left(\left(\lambda + \frac{b}{2} \right)^2 + c - \frac{b^2}{4} \right)}_{\geq 0, \text{ since } b^2 \leq 4c} v$$

The term in the large parentheses above is a positive number. Thus, the equation above implies that $v = 0$. Hence, we have shown that S has no eigenvalues. Let U be a subspace of N with even dimension that is invariant under S and has the largest dimension among all subspaces of N . Note that U could be $\{0\}$. If $U = N$, then we are done.

Therefore, assume that $U \neq N$. By maximality, U is not properly contained in any larger even-dimensional S -invariant subspace of N . Hence, there exists $w \in N$ such that $w \notin U$. Now, let

$$W := \text{span}(w, Sw).$$

Clearly, $W \subseteq N$, and since $w \notin U$, $W \subsetneq U$. We can rewrite (5.6) as $S(Sw) = -bSw - cw$. Using this, we calculate for $x \in W$, written as $x = a_1w + a_2Sw$ for some scalars a_1, a_2 :

$$\begin{aligned} S(x) &= a_1Sw + a_2S^2w = a_1Sw - a_2(bSw + cw) \\ &= (a_1 - a_2b)Sw - a_2cw \\ &\in W \end{aligned}$$

Thus, we see that W is invariant under S (remember, $S = T|_N$). Furthermore, $\dim W = 2$, because otherwise w would be an eigenvalue of S , since w and Sw would be linearly dependent. This means that there would exist nonzero scalars a_1 and a_2 , such that we have a nontrivial relation $a_1w + a_2Sw = 0$. In this case, $-a_1/a_2$ would be an eigenvalue of S with eigenvector w , which is impossible. Now, using 2.43:

$$\dim(U + W) = \dim U + \underbrace{\dim W}_{=2} - \underbrace{\dim(U \cap W)}_{=0} = \dim U + 2,$$

where $U \cap W = \{0\}$, because otherwise, $U \cap W$ would be a one-dimensional S -invariant subspace providing an eigenvector for S , which is impossible. And the dimension of $U \cap W$ can not be 2 either, because then we would have $U \cap W = W$, using 2.39. But then $W \subseteq U$, contradicting the fact that $w \notin U$. Since U was assumed to be maximal among even-dimensional S -invariant subspaces, the existence of $U + W$, with $\dim(U + W) = \dim U + 2$, contradicts the maximality of U . Thus, the assumption that $U \neq N$ was incorrect. Therefore, we conclude that $U = N$. Hence, $N = \text{null}(S^2 + bS + cI) = \text{null}(T|_N^2 + bT|_N + cI) = \text{null}(T^2 + bT + cI)$ has even dimension $\dim N$. $\square$

THM 5.34 (*operators on odd-dimensional vector spaces have eigenvalues*):

Every operator on an odd-dimensional vector space has an eigenvalue.

PROOF: Suppose $\mathbb{F} = \mathbb{R}$ and V is finite-dimensional. Let $n := \dim V$, and suppose n is an odd number.

Let $T \in \mathcal{L}(V)$. We will use induction on n in steps of size two to show that T has an eigenvalue. Note that the desired result holds if $\dim V = 1$, because then every nonzero vector in V is an eigenvector of T .

Now, suppose $n \geq 3$ and the desired result holds for all operators on all odd-dimensional vector spaces of dimension less than n , i.e. with dimension $\leq n - 2$. Let p denote the minimal polynomial of T . If p is a polynomial multiple of $x - \lambda$ for some $\lambda \in \mathbb{R}$, then λ is an eigenvalue of T by theorem 5.27 and we are done. Thus, we can assume that there exists $b, c \in \mathbb{R}$ such that $b^2 < 4c$ and p is a polynomial multiple of $x^2 + bx + c$ by 4.16. By looking at this theorem, we can see that there exists a monic polynomial $q \in \mathcal{P}(\mathbb{R})$ such that $p(x) = q(x)(x^2 + bx + c) \quad \forall x \in \mathbb{R}$. Now,

$$0 = p(T) = (q(T))(T^2 + bT + cI),$$

which means that $q(T)$ equals 0 on $\text{range}(T^2 + bT + cI)$. Or with other words,

$$q(T)|_{\text{range}(T^2 + bT + cI)} = 0.$$

Because $\deg q < \deg p$ and p is the minimal polynomial of T , this implies that

$$\text{range}(T^2 + bT + cI) \neq V. \quad (5.7)$$

The rank-nullity theorem 3.21 tells us that

$$\dim V = \dim \text{null}(T^2 + bT + cI) + \dim \text{range}(T^2 + bT + cI) \quad (5.8)$$

The previous theorem 5.33 tells us that $\dim \text{null}(T^2 + bT + cI)$ is even. Since $\dim V$ is odd, the equation (5.8) above and equation (5.7) shows that $\dim \text{range}(T^2 + bT + cI)$ is odd as well and at least 2 less than $\dim V$. We know that $\text{range}(T^2 + bT + cI)$ is invariant under T by 5.18. Our induction hypothesis now implies that T restricted to $\text{range}(T^2 + bT + cI)$, which is a subset of

V , has an eigenvalue; this also means that T has an eigenvalue. $\square$

EXERCISES ABOUT THE MINIMAL POLYNOMIAL 5B

- ① Suppose $T \in \mathcal{L}(V)$. Prove that 9 is an eigenvalue of $T^2 \iff 3$ or -3 is an eigenvalue of T .

PROOF: $\Leftarrow$ **DIRECTION:** Suppose -3 is an eigenvalue of T . Thus, there exists $v \in V$ such that $Tv = -3v$. If we apply T a second time, we get $TTv = T(-3v) = -3Tv = (-3)(-3)v = 9v$. Therefore, 9 is an eigenvalue of T^2 . The case where we start with 3 as an eigenvalue of T works analogously.

$\Rightarrow$ **DIRECTION:** Let $v \neq 0$ satisfy $T^2v = 9v$. Then

$$0 = (T^2 - 9I)v = (T - 3I)(T + 3I)v.$$

Set $w := (T + 3I)v$. Then $(T - 3I)w = 0$, so $Tw = 3w$ (because $w \in \text{null}(T - 3I)$). There are two cases:

- If $w = 0$, then $(T + 3I)v = 0$. Hence, $Tv = -3v$ and -3 is an eigenvalue of T .
- If $w \neq 0$, then w is an eigenvector of T with eigenvalue 3 .

Thus, T has eigenvalue 3 or -3 , as required. $\square$

- ③ Suppose $n \in \mathbb{N}_{>0}$ and $T \in \mathcal{L}(\mathbb{F}^n)$ is defined by $T(x_1, \dots, x_n) := (x_1 + \dots + x_n, \dots, x_1 + \dots + x_n)$. The matrix $\mathcal{M}(T)$ of T looks like this

$$\mathcal{M}(T) = \begin{pmatrix} 1 & 1 & \cdots & 1 \\ 1 & 1 & \cdots & 1 \\ \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & \cdots & 1 \end{pmatrix}$$

- (a) Find all eigenvalues and eigenvectors of T .
 (b) Find the minimal polynomial of T .

SOLUTION: We want to find all $x \in V$ such that $Tx = \lambda x$. Therefore,

$$x_1 + \dots + x_n = \lambda x_1$$

$$\vdots$$

$$x_1 + \dots + x_n = \lambda x_n$$

If we define $S := x_1 + \dots + x_n$, we can rewrite the system of equations as

$$S = \lambda x_1, \dots, S = \lambda x_n, \tag{5.9}$$

or equivalently: $S = \lambda x_1 = \lambda x_2 = \dots = \lambda x_n$. Solving by x_i for $\lambda \neq 0$ gives

$$x_1 = \frac{S}{\lambda}, x_2 = \frac{S}{\lambda}, \dots, x_n = \frac{S}{\lambda}$$

Summing up all equations for S in (5.9) gives:

$$nS = \lambda x_1 + \dots + \lambda x_n = \lambda S$$

Hence,

$$nS - \lambda S = 0$$

$$S(n - \lambda) = 0. \quad (\text{which is the same as } (S - 0)(n - \lambda) = 0)$$

If $\lambda = n$, then $x_1 = x_2 = \dots = x_n$. Meaning all eigenvectors corresponding to the eigenvalue n are scalar multiples of $(1, 1, \dots, 1) \in \mathbb{F}^n$.

If $\lambda \neq n$, then $S(\lambda - n) = 0$ implies that $S = 0$ and thus, $\lambda = 0$. This means that x belongs to the eigenspace of 0 , if $x_1 + x_2 + x_3 + \dots + x_n = 0$.

THE MINIMAL POLYNOMIAL OF T :

$$T(e_1) = e_1 + \cdots + e_n$$

$$T(e_2) = e_1 + \cdots + e_n$$

$$\vdots$$

$$T(e_n) = e_1 + \cdots + e_n$$

Define $v := e_1 + \cdots + e_n$. Note that $\text{range } T = \text{span}(v)$. Applying T to v gives

$$T(v) = T(e_1 + \cdots + e_n) = T(e_1) + \cdots + T(e_n) = nv$$

$$T^2(v) = nT(v)$$

The last line tells us that $T^2 = nT$ or equivalently $T(T - nI) = 0$. Therefore, the minimal polynomial is

$$p(z) = z(z - n).$$

Because $\text{range } T = \text{span}(v)$, the polynomial is in fact minimal.

- ④ Suppose $\mathbb{F} = \mathbb{C}$, $T \in \mathcal{L}(V)$, $p \in \mathcal{P}(\mathbb{C})$, and $\alpha \in \mathbb{C}$. Prove that α is an eigenvalue of $p(T) \iff \alpha = p(\lambda)$ for some eigenvalue λ of T .

PROOF:

$\Leftarrow$ **DIRECTION:** Suppose $\lambda \in \mathbb{C}$ is an eigenvalue of T with $\alpha = p(\lambda)$. Then there exists a nonzero $v \in V$ such that $Tv = \lambda v$. Repeated application of T yields $T^k v = \lambda^k v$ for all $k \in \mathbb{N}$, and therefore $p(T)v = p(\lambda)v = \alpha v$. Since $v \neq 0$, α is an eigenvalue of $p(T)$.

$\Rightarrow$ **DIRECTION:** Conversely, suppose α is an eigenvalue of $p(T)$. Consider the polynomial $q(z) := p(z) - \alpha \in \mathcal{P}(\mathbb{C})$. Since $\mathbb{F} = \mathbb{C}$, by the Fundamental Theorem of Algebra (second version), $q(z)$ factors completely into linear factors over $\mathbb{C}$:

$$p(z) - \alpha = c(z - \lambda_1)(z - \lambda_2) \cdots (z - \lambda_m),$$

for some non-zero scalar $c \in \mathbb{C}$ and $\lambda_1, \dots, \lambda_m \in \mathbb{C}$. Substituting T into this polynomial equation gives:

$$p(T) - \alpha I = c(T - \lambda_1 I)(T - \lambda_2 I) \cdots (T - \lambda_m I).$$

Since α is an eigenvalue of $p(T)$, the operator $p(T) - \alpha I$ is not injective (and hence not invertible). A product of linear maps is non-invertible if and only if at least one factor is non-invertible. Therefore, there exists some $j \in \{1, \dots, m\}$ such that $T - \lambda_j I$ is not injective. This means λ_j is an eigenvalue of T . Evaluating the polynomial identity at $z = \lambda_j$ yields:

$$p(\lambda_j) - \alpha = c(\lambda_j - \lambda_1) \cdots (\lambda_j - \lambda_j) \cdots (\lambda_j - \lambda_m) = 0 \implies \alpha = p(\lambda_j).$$

Setting $\lambda := \lambda_j$ completes the proof. □

- ⑫ Define $T \in \mathcal{L}(\mathbb{F}^n)$ by $T(x_1, x_2, x_3, \dots, x_n) = (x_1, 2x_2, 3x_3, \dots, nx_n)$. Find the minimal polynomial of T .

SOLUTION: For a standard basis $e_1, \dots, e_n$ we have $T(e_1 + \cdots + e_n) = e_1 + 2e_2 + \cdots + ne_n$. Or equivalently:

$$T(e_1) = e_1$$

$$T(e_2) = 2e_2$$

$$\vdots$$

$$T(e_n) = ne_n.$$

Hence, $e_1, \dots, e_n$ are $\dim \mathbb{F}^n = n$ distinct eigenvectors with corresponding eigenvalues $1, \dots, n$. This makes $Te_1, \dots, Te_n$ linearly independent as well. Using Theorem 5.27, we conclude that

the minimal polynomial of T looks like this:

$$p(z) = (z-1)(z-2)\cdots(z-n) \quad \forall z \in \mathbb{F}. \quad (5.10)$$

We can check if $\forall v \in \mathbb{F}^n$, given as $v = a_1 e_1 + \cdots + a_n e_n$ ($a_i \in \mathbb{F}$), the following equation holds:

$$(T-I)(T-2I)\cdots(T-nI)v = 0.$$

This is true because $a_k e_k \in \text{null}(T - kI)$ for $k = 1, \dots, n$. Therefore,

$$\begin{aligned} (T-I)(T-2I)\cdots(T-nI)v &= (T-I)(T-2I)(T-3I)\cdots(T-nI)(a_1 e_1) \\ &\quad + (T-I)(T-2I)(T-3I)\cdots(T-nI)(a_2 e_2) \\ &\quad + (T-I)(T-2I)(T-3I)\cdots(T-nI)(a_3 e_3) \\ &\quad + \cdots \\ &\quad + (T-I)(T-2I)(T-3I)\cdots(T-nI)(a_n e_n) = 0. \end{aligned}$$

Thus, we have found our minimal polynomial given in equation (5.10). It is easy to conclude that there exists no polynomial of smaller degree that distributes to every basis vector of the sum $a_1 e_1 + \cdots + a_n e_n$.

- 16** Suppose $a_0, a_1, \dots, a_{n-1} \in \mathbb{F}$. Let T be the operator on $\mathbb{F}^n$ whose matrix (with respect to the standard basis) is

$$\mathcal{M}(T) = \begin{matrix} & e_1 & e_2 & \cdots & e_{n-1} & e_n \\ \begin{matrix} e_1 \\ e_2 \\ e_3 \\ \vdots \\ e_{n-1} \\ e_n \end{matrix} & \begin{pmatrix} 0 & & & & -a_0 \\ 1 & 0 & & & -a_1 \\ & 1 & \ddots & & -a_2 \\ & & \ddots & & \vdots \\ & & & 0 & -a_{n-2} \\ & & & 1 & -a_{n-1} \end{pmatrix} \end{matrix}.$$

Note that all the other entries of this matrix are zero. Show that the minimal polynomial of T is the polynomial

$$a_0 + a_1 z + \cdots + a_{n-1} z^{n-1} + z^n.$$

SOLUTION: Looking at the matrix $\mathcal{M}(T)$ and having a glance at Theorem 3.31, we observe that

$$\begin{aligned} T e_1 &= e_2 \\ T e_2 &= e_3 = T(T e_1) = T^2 e_1 \\ T e_3 &= e_4 = T(T e_2) = T^3 e_1 \\ T e_4 &= e_5 = T(T e_3) = T^4 e_1 \\ &\vdots \\ T e_{n-1} &= e_n = T(T e_{n-2}) = T^{n-1} e_1 \\ T e_n &= -a_0 e_1 - a_1 e_2 - a_2 e_3 - \cdots - a_{n-2} e_{n-1} - a_{n-1} e_n = T(T^{n-1} e_1) = T^n e_1. \end{aligned}$$

We can rewrite the last expression as

$$-T^n(e_1) = a_0 e_1 + a_1 T e_1 + a_2 T^2 e_1 + \cdots + a_{n-2} T^{n-2} e_1 + a_{n-1} T^{n-1} e_1$$

The list $e_1, T e_1, T^2 e_1, T^3 e_1, \dots, T^{n-1} e_1$, which equals the list $e_1, e_2, e_3, \dots, e_n$, is linearly independent. Therefore, no non-zero polynomial of degree strictly less than n can annihilate e_1 . Hence, the minimal polynomial of T is

$$a_0 + a_1 z + \cdots + a_{n-1} z^{n-1} + z^n.$$

UPPER-TRIANGULAR MATRICES (SECTION 5C)

THM 5.39 (conditions for upper-triangular matrix):

If $T \in \mathcal{L}(V)$ and $v_1, \dots, v_n$ is a basis of V , then the following are equivalent.

- (a) The matrix $\mathcal{M}(T)$ of T with respect to $v_1, \dots, v_n$ is upper triangular.
- (b) $\text{span}(v_1, \dots, v_k)$ is invariant under T for every $k = 1, \dots, n$.
- (c) $Tv_k \in \text{span}(v_1, \dots, v_k)$ for every $k = 1, \dots, n$.

PROOF: We prove (a) $\implies$ (b) $\implies$ (c) $\implies$ (a).

STEP 1: Suppose (a) holds and $j, k \in \{1, \dots, n\}$ with $j \leq k$. The matrix of T looks like this:

$$\mathcal{M}(T) = \begin{matrix} & v_1 & \cdots & v_j & \cdots & v_n \\ \begin{matrix} v_1 \\ \vdots \\ v_j \\ \vdots \\ v_n \end{matrix} & \begin{pmatrix} A_{1,1} & \cdots & A_{1,j} & \cdots & A_{1,n} \\ & \ddots & \vdots & \ddots & \vdots \\ & & A_{j,j} & \cdots & A_{j,n} \\ & & & \ddots & \vdots \\ & & & & A_{n,n} \end{pmatrix} \end{matrix}.$$

By definition of the matrix entries (3.31), we have that

$$Tv_j = A_{1,j}v_1 + A_{2,j}v_2 + \cdots + A_{j,j}v_j = \sum_{\ell=1}^j A_{\ell,j}v_\ell.$$

Hence, the upper-triangular structure of $\mathcal{M}(T)$ implies

$$Tv_j \in \text{span}(v_1, \dots, v_j).$$

Because

$$\text{span}(v_1, \dots, v_j) \subseteq \text{span}(v_1, \dots, v_k),$$

it's immediate that

$$Tv_j \in \text{span}(v_1, \dots, v_k).$$

Since we had $j \in \{1, \dots, k\}$ and so v_j could be any basis vector in the list $v_1, \dots, v_k$, $\text{span}(v_1, \dots, v_k)$ is invariant under T , completing the proof that (a) implies (b).

STEP 2: Now, suppose (b) holds. Hence, we assume $\text{span}(v_1, \dots, v_k)$ is invariant under T for each $k \in \{1, \dots, n\}$. In particular

$$Tv_k \in \text{span}(v_1, \dots, v_k) \quad \forall k \in \{1 \dots n\}.$$

Thus, (b) implies (c).

STEP 3: Finally, suppose (c) holds. Hence, $Tv_k \in \text{span}(v_1, \dots, v_k)$ for $k = 1, \dots, n$. This implies that there exist $a_k, \dots, a_n \in \mathbb{F}$, such that

$$Tv_k = a_1v_1 + \cdots + a_kv_k = a_1v_1 + \cdots + a_kv_k + 0v_{k+1} + \cdots + 0v_n.$$

Hence, all the entries below the diagonal of $\mathcal{M}(T)$ are 0, because the list $v_1, \dots, v_n$ is linearly independent and we cannot choose another representation for Tv_k in respect to that list. Thus, $\mathcal{M}(T)$ is an upper-triangular matrix, completing the proof that (c) implies (a). After all, we have shown that (a) $\implies$ (b) $\implies$ (c) $\implies$ (a). $\square$

THM 5.40 (equation satisfied by operator with upper-triangular matrix):

Suppose $T \in \mathcal{L}(V)$ and V has a basis with respect to which T has an upper-triangular matrix $\mathcal{M}(T)$ with diagonal entries $\lambda_1, \dots, \lambda_n$. Then

$$(T - \lambda_1 I) \cdots (T - \lambda_n I) = 0.$$

Note that this theorem does not mention anything about eigenvalues.

PROOF: Let $A := \mathcal{M}(T)$. Let $v_1, \dots, v_n$ denote a basis of V with respect to which T has an upper-triangular matrix with diagonal entries $\lambda_1, \dots, \lambda_n$.

$$A = \mathcal{M}(T) = \begin{pmatrix} \lambda_1 & & * \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix} = \begin{pmatrix} \lambda_1 & A_{1,2} & \cdots & A_{1,n} \\ & \lambda_2 & \cdots & A_{2,n} \\ & & \ddots & \vdots \\ 0 & & & \lambda_n \end{pmatrix}$$

Then

$$Tv_1 = \lambda_1 v_1 \iff (T - \lambda_1 I)v_1 = 0, \quad (5.11)$$

which implies that

$$\begin{aligned} (T - \lambda_1 I)(T - \lambda_2 I)v_1 &= 0, \\ (T - \lambda_1 I)(T - \lambda_2 I)(T - \lambda_3 I)v_1 &= 0, \\ \dots \text{ or more generally speaking:} \\ (T - \lambda_1 I) \cdots (T - \lambda_m I)v_1 &= 0, \text{ for } m = 1, \dots, n. \end{aligned}$$

(Using commutativity for linear maps after multiplying both sides of the equation with anything.)

Note that $(T - \lambda_2 I)v_2 \in \text{span}(v_1)$ because $Tv_2 = A_{1,2}v_1 + \lambda_2 v_2$. Thus using (5.11)

$$\begin{aligned} (T - \lambda_1 I)(T - \lambda_2 I)v_2 &= (T - \lambda_1 I)(A_{1,2}v_1) \\ &= A_{1,2}(T - \lambda_1 I)v_1 \\ &= 0, \end{aligned} \quad (5.12)$$

which implies that

$$\begin{aligned} (T - \lambda_1 I)(T - \lambda_2 I)(T - \lambda_3 I)v_2 &= 0, \\ (T - \lambda_1 I)(T - \lambda_2 I)(T - \lambda_3 I)(T - \lambda_4 I)v_2 &= 0, \\ \dots \text{ or more generally speaking:} \\ (T - \lambda_1 I) \cdots (T - \lambda_m I)v_2 &= 0, \text{ for } m = 2, \dots, n. \end{aligned}$$

(Using commutativity of polynomial factors of T .)

Note that $(T - \lambda_3 I)v_3 \in \text{span}(v_1, v_2)$ because $Tv_3 = A_{1,3}v_1 + A_{2,3}v_2 + \lambda_3 v_3$.

Thus using (5.11) and (5.12), we get

$$\begin{aligned} (T - \lambda_1 I)(T - \lambda_2 I)(T - \lambda_3 I)v_3 &= (T - \lambda_1 I)(T - \lambda_2 I)(A_{1,3}v_1 + A_{2,3}v_2) \\ &= A_{1,3}(T - \lambda_2 I)(T - \lambda_1 I)v_1 + A_{2,3}(T - \lambda_1 I)(T - \lambda_2 I)v_2 \\ &= A_{1,3}(T - \lambda_2 I)(0) + A_{2,3}(0) \\ &= 0, \end{aligned}$$

which implies that

$$(T - \lambda_1 I)(T - \lambda_2 I) \cdots (T - \lambda_m I)v_3 = 0, \text{ for } m = 3, \dots, n.$$

Continuing this pattern, we see that

$$(T - \lambda_1 I) \cdots (T - \lambda_n I)v_k = 0 \quad \forall k \in \{1 \dots n\}.$$

Thus $(T - \lambda_1 I) \cdots (T - \lambda_n I)$ is the 0 operator because it is 0 on each vector in a basis of V . $\square$

EXAMPLE (octave, matlab):

To test this in octave or matlab, type:

```
1 A=rand(4,4); I=eye(4); T=triu(A);
2 ZERO=(T-T(1,1)*I)*(T-T(2,2)*I)*(T-T(3,3)*I)*(T-T(4,4)*I)
```

which should yield a 4-by-4 zero-matrix.

THM 5.41 (*determination of eigenvalues from upper-triangular matrix*):

Suppose $T \in \mathcal{L}(V)$ has an upper-triangular matrix with respect to some basis of V . Then the eigenvalues of T are precisely the entries on the diagonal of that upper-triangular matrix.

PROOF: Let $v_1, \dots, v_n$ denote a basis of V with respect to which T has an upper-triangular matrix with diagonal entries $\lambda_1, \dots, \lambda_n$.

$$\mathcal{M}(T) = \begin{pmatrix} \lambda_1 & \cdots & * \\ & \ddots & \vdots \\ 0 & & \lambda_n \end{pmatrix}$$

Because $Tv_1 = \lambda_1 v_1$, we see that λ_1 is an eigenvalue of T . Suppose $k \in \{2, \dots, n\}$.

Then $(T - \lambda_k I)v_k \in \text{span}(v_1, \dots, v_{k-1})$, because

$$Tv_k = \mathcal{M}(T)_{1,k}v_1 + \mathcal{M}(T)_{2,k}v_2 + \cdots + \mathcal{M}(T)_{k-1,k}v_{k-1} + \lambda_k v_k.$$

Thus $T - \lambda_k I$ maps $\text{span}(v_1, \dots, v_k)$ into $\text{span}(v_1, \dots, v_{k-1})$.

Because

$$\dim \text{span}(v_1, \dots, v_k) = k \text{ and } \dim \text{span}(v_1, \dots, v_{k-1}) = k - 1,$$

this implies that the restriction of $T - \lambda_k I$ to $\text{span}(v_1, \dots, v_k)$ is not injective by Theorem 3.22.

Thus

$$\exists v \in \text{span}(v_1, \dots, v_k), v \neq 0 \text{ such that } (T - \lambda_k I)v = 0.$$

Thus, λ_k is an eigenvalue of T with eigenvector v . Hence we have shown that every entry on the diagonal of $\mathcal{M}(T)$ is an eigenvalue of T .

To prove T has no other eigenvalues, let

$$q(z) := (z - \lambda_1) \cdots (z - \lambda_n).$$

$$\implies q(T) = 0 \text{ by Theorem 5.40}$$

Hence q is a polynomial multiple of the minimal polynomial of T by Theorem 5.29. Thus every zero of the minimal polynomial (i.e. every eigenvalue of T) is a zero of q , which means every eigenvalue of T is in $\{\lambda_1, \dots, \lambda_n\}$. Because the zeros of the minimal polynomial of T are the eigenvalues of T by 5.27, this implies that every eigenvalue of T is a zero of q . Hence the eigenvalues of T are all contained in the list $\lambda_1, \dots, \lambda_n$. $\square$

THM 5.44 (*necessary and sufficient condition to have an upper-triangular-matrix*):

Suppose $T \in \mathcal{L}(V)$, $\dim V \neq \infty$. Then T has an upper-triangular matrix $\mathcal{M}(T)$ in respect to some basis of $V \iff$ The minimal polynomial of T equals $(z - \lambda_1) \cdots (z - \lambda_m)$, where $\lambda_1, \dots, \lambda_m \in \mathbb{F}$.

PROOF: $\implies$ **DIRECTION:** Suppose $T \in \mathcal{L}(V)$ has an upper-triangular matrix with respect to some basis of V . Let $\alpha_1, \dots, \alpha_n$ denote the diagonal entries of that matrix. I.e.,

$$\mathcal{M}(T) = \begin{pmatrix} \alpha_1 & \cdots & * \\ & \ddots & \vdots \\ 0 & & \alpha_n \end{pmatrix}.$$

Let $q \in \mathcal{P}(\mathbb{F})$ such that

$$q(z) := (z - \alpha_1) \cdots (z - \alpha_n).$$

Then $q(T) = 0$, by 5.40. Hence, q is a polynomial multiple of the minimal polynomial of T , by 5.29. Therefore, the minimal polynomial p of T equals

$$p(z) = (z - \lambda_1) \cdots (z - \lambda_m) \quad \forall z \in \mathbb{F}, \quad (\lambda_1, \dots, \lambda_m \in \mathbb{F})$$

and we have

$$\{\lambda_1, \dots, \lambda_m\} \subseteq \{\alpha_1, \dots, \alpha_n\}.$$

⇐ DIRECTION: Let $p \in \mathcal{P}(\mathbb{F})$ be the minimal polynomial of T such that

$$p(z) = (z - \lambda_1) \cdots (z - \lambda_m) \quad \forall z \in \mathbb{F}, \quad (\lambda_1, \dots, \lambda_m \in \mathbb{F}) \quad (5.13)$$

If $m = 1$, then $z - \lambda_1$ is the minimal polynomial of T . Thus, $T = \lambda_1 I$, which implies that the matrix of T is the identity matrix times a scalar.

$$\mathcal{M}(T) = \lambda_1 [I_{\dim V}]$$

Thus, $\mathcal{M}(T)$ is upper triangular. Therefore we are ready to use induction over m . Now, suppose $m > 1$ and the result holds for all smaller integers / polynomials. Let

$$U := \text{range}(T - \lambda_m I) \quad (5.14)$$

In this case, U is invariant under T , because of 5.18 where $p(z)$ would be $z - \lambda_m$. Thus $T|_U$ is an operator on U . If $u \in U$, then $u = (T - \lambda_m I)v$ for some $v \in V$. For this arbitrary u and corresponding v , it holds that

$$\begin{aligned} (T - \lambda_1 I) \cdots (T - \lambda_{m-1} I)u &= (T - \lambda_1 I) \cdots (T - \lambda_{m-1} I)(T - \lambda_m I)v \\ &= 0, \end{aligned}$$

because of (5.13). Hence, $(T - \lambda_1 I) \cdots (T - \lambda_{m-1} I)$ is a polynomial multiple of the minimal polynomial of $T|_U$, by 5.29. With other words: It is the product of at most $m - 1$ terms of the form $(z - \lambda_k)$ for $k = 1, \dots, m - 1$. By the induction hypothesis, there is a basis $u_1, \dots, u_M$ of U with respect to which $T|_U$ has an upper-triangular matrix. Using ??, we know that

$$\forall k \in \{1, \dots, M\} : Tu_k = (T|_U)(u_k) \in \text{span}(u_1, \dots, u_k). \quad (5.15)$$

Extend $u_1, \dots, u_M$ to a basis $u_1, \dots, u_M, v_1, \dots, v_N$ of V . If $k \in \{1, \dots, N\}$, then

$$Tv_k = Tv_k - \lambda_m v_k + \lambda_m v_k = (T - \lambda_m I)v_k + \lambda_m v_k. \quad (5.16)$$

The definition (5.14) of U shows that $(T - \lambda_m I)v_k \in U = \text{span}(u_1, \dots, u_M)$. Thus the equation (5.16) above shows that

$$\forall k \in \{1, \dots, N\} : Tv_k \in \text{span}(u_1, \dots, u_M, v_1, \dots, v_k) \quad (5.17)$$

Using theorem ?? together with (5.15) and (5.17), implies that $T \in \mathcal{L}(V)$ has an upper-triangular matrix with respect to the basis $u_1, \dots, u_M, v_1, \dots, v_N$ of V . $\square$

THM 5.47 (sufficient condition about the field for every operator on V to have an upper-triangular matrix):

Let $\mathbb{F} = \mathbb{C}$. Let $T \in \mathcal{L}(V)$, $\dim V \neq \infty$. Then T has an upper-triangular matrix with respect to some basis of V .

PROOF: This follows from the previous theorem and 5.44 and the second version of the fundamental theorem of algebra 4.13. Because $\mathbb{F} = \mathbb{C}$, the minimal polynomial of T can be expressed as $p \in \mathcal{P}(\mathbb{C})$, where

$$p(z) := (z - \lambda_1) \cdots (z - \lambda_m) \quad \forall z \in \mathbb{C}.$$

Hence, T has an upper-triangular matrix $\mathcal{M}(T)$. $\square$

EXERCISES ABOUT UPPER-TRIANGULAR MATRICES 5C

- 10** Suppose $T \in \mathcal{L}(V)$ and $v_1, \dots, v_n$ is a basis of V . Show that the following are equivalent.
- (a) The matrix $\mathcal{M}(T)$ of T with respect to $v_1, \dots, v_n$ is lower triangular.
 - (b) $\text{span}(v_k, \dots, v_n)$ is invariant under T for each $k = 1, \dots, n$.
 - (c) $Tv_k \in \text{span}(v_k, \dots, v_n)$ for each $k = 1, \dots, n$.

PROOF: Like in 5.39, we will prove that (a) $\implies$ (b) $\implies$ (c) $\implies$ (a).

STEP 1: Suppose (a) holds and fix $k \in \{1, \dots, n\}$. Consider $j \in \{k, \dots, n\}$ (i.e., $k \leq j$). The matrix $\mathcal{M}(T)$ of T looks as follows.

$$\mathcal{M}(T) = \begin{matrix} & v_1 & \cdots & v_j & \cdots & v_n \\ \begin{matrix} v_1 \\ \vdots \\ v_j \\ \vdots \\ v_n \end{matrix} & \begin{pmatrix} A_{1,1} & & & & \\ & \ddots & & & \\ & & A_{j,j} & & \\ & & \vdots & \ddots & \\ & & A_{n,j} & \cdots & A_{n,n} \end{pmatrix} \end{matrix}.$$

We have

$$Tv_j = A_{j,j}v_j + A_{j+1,j}v_{j+1} + \cdots + A_{n,j}v_n = \sum_{\ell=1}^n A_{\ell,j}v_\ell.$$

Therefore, $Tv_j \in \text{span}(v_j, \dots, v_n)$. Because

$$\text{span}(v_j, \dots, v_n) \subseteq \text{span}(v_k, \dots, v_n),$$

it's immediate that $Tv_j \in \text{span}(v_k, \dots, v_n)$. Since v_j could be any basis vector in the list $v_k, \dots, v_n$, $\text{span}(v_k, \dots, v_n)$ is invariant under T , completing the proof that (a) implies (b).

STEP 2: Now, suppose (b) holds. Hence, we assume $\text{span}(v_k, \dots, v_n)$ is invariant under T for each $k = 1, \dots, n$. In particular,

$$Tv_j \in \text{span}(v_j, \dots, v_n) \quad \forall j \in \{1, \dots, n\}.$$

Thus, (b) implies (c)

STEP 3: Finally, suppose (c) holds. Therefore, $Tv_k \in \text{span}(v_k, \dots, v_n)$ for $k = 1, \dots, n$. This implies that for a fixed k , there exist scalars $a_k, \dots, a_n \in \mathbb{F}$ such that

$$Tv_k = a_kv_k + \cdots + a_nv_n = 0v_1 + 0v_2 + \cdots + 0v_{k-1} + a_kv_k + \cdots + a_nv_n.$$

This implies that all the entries above the diagonal of $\mathcal{M}(T)$ are 0, because the list $v_1, \dots, v_n$ is linearly independent and we cannot choose another representation for Tv_k in respect to that list. Therefore, the scalars $a_k, \dots, a_n$ coincide with the entries of $\mathcal{M}(T)$. Thus, $\mathcal{M}(T)$ is a lower-triangular matrix, completing the proof that (c) implies (a). $\square$

DIAGONALIZABLE OPERATORS (SECTION 5D)

DIAGONAL MATRICES (SUB-SECTION 5D1)

DEF 5.48 (*diagonal matrix*):

A “*diagonal matrix*” is a square matrix that is 0 everywhere except possibly on the diagonal.

DEF 5.50 (*diagonalizable*):

An operator on V is called “*diagonalizable*” if the operator has a diagonal matrix with respect to some basis of V .

DEF 5.52 (*eigenspace, $E(\lambda, T)$*):

Let $T \in \mathcal{L}(V)$ and $\lambda \in \mathbb{F}$. The “*eigenspace*” of T corresponding to λ is the subspace $E(\lambda, T)$ of V defined by

$$E(\lambda, T) := \text{null}(T - \lambda I) = \{v \in V \mid Tv = \lambda v\}.$$

THM 5.54 (*sum of eigenspaces is a direct sum*):

Suppose $T \in \mathcal{L}(V)$ and $\lambda_1, \dots, \lambda_m$ are distinct eigenvalues of T . Then

$$E(\lambda_1, T) + \dots + E(\lambda_m, T)$$

is a direct sum (i.e., $\sum_{i=1}^m E(\lambda_i, T) = \bigoplus_{i=1}^m E(\lambda_i, T)$).

Furthermore, if V is finite-dimensional, then

$$\begin{aligned} \dim E(\lambda_1, T) + \dots + \dim E(\lambda_m, T) &= \dim(E(\lambda_1, T) \oplus \dots \oplus E(\lambda_m, T)) \\ &\leq \dim V. \end{aligned}$$

PROOF: Let $\lambda_1, \dots, \lambda_m$ be distinct eigenvalues. Suppose $v_1 + \dots + v_m = 0$, where each $v_k \in E(\lambda_k, T)$ for $k = 1, \dots, m$. This means $\forall k \in \{1, \dots, m\} : v_k = 0$, because eigenvectors corresponding to distinct eigenvalues are linearly independent by 5.11. Thus, $E(\lambda_1, T) + \dots + E(\lambda_m, T)$ is a direct sum by 1.45. By 3.94, the dimensions of a direct sum add up, and by 2.37, the dimension of a subspace is always $\leq$ to the space itself (if it is finite-dimensional). Therefore,

$$\dim \left(\underbrace{E(\lambda_1, T) \oplus \dots \oplus E(\lambda_m, T)}_{\text{subspace of } V} \right) = \dim E(\lambda_1, T) + \dots + \dim E(\lambda_m, T) \leq \dim V$$

□

CONDITIONS FOR DIAGONALIZABILITY (SUB-SECTION 5D2)

THM 5.55 (*conditions equivalent to diagonalizability*):

Let $\lambda_1, \dots, \lambda_m$ denote the distinct (i.e., $\lambda_j \neq \lambda_k$) eigenvalues of $T \in \mathcal{L}(V)$. Then the following are equivalent:

- (a) T is diagonalizable.
- (b) V has a basis consisting of eigenvectors of T .
- (c) $V = E(\lambda_1, T) \oplus \dots \oplus E(\lambda_m, T)$.
- (d) $\dim V = \dim E(\lambda_1, T) + \dots + \dim E(\lambda_m, T)$.

PROOF: We prove (a) $\iff$ (b), (b) $\implies$ (c), (c) $\implies$ (d), and then (d) $\implies$ (b). Let $\lambda_1, \dots, \lambda_m$ be the distinct eigenvalues of $T \in \mathcal{L}(V)$ for the whole proof.

STEP 1: Suppose (a) holds. The matrix of T is diagonal with respect to some basis $v_1, \dots, v_n$ (where $n = \dim V$). This means that each column contains only a diagonal entry, which is the

case $\iff Tv_k = \lambda_k v_k$ for each $k = 1, \dots, n$. Thus $v_1, \dots, v_n$ are eigenvectors of T . Thus (a) and (b) are equivalent, i.e. (a) $\iff$ (b).

STEP 2: Now, suppose (b) holds; thus V has a basis consisting of eigenvectors of T . Hence, every vector in V is a linear combination of such basis, which implies

$$V = E(\lambda_1, T) + \dots + E(\lambda_m, T).$$

By Theorem 5.54, distinctness of all λ_j 's implies that this is a direct sum (i.e., $V = \bigoplus_{j=1}^m E(\lambda_j, T)$). This proves that (b) $\implies$ (c).

STEP 3: (c) $\implies$ (d) follows immediately from Theorem 3.94, which states that the dimensions of a direct sum add up (i.e., $\dim\left(\bigoplus_{j=1}^m E(\lambda_j, T)\right) = \sum_{j=1}^m \dim E(\lambda_j, T)$).

STEP 4: Finally, suppose (d) holds; thus

$$\dim V = \dim E(\lambda_1, T) + \dots + \dim E(\lambda_m, T).$$

Choose a basis of each $E(\lambda_k, T)$ (for $k = 1, \dots, m$); put all these together to form a list

$$v_1, \dots, v_n \tag{5.18}$$

of eigenvectors of T , where $n = \dim V$. We want to show that (5.18) forms a basis of V . To this end, suppose for some $a_1, \dots, a_n \in \mathbb{F}$:

$$a_1 v_1 + \dots + a_n v_n = 0. \tag{5.19}$$

Now, for every $k = 1, \dots, m$, let

$$u_k := \sum_{\substack{v_j \in E(\lambda_k, T), \\ v_j \text{ is a basis vector of } E(\lambda_k, T)}} a_j v_j$$

denote the k -th sum of the terms $a_j v_j$, where the v_j 's belong to the basis of $E(\lambda_k, T)$. Thus $u_k \in E(\lambda_k, T)$. Therefore, we have

$$u_1 + \dots + u_m = 0,$$

as this is the same sum we had earlier (5.19). By Theorem 5.54, the sum of eigenspaces is a direct sum, which forces

$$u_k = 0 \quad \forall k \in \{1, \dots, m\}.$$

This implies that all the a 's must be 0, because the v 's in each eigenspace are linearly independent bases. Therefore, $v_1, \dots, v_n$ is a linearly independent list, and hence a basis of V , by Theorem 2.38 ($n = \dim V$). Now, we have shown that (d) $\implies$ (b), completing the proof. $\square$

THM 5.58 (*enough eigenvalues implies diagonalizability*):

If $T \in \mathcal{L}(V)$ has $\dim V$ distinct eigenvalues, then T is diagonalizable.

PROOF: Let $\lambda_1, \dots, \lambda_{\dim V}$ be the distinct eigenvalues of T with corresponding eigenvectors $v_1, \dots, v_{\dim V}$, which are linearly independent by Theorem 5.11. So we have a basis consisting of $\dim V$ eigenvectors. Hence, T is diagonalizable by Theorem 5.55. $\square$

THM 5.62 (*necessary and sufficient condition for diagonalizability*):

Let $T \in \mathcal{L}(V)$, $\dim V \neq \infty$. Then T is diagonalizable $\iff$ the minimal polynomial of T equals

$$(z - \lambda_1) \cdots (z - \lambda_m),$$

where $\lambda_1, \dots, \lambda_m \in \mathbb{F}$ and $\lambda_i \neq \lambda_j$ for $i, j \in \{1, \dots, m\}$.

PROOF: For this proof, let $T \in \mathcal{L}(V)$ be finite-dimensional ($\dim V \neq \infty$).

$\implies$ **DIRECTION:** Suppose T is diagonalizable. Thus, there exists a basis $v_1, \dots, v_n$ of V consisting of eigenvectors of T . Let $\lambda_1, \dots, \lambda_m$ be said distinct eigenvalues of T . Therefore, for every v_j we

have $(T - \lambda_k I)v_j = 0$. Thus,

$$(T - \lambda_1 I) \cdots (T - \lambda_m I)v_j = 0,$$

which implies the minimal polynomial equals $(z - \lambda_1) \cdots (z - \lambda_m)$. (Why?)

$\Leftarrow$ **DIRECTION:** Suppose the minimal polynomial of T equals $(z - \lambda_1) \cdots (z - \lambda_m)$ for some $\lambda_1, \dots, \lambda_m \in \mathbb{F}$ with $\lambda_i \neq \lambda_j$. Write

$$(T - \lambda_1 I) \cdots (T - \lambda_{m-1} I)(T - \lambda_m I) = 0. \quad (5.20)$$

Suppose $m = 1$. Then $T - \lambda_1 I = 0 \iff T = \lambda_1 I$, so T is diagonalizable. This is our base case for induction over $m \in \mathbb{N}$. Now, suppose $m > 1$ that and the desired result holds for all values smaller than m . Like in the proof of theorem 5.44, we observe that $\text{range}(T - \lambda_m I)$ is invariant under any polynomial of T (using 5.18 with $p(z) := z - \lambda_m$). Now, define

$$U := \text{range}(T - \lambda_m I).$$

This means that $T|_U$ is an operator on U . If $u \in \text{range}(T - \lambda_m I)$, then $u = (T - \lambda_m I)v$ for some $v \in V$. Together with (5.20), we have that

$$\begin{aligned} (T - \lambda_1 I) \cdots (T - \lambda_{m-1} I)u &= (T - \lambda_1 I) \cdots (T - \lambda_m I)v \\ &= 0. \end{aligned} \quad (5.21)$$

Since $u \in U$ was arbitrary, $q(z) := (z - \lambda_1) \cdots (z - \lambda_{m-1})$ is a polynomial such that $q(T)(u) = 0$ for every $u \in U$. By 5.29, $q(z)$ is a polynomial multiple of the minimal polynomial of $T|_U$. By our induction hypothesis, there is a basis of U consisting of eigenvectors of $T|_U$ and therefore, as well as eigenvectors of T . Suppose

$$x \in U \cap \text{null}(T - \lambda_m I) = U \cap E(\lambda_m, T).$$

Hence, $Tx = \lambda_m x$, because $Tx - \lambda_m Ix = (T - \lambda_m I)x = 0$. Since $x \in U$, equation (5.21) implies that

$$\begin{aligned} 0 &= (T - \lambda_1 I) \cdots (T - \lambda_{m-1} I)x \\ &= (\lambda_m I - \lambda_1 I) \cdots (\lambda_m I - \lambda_{m-1} I)x \\ &= (\lambda_m - \lambda_1) \cdots (\lambda_m - \lambda_{m-1})x \end{aligned}$$

Because $\forall i \neq j : \lambda_j \neq \lambda_k$, this can only happen if $x = 0$. Therefore,

$$U \cap \text{null}(T - \lambda_m I) = \{0\}.$$

Thus, $U + \text{null}(T - \lambda_m I)$ is a direct sum by 1.46. That is,

$$U + \text{null}(T - \lambda_m I) = U \oplus \text{null}(T - \lambda_m I)$$

Since $(T - \lambda_m I) \in \mathcal{L}(V)$ we can use the Rank Nullity Theorem 3.21 to conclude that

$$\dim V = \dim U + \dim \text{null}(T - \lambda_m I).$$

3.94 states that a sum is a direct sum $\iff$ the dimensions add up to the dimension of the sum.

This means that we can rewrite the equation above to

$$\dim V = \dim(U \oplus \text{null}(T - \lambda_m I))$$

According to 2.39, together with the fact that $U \oplus \text{null}(T - \lambda_m I) \subseteq V$, this implies

$$V = U \oplus \text{null}(T - \lambda_m I) = U \oplus E(\lambda_m, T).$$

Every vector in $\text{null}(T - \lambda_m I)$ is an eigenvector of T with eigenvalue λ_m . Earlier in this proof we saw that there is basis of $U = \text{range}(T - \lambda_m I)$ consisting of eigenvectors of T (using our induction hypothesis on this smaller subspace of V). Adjoining to that basis a basis of $E(\lambda_m, T) = \text{null}(T - \lambda_m I)$ gives a basis of V consisting of eigenvectors of T . The matrix $\mathcal{M}(T)$ of T with respect to this basis is a diagonal matrix, as desired, using 5.55 (according to that theorem, V has a basis consisting of eigenvectors of $T \in \mathcal{L}(V) \iff$ it is diagonalizable). As a side node, this theorem states that $V = E(\lambda_1, T) \oplus \cdots \oplus E(\lambda_m, T)$, which correlates with $V = U \oplus \text{null}(T - \lambda_m I)$. $\square$

THM 5.65 (*restriction of diagonalizable operator to invariant subspace*):

Let $T \in \mathcal{L}(V)$. Suppose T is diagonalizable and U is a subspace of V that is invariant under T . Then

$T|_U$ is a diagonalizable operator on U

PROOF: Diagonalizability of $T \iff_{(5.62)}$ the minimal polynomial of T equals

$$(z - \lambda_1) \cdots (z - \lambda_m) \text{ for } \lambda_j \neq \lambda_k.$$

By 5.31, the minimal polynomial of T is a polynomial multiple of the minimal polynomial of $T|_U$. Hence the minimal polynomial of $T|_U$ has the form required by 5.62, which shows that $T|_U$ is diagonalizable. It consists of any factor of the form $(z - \lambda_k)$. $\square$

EXERCISES ABOUT DIAGONALIZABLE OPERATORS 5D

- ② Suppose $T \in \mathcal{L}(V)$ has a diagonal matrix $A = \mathcal{M}(T)$ with respect to some basis of V . Prove that if $\lambda \in \mathbb{F}$, then λ appears on the diagonal of A precisely $\dim E(\lambda, T)$ times.

KI Bemerkung: Es gab hier drei verschiedene, teilweise fehlerhafte oder unvollständige Beweise für dieselbe Aufgabenstellung (die zudem als Exercise 2 und 3 dupliziert war). Ich habe diese zusammengefasst und den besten, ausführlichsten Beweis behalten sowie die Makros (wie `\linmap` statt `\op`) vereinheitlicht.

PROOF: If λ^* is not an eigenvalue of T , then

$$\dim E(\lambda^*, T) = \dim\{v \in V : Tv = \lambda^*v\} = \dim\{0\} = 0,$$

and therefore λ^* does not appear among the diagonal entries of A .

Now assume λ^* is an eigenvalue. Let $\mathcal{B} = \{v_1, \dots, v_\ell\}$ be a basis of V with respect to which $A = \mathcal{M}(T)$ is diagonal; write

$$A = \text{diag}(\mu_1, \dots, \mu_\ell),$$

so that for each j we have $Tv_j = \mu_j v_j$. In particular every v_j is an eigenvector of T (possibly for different eigenvalues).

Define the subset of basis vectors corresponding to the eigenvalue λ^* by

$$S = \{v_j \in \mathcal{B} \mid \mu_j = \lambda^*\}.$$

We first observe that $\text{span}(S) \subseteq E(\lambda^*, T)$, since each vector in S is by definition a λ^* -eigenvector and linear combinations of λ^* -eigenvectors remain in the λ^* -eigenspace.

For the reverse inclusion, let $w \in E(\lambda^*, T)$. Express w in the basis $\mathcal{B}$:

$$w = \sum_{j=1}^{\ell} c_j v_j \quad (c_j \in \mathbb{F}).$$

Applying T to both sides and using $Tv_j = \mu_j v_j$ gives

$$\lambda^* \sum_{j=1}^{\ell} c_j v_j = Tw = \sum_{j=1}^{\ell} c_j \mu_j v_j.$$

By uniqueness of coordinates with respect to the basis $\mathcal{B}$, the coefficient of v_j on the left must equal the coefficient of v_j on the right for each j , hence

$$(\lambda^* - \mu_j)c_j = 0 \quad \text{for every } j = 1, \dots, \ell.$$

Consequently $c_j = 0$ whenever $\mu_j \neq \lambda^*$, so every nonzero coefficient c_j corresponds to some $v_j \in S$. Thus w is a linear combination of vectors from S , meaning $w \in \text{span}(S)$. This proves the reverse inclusion and therefore

$$\text{span}(S) = E(\lambda^*, T).$$

Since S is a subset of the basis $\mathcal{B}$, its vectors are linearly independent; hence the number of elements of S equals the dimension of its span. Therefore

$$|S| = \dim(\text{span}(S)) = \dim E(\lambda^*, T).$$

Finally, by definition $|S|$ is exactly the number of diagonal entries of A equal to λ^* . We conclude that the scalar λ^* appears on the diagonal of A precisely $\dim E(\lambda^*, T)$ times. $\square$

COMMUTING OPERATORS (SECTION 5E)

DEF 5.66:

Two operators or matrices A and B “*commute*” if $ST = TS$

INNER PRODUCT SPACES (CHAPTER 6)

INNER PRODUCTS AND NORMS (SECTION 6A)

INNER PRODUCTS (SUB-SECTION 6A1)

We define the norm of $x = (x_1, \dots, x_n) \in \mathbb{R}^n$ by

$$\|x\| = \sqrt{x_1^2 + \dots + x_n^2}$$

The norm is not linear. The dot product is.

DEF 6.1 (dot product):

The “dot product” of $x, y \in \mathbb{R}^n$ is defined by

$$x \cdot y := x_1 y_1 + \dots + x_n y_n$$

The dot product on $\mathbb{R}^n$ has the following properties.

- $x \cdot x = \|x\|^2 \quad \forall x \in \mathbb{R}^n$
- $x \cdot x = 0 \iff x = 0$
- $y \in \mathbb{R}^n$: The map from $\mathbb{R}^n$ to $\mathbb{R}$ that sends $x \in \mathbb{R}^n$ to $x \cdot y$ is linear.
 $T : \mathbb{R}^n \rightarrow \mathbb{R}, T(x) = x \cdot y$. Then T is linear for a fixed $y \in \mathbb{R}^n$.
- $x \cdot y = y \cdot x$.

For $z = (z_1, \dots, z_n) \in \mathbb{C}^n$, we define the norm of z by

$$\begin{aligned} \|z\| &:= \sqrt{|z_1|^2 + \dots + |z_n|^2} = \sqrt{z_1 \bar{z}_1 + \dots + z_n \bar{z}_n} \\ &\implies \|z\|^2 = z_1 \bar{z}_1 + \dots + z_n \bar{z}_n \end{aligned}$$

DEF 6.2 (inner product):

The “inner product” on V is a function that takes each ordered pair (u, v) of elements of V to a number $\langle u, v \rangle \in \mathbb{F}$ and has the following properties:

- **POSITIVITY:** $\langle v, v \rangle \geq 0 \quad \forall v \in V$
- **DEFINITENESS:** $\langle v, v \rangle = 0 \iff v = 0$
- **ADDITIVITY IN FIRST SLOT:** $\langle u + v, w \rangle = \langle u, w \rangle + \langle v, w \rangle \quad \forall u, v, w \in V$
- **HOMOGENEITY IN FIRST SLOT:** $\langle \lambda u, v \rangle = \lambda \langle u, v \rangle \quad \forall \lambda \in \mathbb{F} \text{ and } \forall u, v \in V$.
- **CONJUGATE SYMMETRY:** $\langle u, v \rangle = \overline{\langle v, u \rangle} \quad \forall u, v \in V$

EXAMPLE 6.3:

$\langle w, z \rangle := c_1 w_1 \bar{z}_1 + \dots + c_n w_n \bar{z}_n$ (for positive constants $c_1, \dots, c_n$).

DEF 6.4 (inner product space):

An “inner product space” is a vector space V along with an inner product on V .

THM 6.6 (basic properties of an inner product):

$\forall u, v, w \in V$ and $\forall \lambda \in \mathbb{F}$:

- (a) For each fixed $u \in V$, the map $x \in V \mapsto \langle x, u \rangle \in \mathbb{F}$ is a linear map from V to $\mathbb{F}$.
- (b) $\langle 0, v \rangle = 0$
- (c) $\langle v, 0 \rangle = 0$
- (d) $\langle u, v + w \rangle = \langle u, v \rangle + \langle u, w \rangle$
- (e) $\langle u, \lambda v \rangle = \bar{\lambda} \langle u, v \rangle$

NORMS (SUB-SECTION 6A2)

DEF 6.7 (*norm*):

$$\|v\| := \sqrt{\langle v, v \rangle}, \quad v \in V$$

EXAMPLE 6.8:

For $f \in C[-1, 1]$ and with inner product given as in 6.3(c) we have $\|f\| = \sqrt{\int_{-1}^1 f^2}$

THM 6.9 (*basic properties of the norm*):

Suppose $v \in V$

- (a) $\|v\| = 0 \iff v = 0$
- (b) $\|\lambda v\| = |\lambda| \|v\| \quad \lambda \in \mathbb{F}.$

PROOF: Let $\lambda \in \mathbb{F}$ and $v \in V$. Then

- (a) $\langle v, v \rangle = 0 \iff v = 0$
- (b) $\|\lambda v\|^2 = \langle \lambda v, \lambda v \rangle = \lambda \bar{\lambda} \langle v, v \rangle = |\lambda|^2 \|v\|^2$

□

DEF 6.10 (*orthogonal*):

$u, v \in V$ are called “orthogonal” or “perpendicular” if

$$\langle u, v \rangle = 0. \text{ One also writes } u \perp v.$$

THM 6.11 (*orthogonality and 0*):

- (a) 0 is orthogonal to every vector in V .
- (b) 0 is the only vector that is orthogonal to itself.

THM 6.12 (*Pythagorean theorem*):

If $u, v \in V$ are orthogonal, then

$$\|u + v\|^2 = \|u\|^2 + \|v\|^2.$$

In this version of the theorem, u and v can equal to 0.

PROOF: $\|u + v\|^2 = \langle u + v, u + v \rangle = \underbrace{\langle u, u \rangle}_{=0} + \underbrace{\langle u, v \rangle + \langle v, u \rangle}_{=0} + \langle v, v \rangle = \|u\|^2 + \|v\|^2$

□

THM 6.13 (*an orthogonal decomposition*):

Suppose $u, v \in V, v \neq 0$. Set $c := \frac{\langle u, v \rangle}{\|v\|^2}$ and $w := u - \frac{\langle u, v \rangle}{\|v\|^2} v$. Then

$$u = cv + w = \frac{\langle u, v \rangle}{\|v\|^2} v + \left(u - \frac{\langle u, v \rangle}{\|v\|^2} v \right) \text{ and } \langle w, v \rangle = 0$$

THM 6.14 (*Cauchy-Schwarz inequality*):

Suppose $u, v \in V$. Then

$$|\langle u, v \rangle| \leq \|u\| \|v\|.$$

Equality holds $\iff$ one of u, v is a scalar multiple of the other.

Proof. If $v = 0$, then both sides equal 0, so the inequality holds. Thus suppose $v \neq 0$.

By the orthogonal decomposition (Theorem 6.13), we can write

$$u = \frac{\langle u, v \rangle}{\|v\|^2} v + w, \quad \text{where } \langle w, v \rangle = 0.$$

Since $\frac{\langle u, v \rangle}{\|v\|^2}v$ and w are orthogonal, the Pythagorean theorem (Theorem 6.12) gives:

$$\begin{aligned}\|u\|^2 &= \left\| \frac{\langle u, v \rangle}{\|v\|^2}v \right\|^2 + \|w\|^2 \\ &= \left| \frac{\langle u, v \rangle}{\|v\|^2} \right|^2 \|v\|^2 + \|w\|^2 \\ &= \frac{|\langle u, v \rangle|^2}{\|v\|^4} \|v\|^2 + \|w\|^2 \\ &= \frac{|\langle u, v \rangle|^2}{\|v\|^2} + \|w\|^2 \\ &\geq \frac{|\langle u, v \rangle|^2}{\|v\|^2}.\end{aligned}$$

Multiplying both sides by $\|v\|^2$ gives $\|u\|^2 \|v\|^2 \geq |\langle u, v \rangle|^2$. Taking square roots yields

$$|\langle u, v \rangle| \leq \|u\| \|v\|.$$

Equality holds if and only if $w = 0$, which occurs if and only if u is a scalar multiple of v . $\square$

THM 6.17 (triangle inequality):

Suppose $u, v \in V$. Then

$$\|u + v\| \leq \|u\| + \|v\|.$$

This inequality is an equality $\iff$ one of u, v is a non-negative real multiple of the other (i.e., $u = cv$ with $c \geq 0$ or $v = cu$ with $c \geq 0$).

PROOF: Let $u, v \in V$, where both might equal to 0. Thus

$$\begin{aligned}\|u + v\|^2 &= \langle u + v, u + v \rangle \\ &= \langle u, u + v \rangle + \langle v, u + v \rangle \\ &= \langle u, u \rangle + \langle u, v \rangle + \langle v, u \rangle + \langle v, v \rangle \\ &= \langle u, u \rangle + \langle v, v \rangle + \langle u, v \rangle + \overline{\langle u, v \rangle} \quad \text{conjugate symmetry} \\ &= \|u\|^2 + \|v\|^2 + 2\operatorname{Re} \langle u, v \rangle \quad z + \bar{z} = 2\operatorname{Re} z \\ &\leq \|u\|^2 + \|v\|^2 + 2|\langle u, v \rangle| \quad z = \operatorname{Re} z + (\operatorname{Im} z)i \\ &\leq \|u\|^2 + \|v\|^2 + 2\|u\|\|v\| \quad \text{Cauchy-Schwarz} \\ &= (\|u\| + \|v\|)^2\end{aligned}$$

We have equality in the triangle inequality $\iff \langle u, v \rangle = \|u\| \cdot \|v\| \iff u = \lambda v$. $\square$

ORTHONORMAL BASES (SECTION 6B)

ORTHONORMAL LISTS AND THE GRAM-SCHMIDT PROCEDURE (SUB-SECTION 6B1)

DEF 6.22 (orthonormal):

A list of vectors is called “*orthonormal*” if each vector in the list has norm 1 and is orthogonal to all the other vectors in the list. Since the square root of 1 is 1, we can use the following definition as well:

A list $e_1, \dots, e_m$ of vectors in V is “*orthonormal*” if $\forall j, k \in \{1, \dots, m\}$

$$\langle e_j, e_k \rangle = \begin{cases} 1, & \text{if } j = k \\ 0, & \text{if } j \neq k \end{cases}$$

Recall that $\|v\| \equiv \sqrt{\langle v, v \rangle}$.

THM 6.24 (*norm of an orthonormal linear combination*):

If $e_1, \dots, e_m$ is an orthonormal list of vectors in V , Then

$$\|a_1 e_1 + \dots + a_m e_m\|^2 = |a_1|^2 + \dots + |a_m|^2 \quad \forall a_1, \dots, a_m \in \mathbb{F}.$$

PROOF: Because $\forall k : \|e_k\| = 1$, this follows from repeated applications of the Pythagorean theorem

6.12

$$\begin{aligned} \|a_1 e_1 + \dots + a_m e_m\|^2 &= \|(a_1 e_1 + \dots + a_{m-1} e_{m-1}) + a_m e_m\|^2 \\ &= \|(a_1 e_1 + \dots + a_{m-1} e_{m-1})\|^2 + \|a_m e_m\|^2 \\ &= \|(a_1 e_1 + \dots + a_{m-2} e_{m-2}) + a_{m-1} e_{m-1}\|^2 + |a_m|^2 \\ &= \|(a_1 e_1 + \dots + a_{m-2} e_{m-2})\|^2 + \|a_{m-1} e_{m-1}\|^2 + |a_m|^2 \\ &= \|(a_1 e_1 + \dots + a_{m-3} e_{m-3}) + a_{m-2} e_{m-2}\|^2 + |a_{m-1}|^2 + |a_m|^2 \\ &= \dots \\ &= |a_1|^2 + |a_2|^2 + \dots + |a_{m-1}|^2 + |a_m|^2 \end{aligned}$$

□

THM 6.25 (*linear independence*):

Every orthonormal list of vectors is linearly independent.

PROOF: Suppose $e_1, \dots, e_m$ is an orthonormal list of vectors in V and $a_1, \dots, a_m \in \mathbb{F}$ are such that

$$a_1 e_1 + \dots + a_m e_m = 0.$$

$\implies |a_1|^2 + \dots + |a_m|^2 = 0$ by 6.24, which means that all the a_k 's are 0. This is all that is needed to show. □

THM 6.26 (*Bessel's inequality*):

Suppose $e_1, \dots, e_m$ is an orthonormal list of vectors in V . If $v \in V$ then

$$|\langle v, e_1 \rangle|^2 + \dots + |\langle v, e_m \rangle|^2 \leq \|v\|^2.$$

PROOF: Suppose $v \in V$. Then

$$v = \underbrace{\langle v, e_1 \rangle e_1 + \dots + \langle v, e_m \rangle e_m}_u + \underbrace{(v - (\langle v, e_1 \rangle e_1 + \dots + \langle v, e_m \rangle e_m))}_w$$

Let u and w be defined as in the equation above. If $k \in \{1, \dots, m\}$, then

$$\begin{aligned} \langle w, e_k \rangle &= \langle v - (\langle v, e_1 \rangle e_1 + \dots + \langle v, e_m \rangle e_m), e_k \rangle \\ &= \langle v, e_k \rangle - \langle v, e_1 \rangle \langle e_1, e_k \rangle - \langle v, e_2 \rangle \langle e_2, e_k \rangle - \dots - \langle v, e_m \rangle \langle e_m, e_k \rangle \\ &= \langle v, e_k \rangle - \langle v, e_k \rangle \langle e_k, e_k \rangle \\ &= 0. \text{ Hence} \end{aligned}$$

$$\begin{aligned} \langle w, u \rangle &= \langle w, \langle v, e_1 \rangle e_1 + \dots + \langle v, e_m \rangle e_m \rangle \\ &= \langle w, \langle v, e_1 \rangle e_1 \rangle + \dots + \langle w, \langle v, e_m \rangle e_m \rangle \\ &= \overline{\langle v, e_1 \rangle} \langle w, e_1 \rangle + \dots + \overline{\langle v, e_m \rangle} \langle w, e_m \rangle \\ &= 0. \end{aligned}$$

Since this means that u and w are orthogonal, we can use the Pythagorean theorem Theorem 6.12 to conclude that

$$\begin{aligned} \|v\|^2 &= \|u\|^2 + \|w\|^2 \\ &\geq \|u\|^2 \\ &= |\langle v, e_1 \rangle|^2 + \dots + |\langle v, e_m \rangle|^2, \end{aligned}$$

where the last line comes from Theorem 6.24. $\square$

DEF 6.27 (*orthonormal basis*):

An “orthonormal basis” of V is an orthonormal list of vectors in V that is also a basis of V .

THM 6.28 (*orthonormal lists of the right length are orthonormal bases*):

Suppose V is finite-dimensional. Then every orthonormal list of vectors in V of length $\dim V$ is an orthonormal basis of V .

PROOF: By Theorem 6.25, every orthonormal list is linearly independent. Every linearly independent list in V of length $\dim V$ is a basis of V (by Theorem 2.38). Hence the list is an orthonormal basis of V . $\square$

THM 6.30 (*writing a vector as a linear combination of an orthonormal basis*):

Suppose $e_1, \dots, e_n$ is an orthonormal basis of V and $u, v \in V$. Then

- (a) $v = \langle v, e_1 \rangle e_1 + \dots + \langle v, e_n \rangle e_n$;
- (b) $\|v\|^2 = |\langle v, e_1 \rangle|^2 + \dots + |\langle v, e_n \rangle|^2$;
- (c) $\langle u, v \rangle = \langle u, e_1 \rangle \overline{\langle v, e_1 \rangle} + \dots + \langle u, e_n \rangle \overline{\langle v, e_n \rangle}$;

PROOF: Let $v = a_1 e_1 + \dots + a_n e_n$. Then $\forall k \in \{1, \dots, n\}$ we have

$$\begin{aligned} \langle v, e_k \rangle &= \langle a_1 e_1 + \dots + a_n e_n, e_k \rangle \\ &= \langle a_1 e_1, e_k \rangle + \dots + \langle a_n e_n, e_k \rangle \\ &= \langle a_k e_k, e_k \rangle \\ &= a_k \langle e_k, e_k \rangle \\ &= a_k \end{aligned}$$

Thus (a) holds.

Now (b) follows immediately from (a) and 6.24.

Using conjugate linearity

$$\begin{aligned} \langle u, v \rangle &= \langle u, \langle v, e_1 \rangle e_1 + \dots + \langle v, e_n \rangle e_n \rangle \\ &= \langle u, \langle v, e_1 \rangle e_1 \rangle + \dots + \langle u, \langle v, e_n \rangle e_n \rangle \\ &= \overline{\langle v, e_1 \rangle} \langle u, e_1 \rangle + \dots + \overline{\langle v, e_n \rangle} \langle u, e_n \rangle \\ &= \langle u, e_1 \rangle \overline{\langle v, e_1 \rangle} + \dots + \langle u, e_n \rangle \overline{\langle v, e_n \rangle} \end{aligned}$$

Therefore (c) holds. $\square$

LINEAR FUNCTIONALS ON INNER PRODUCT SPACES (SUB-SECTION 6B2)

ORTHOGONAL COMPLEMENTS AND MINIMIZATION PROBLEMS (SECTION 6C)

ORTHOGONAL COMPLEMENTS (SUB-SECTION 6C1)

DEF 6.46 (*orthogonal complement, $U^\perp$*):

If $U \subset V$, then the “orthogonal complement” of U , denoted by $U^\perp$, is defined as follows:

$$U^\perp := \{v \in V \mid \langle u, v \rangle = 0 \quad \forall u \in U\}.$$

THM 6.48 (*properties of the orthogonal complement*):

- (a) If $U \subseteq V$, then $U^\perp$ is a subspace of V .

- (b) $\{0\}^\perp = V$.
 (c) $V^\perp = \{0\}$.
 (d) If U is a subset of V , then $U \cap U^\perp \subseteq \{0\}$.
 (e) If G and H are subsets of V and $G \subseteq H$, then

$$H^\perp \subseteq G^\perp. \text{ This is not a typo.}$$

PROOF:

- (a) We have to check for additive identity, closedness under addition and closedness under scalar multiplication.

- $\langle u, 0 \rangle = 0 \quad \forall u \in U \implies 0 \in U^\perp$.
- Let $v, w \in U^\perp, u \in U$. Then $\langle u, v+w \rangle = \langle u, v \rangle + \langle u, w \rangle = 0 + 0 = 0 \implies v+w \in U^\perp$.
- Let $\lambda \in \mathbb{F}, w \in U^\perp, u \in U$. Then $\langle u, \lambda w \rangle = \bar{\lambda} \langle u, w \rangle = \bar{\lambda} 0 = 0 \implies \lambda w \in U^\perp$.

Therefore $U^\perp$ is a subspace of V .

- (b) Let $v \in V$. $\implies \langle 0, v \rangle = 0 \implies v \in \{0\}^\perp$. Since v was arbitrary, $\{0\}^\perp = V$.
 (c) Let $v \in V^\perp \implies \langle v, v \rangle = 0 \implies v = 0 \implies V^\perp = \{0\}$.
 (d) Suppose $U \subseteq V$ and $u \in U \cap U^\perp \implies \langle u, u \rangle = 0 \implies u = 0 \implies U \cap U^\perp \subseteq \{0\}$.
 (e) Suppose $G \subseteq H \subseteq V$.

$$\begin{aligned} \text{Let } v \in H^\perp &\implies \langle u, v \rangle = 0 \quad \forall u \in H \\ &\implies \langle u, v \rangle = 0 \quad \forall u \in G \\ &\implies v \in G^\perp. \text{ Thus } H^\perp \subseteq G^\perp. \end{aligned}$$

□

THM 6.49 (*direct sum of a subspace and its orthogonal complement*):

Suppose U is a finite dimensional subspace of V . Then

$$V = U \oplus U^\perp. \quad (v \in V \text{ can be uniquely written as } v = u + w, \text{ where } u \in U, w \in U^\perp).$$

PROOF: Let $v \in V$ and $e_1, \dots, e_m$ be an orthonormal basis of U . Like in 6.26, we have that

$$v = \underbrace{\langle v, e_1 \rangle e_1 + \dots + \langle v, e_m \rangle e_m}_u + \underbrace{v - \langle v, e_1 \rangle e_1 - \dots - \langle v, e_m \rangle e_m}_w.$$

and we let u and w be defined as above.

$$\forall k \in \{1, \dots, m\} : e_k \in U. \implies u \in U, \text{ because it is a linear combination of } e_1, \dots, e_m.$$

Again like in 6.26, $\forall k \in \{1, \dots, m\}$:

$$\begin{aligned} \langle w, e_k \rangle &= \left\langle v - \underbrace{\langle v, e_1 \rangle e_1 + \dots + \langle v, e_m \rangle e_m}_{\in \mathbb{F}} \right\rangle e_k \\ &= \langle v, e_k \rangle - \underbrace{\langle v, e_1 \rangle \langle e_1, e_k \rangle}_{\in \mathbb{F}} - \underbrace{\langle v, e_2 \rangle \langle e_2, e_k \rangle}_{\in \mathbb{F}} - \dots - \underbrace{\langle v, e_m \rangle \langle e_m, e_k \rangle}_{\in \mathbb{F}} \\ &= \langle v, e_k \rangle - \langle v, e_k \rangle \langle e_k, e_k \rangle \\ &= 0. \end{aligned}$$

$\implies w \perp x$, whenever $x \in \text{span}(e_1, \dots, e_m)$. This is very easy to verify using linearity of the inner product in the second slot. Since we had $\forall k \in \{1, \dots, m\} : e_k \in U$,

$$\implies w \in U^\perp.$$

$$\implies v = u + w, \text{ where } u \in U \text{ and } w \in U^\perp.$$

□

THM 6.51 (*dimension of orthogonal complement*):

Suppose V is finite-dimensional and U is a subspace of V . Then

$$\dim U^\perp = \dim V - \dim U$$

PROOF: Because V is a subspace of itself, this follows immediately from 6.49 and 3.94. $\square$

THM 6.52 (*orthogonal complement of the orthogonal complement*):

Suppose U is a finite-dimensional subspace of V . Then

$$U = (U^\perp)^\perp$$

PROOF: We proof this theorem by showing that $U \subseteq (U^\perp)^\perp$ and $(U^\perp)^\perp \subseteq U$.

STEP 1: Let $u \in U$. $\implies \langle u, w \rangle = 0 \ \forall w \in U^\perp$. Therefore for the same w 's using conjugate symmetry

$$\langle u, w \rangle = \overline{\langle w, u \rangle} = 0 \implies \overline{\langle w, u \rangle} = \overline{0} \implies \langle w, u \rangle = 0.$$

Because u is orthogonal to every vector w in $U^\perp$, we have $u \in (U^\perp)^\perp$. Hence $U \subseteq (U^\perp)^\perp$.

STEP 2: Let $v \in (U^\perp)^\perp \subseteq V$. By 6.49 we can write v as a sum like this:

$$v = u + w, \text{ where } u \in U \text{ and } w \in U^\perp.$$

Hence $v - u = w \in U^\perp$. Because $v \in (U^\perp)^\perp$ and $u \in U \subseteq (U^\perp)^\perp$ from step 1, we have

$$v - u = w \in U^\perp \cap (U^\perp)^\perp \subseteq \{0\}.$$

$$\implies v = u. \implies v \in U. \implies (U^\perp)^\perp \subseteq U, \text{ because } v \text{ was arbitrary.}$$

$\square$

THM 6.54:

Suppose U is a finite-dimensional subspace of V . Then

$$U^\perp = \{0\} \iff U = V.$$

PROOF: “ $\implies$ -direction:” $U^\perp = \{0\}$. Then by first using 6.52 and then using 6.48 (c), $U = (U^\perp)^\perp = \{0\}^\perp = V$.

“ $\impliedby$ -direction:” Only using 6.48 (c): If $U = V$, then $U^\perp = V^\perp = \{0\}$. $\square$

DEF 6.55 (*orthogonal projection, P_U*):

Suppose $U \subseteq V$, $\dim U \neq \infty$. The “orthogonal projection” of V onto U is the operator $P_U \in \mathcal{L}(V)$ defined as follows:

$\forall v \in V$, we can write $v = u + w$, where $u \in U$ and $w \in U^\perp$ (Because $V = U \oplus U^\perp$). Then let

$$P_U v := u.$$

EXAMPLE 6.56 (*orthogonal projection onto one-dimensional subspace*):

Suppose $u \in V, u \neq 0$ and $U := \text{span}(u)$ is a one-dimensional subspace of V . If $v \in V$, then

$$v = \underbrace{\frac{\langle v, u \rangle}{\|u\|^2} u}_{\in \text{span}(u) = U} + \underbrace{\left(v - \frac{\langle v, u \rangle}{\|u\|^2} u \right)}_{\in U^\perp}$$

Therefore

$$P_U v = \frac{\langle v, u \rangle}{\|u\|^2} u \quad \forall v \in V.$$

Notice that $P_U u = u$ and $P_U v = 0 \iff v \in \{u\}^\perp$. Recall $\langle u, u \rangle = \|u\|^2$.

THM 6.57 (*properties orthogonal projection P_U*):

Suppose $U \subseteq V$, $\dim U \neq \infty$. Then

- (a) $P_U \in \mathcal{L}(V)$;
- (b) $P_U u = u \quad \forall u \in U$;
- (c) $P_U w = 0 \quad \forall w \in U^\perp$;
- (d) $\text{range } P_U = U$;
- (e) $\text{null } P_U = U^\perp$;

- (f) $v - P_U v \in U^\perp \quad \forall v \in V$;
- (g) $P_U^2 = P_U$;
- (h) $\|P_U v\| \leq \|v\| \quad \forall v \in V$;
- (i) If $e_1, \dots, e_m$ is an orthonormal basis of U and $v \in V$, then

$$P_U v = \langle v, e_1 \rangle e_1 + \dots + \langle v, e_m \rangle e_m.$$

MINIMIZATION PROBLEMS (SUB-SECTION 6C2)

Given a subspace $U \subseteq V$ and $v \in V$, find a $u \in U$ such that $\|v - u\|$ is as small as possible.

THM 6.58 (*minimizing distance to a subspace*):

Suppose U is a finite-dimensional subspace of V , $v \in V$, and $u \in U$. Then

$$\|v - P_U v\| \leq \|v - u\|.$$

Furthermore, the inequality above is an equality $\iff u = P_U v$.

PROOF: For this proof, visualize that $v - P_U v$ is pointing from $P_U v$ to v and $P_U v - u$ is pointing from u to the projection of v , namely $P_U v$ which lies on the vector space U .

We also have the properties that $v - P_U v \in U^\perp$ and $P_U v - u \in U$. Therefore, we can use the Pythagorean theorem Theorem 6.12 after adding a always-positive term to the left hand side of the inequality.

$$\begin{aligned} \|v - P_U v\|^2 &\leq \|v - P_U v\|^2 + \underbrace{\|P_U v - u\|^2}_{\geq 0} \\ &= \|(v - P_U v) + (P_U v - u)\|^2 \\ &= \|v - u\|^2. \end{aligned}$$

The inequality proved above is an equality $\iff \|P_U v - u\|^2 = 0 \iff u = P_U v$. □

PSEUDOINVERSE (SUB-SECTION 6C3)

If $T \in \mathcal{L}(V, W)$ is not invertible, instead of solving the equation $Tv = w$, which is the same as $Tv - w = 0$, we can try to find $v \in V$ such that $\|Tv - w\|$ is as small as possible.

THM 6.59 (*restriction of a linear map to obtain a one-to-one and onto map*):

Suppose $\dim V \neq 0$ and $T \in \mathcal{L}(V, W)$. Then $T|_{(\text{null } T)^\perp}$ is an injective map of $(\text{null } T)^\perp$ onto $\text{range } T$.

PROOF: We will first prove injectivity and then surjectivity. Suppose $v \in (\text{null } T)^\perp$ such that $T|_{(\text{null } T)^\perp} v = 0$. Hence $Tv = 0$ and $v \in (\text{null } T) \cap (\text{null } T)^\perp$, which implies that $v = 0$ by Theorem 6.48, because $(\text{null } T) \cap (\text{null } T)^\perp \subseteq \{0\}$.

$$\implies \text{null } T|_{(\text{null } T)^\perp} = \{0\},$$

which Theorem 3.15 implies that $T|_{(\text{null } T)^\perp}$ is injective.

To prove that $\text{range } T|_{(\text{null } T)^\perp}$ is onto $\text{range } T$, we will show that $\text{range } T|_{(\text{null } T)^\perp} \subseteq \text{range } T$ and then $\text{range } T \subseteq \text{range } T|_{(\text{null } T)^\perp}$. Because this will imply that $\text{range } T = \text{range } T|_{(\text{null } T)^\perp}$ and therefore surjectivity.

PROOF OF $\text{range } T|_{(\text{null } T)^\perp} \subseteq \text{range } T$: The fact that $\text{range } T|_{(\text{null } T)^\perp} \subseteq \text{range } T$ is actually obvious, because $\text{range } T|_{(\text{null } T)^\perp}$ is just T on a restricted domain $(\text{null } T)^\perp$.

PROOF OF $\text{range } T \subseteq \text{range } T|_{(\text{null } T)^\perp}$: Now suppose $w \in \text{range } T \implies \exists v \in V$ such that $w = Tv$.

By Theorem 6.49, there exists $u \in \text{null } T$ and $x \in (\text{null } T)^\perp$ such that

$$v = u + x \iff x = v - u.$$

$$\implies T|_{(\text{null } T)^\perp} x = T(x) = T(v - u) = T(v) - T(u) = T(v) - 0 = w,$$

which shows that $w \in \text{range } T|_{(\text{null } T)^\perp}$ and thus $\text{range } T \subseteq \text{range } T|_{(\text{null } T)^\perp}$.

We conclude: $\text{range } T = \text{range } T|_{(\text{null } T)^\perp}$. □

DEF 6.68:

Suppose $\dim V \neq \infty$ and $T \in \mathcal{L}(V, W)$. The “pseudoinverse” $T^\dagger \in \mathcal{L}(W, V)$ of T is defined by

$$T^\dagger w := \left(T|_{(\text{null } T)^\perp} \right)^{-1} P_{\text{range } T} w \quad \forall w \in W.$$

Theorem 6.59 makes sure that $\left(T|_{(\text{null } T)^\perp} \right)^{-1} \in \mathcal{L}(W, (\text{null } T)^\perp)$ is well defined.

Recall that

- By 6.57 (b): $P_{\text{range } T} w = w$ if $w \in \text{range } T$.
- By 6.57 (c): $P_{\text{range } T} w = 0$ if $w \in (\text{range } T)^\perp$.

Since inverse maps are always injective and onto, we have that

- $w \in \text{range } T \implies T^\dagger w$ is the unique element of $(\text{null } T)^\perp$ such that $T(T^\dagger w) = w$.
- $w \in (\text{range } T)^\perp \implies T^\dagger w = 0$.

THM 6.69 (algebraic properties of the pseudoinverse):

Suppose $\dim V \neq \infty$ and $T \in \mathcal{L}(V, W)$.

- (a) If T is invertible, then $T^\dagger = T^{-1}$.
- (b) $TT^\dagger = P_{\text{range } T}$ = the orthogonal projection of W onto $\text{range } T$.
- (c) $T^\dagger T = P_{(\text{null } T)^\perp}$ = the orthogonal projection of V onto $(\text{null } T)^\perp$.

PROOF: Suppose $T \in \mathcal{L}(V, W)$, $\dim V \neq \infty$ is invertible. This means $\text{null } T = \{0\}$. Hence by 6.48 (b) we have that

$$(\text{null } T)^\perp = \{0\}^\perp = V. \implies T|_{(\text{null } T)^\perp} = T|_V = T. \quad (6.1)$$

Invertibility also implies that $\text{range } T = W$. Hence

$$P_{\text{range } T} = I_W \quad (6.2)$$

Equation (6.1) and (6.2) imply that $\forall w \in W$:

$$T^\dagger w = \left(T|_{(\text{null } T)^\perp} \right)^{-1} P_{\text{range } T} w = T^{-1} I_W w = T^{-1} w.$$

Therefore we have shown that $T^\dagger = T^{-1}$, which proves (a).

PROOF OF (b): By 6.49, we have that $W = \text{range } T \oplus (\text{range } T)^\perp$.

So suppose $w \in \text{range } T$. Then $P_{\text{range } T} w = w$. Hence

$$TT^\dagger w = T \left(T|_{(\text{null } T)^\perp} \right)^{-1} w = P_{\text{range } T} w.$$

Now suppose $w \in (\text{range } T)^\perp$. We know from before that $T^\dagger w = 0$. Now

$$TT^\dagger w = 0 = P_{\text{range } T} w.$$

Thus $TT^\dagger$ and $P_{\text{range } T}$ both agree on $\text{range } T$ and on $(\text{range } T)^\perp$. Hence these two linear maps are equal. So we have proven (b).

PROOF OF (c): By 6.49, it is true that $V = \text{null } T \oplus (\text{null } T)^\perp$.

So let $v \in (\text{null } T)^\perp$. Because $Tv \in \text{range } T$, it is the case that $P_{\text{range } T}(Tv) = Tv$ and therefore

$$T^\dagger(Tv) = \left(T|_{(\text{null } T)^\perp} \right)^{-1} P_{\text{range } T}(Tv) = \left(T|_{(\text{null } T)^\perp} \right)^{-1} (Tv) = v = P_{(\text{null } T)^\perp} v$$

Now let $v \in \text{null } T$. Then

$$T^\dagger T v = 0 = P_{(\text{null } T)^\perp} v,$$

because of 6.57 (c). Thus $T^\dagger T$ and $P_{(\text{null } T)^\perp}$ agree on $(\text{null } T)^\perp$ and $\text{null } T$. Hence these two linear maps are equal. So we have proven (c). $\square$

THM 6.70 (*pseudoinverse provides best approximate solution or best solution*):

Suppose $\dim V \neq \infty$, $T \in \mathcal{L}(V, W)$ and $w \in W$.

(a) If $v \in V$, then

$$\|T(T^\dagger w) - w\| \leq \|Tv - w\|,$$

with equality $\iff v \in T^\dagger w + \text{null } T$.

(b) If $v \in T^\dagger w + \text{null } T$, then

$$\|T^\dagger w\| \leq \|v\|,$$

with equality $\iff v = T^\dagger w$.

PROOF: Suppose $v \in V$. Then with some algebra

$$Tv - w = \underbrace{(Tv - TT^\dagger w)}_{\in \text{range } T} + \underbrace{(TT^\dagger w - w)}_{\in (\text{range } T)^\perp} \quad (6.3)$$

This is because from 6.69 (b) we know that

$$TT^\dagger = P_{\text{range } T}.$$

Thus $Tv - TT^\dagger w \in \text{range } T$. From 6.57 (f) we know that

$$w - TT^\dagger w = w - P_{\text{range } T} w \in (\text{range } T)^\perp \quad \forall w \in W.$$

Of course, because of closedness, the same goes for its negative $TT^\dagger w - w \in (\text{range } T)^\perp \quad \forall w \in W$.

This means, we can use the Pythagorean theorem 6.12 and (6.3) becomes

$$\begin{aligned} \|Tv - w\|^2 &= \|(Tv - TT^\dagger w) + (TT^\dagger w - w)\|^2 \\ &= \underbrace{\|(Tv - TT^\dagger w)\|^2}_{\geq 0} + \|(TT^\dagger w - w)\|^2 \\ &\geq \|(TT^\dagger w - w)\|^2, \end{aligned}$$

with equality only if $Tv - TT^\dagger w = 0 \iff v - T^\dagger w \in \text{null } T$, which is equivalent to $v \in T^\dagger w + \text{null } T$, completing the proof of (a).

PROOF OF (b): Suppose $v \in T^\dagger w + \text{null } T$. This is equivalent to $v - T^\dagger w \in \text{null } T$. Now

$$v = (v - T^\dagger w) + T^\dagger w.$$

The definition of $T^\dagger$ implies that $T^\dagger w \in (\text{null } T)^\perp$. This is because $\left(T|_{(\text{null } T)^\perp}\right)^{-1}$ is a map from W to $(\text{null } T)^\perp \subseteq V$. Thus the Pythagorean theorem implies that $\|T^\dagger w\| \leq \|v\|$, with equality $\iff v = T^\dagger w$. $\square$

OPERATORS ON INNER PRODUCT SPACES (CHAPTER 7)

$\mathbb{F}$ denotes $\mathbb{R}$ or $\mathbb{C}$. V and W are nonzero finite-dimensional inner product spaces over $\mathbb{F}$.

SELF-ADJOINT AND NORMAL OPERATORS (SECTION 7A)

ADJOINTS (SUB-SECTION 7A1)

DEF 7.1 (*adjoint, T^**):

Suppose $T \in \mathcal{L}(V, W)$. The “adjoint” of T is the function $T^* : W \rightarrow V$ such that

$$\underbrace{\langle Tv, w \rangle}_{\text{takes place in } W} = \underbrace{\langle v, T^*w \rangle}_{\text{takes place in } V} \quad \forall v \in V \text{ and } \forall w \in W.$$

RECALL: RIESZ REPRESENTATION THEOREM: For a linear functional $\varphi \in \mathcal{L}(V, \mathbb{F})$ it holds that

$$\exists! v \in V : \varphi(u) = \langle u, v \rangle \quad \forall u \in V.$$

EXAMPLE 7.2 (*adjoint linear map from $\mathbb{R}^3$ to $\mathbb{R}^2$*):

Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$, $T(x_1, x_2, x_3) := (x_2 + 3x_3, 2x_1)$.

$$\begin{aligned} \text{COMPUTE: } \langle T(x_1, x_2, x_3), (y_1, y_2) \rangle &= \langle (x_2 + 3x_3, 2x_1), (y_1, y_2) \rangle \\ &= x_2 y_1 + 3x_3 y_1 + 2x_1 y_2 \\ &= \langle (x_1, x_2, x_3), (2y_2, y_1, 3y_1) \rangle \end{aligned}$$

$$\implies T^*(y_1, y_2) = (2y_2, y_1, 3y_1).$$

EXAMPLE 7.3 (*adjoint of a linear map with range of dimension at most 1*):

Fix $u \in V$ and $x \in W$. Define $T \in \mathcal{L}(V, W)$ by

$$Tv := \langle v, u \rangle x \quad \forall v \in V.$$

Let $\tilde{v} \in V$ and $\tilde{w} \in W$. Then

$$\begin{aligned} \text{COMPUTE: } \langle T\tilde{v}, \tilde{w} \rangle &= \langle \langle \tilde{v}, u \rangle x, \tilde{w} \rangle = \langle \tilde{v}, u \rangle \langle x, \tilde{w} \rangle \\ &= \langle x, \tilde{w} \rangle \langle \tilde{v}, u \rangle = \overline{\langle \tilde{w}, x \rangle} \langle \tilde{v}, u \rangle \\ &= \left\langle \tilde{v}, \overline{\langle \tilde{w}, x \rangle} u \right\rangle = \left\langle \tilde{v}, \langle \tilde{w}, x \rangle u \right\rangle \end{aligned}$$

$$\implies T^*\tilde{w} = \langle \tilde{w}, x \rangle u.$$

THM 7.4 (*adjoint of a linear map is a linear map*):

If $T \in \mathcal{L}(V, W)$, then $T^* \in \mathcal{L}(W, V)$.

PROOF: Let $T \in \mathcal{L}(V, W)$. If $v \in V$ and $w_1, w_2 \in W$, then

$$\begin{aligned} \langle v, T^*(w_1 + w_2) \rangle &= \langle Tv, w_1 + w_2 \rangle \\ &= \langle Tv, w_1 \rangle + \langle Tv, w_2 \rangle \\ &= \langle v, T^*w_1 \rangle + \langle v, T^*w_2 \rangle \\ &= \langle v, T^*(w_1) + T^*(w_2) \rangle \\ \implies T^*(w_1 + w_2) &= T^*(w_1) + T^*(w_2). \end{aligned}$$

If $v \in V$, $\lambda \in \mathbb{F}$, and $w \in W$, then

$$\begin{aligned}\langle v, T^*(\lambda w) \rangle &= \langle Tv, \lambda w \rangle = \overline{\lambda} \langle Tv, w \rangle \\ &= \overline{\lambda} \langle v, T^*w \rangle \\ &= \langle v, \lambda T^*w \rangle.\end{aligned}$$

$$\implies T^*(\lambda w) = \lambda T^*w.$$

Thus T^* is a linear map as desired. □

THM 7.5 (*properties of the adjoint*):

Suppose $T \in \mathcal{L}(V, W)$. Then

- (a) $(S + T)^* = S^* + T^* \quad \forall S \in \mathcal{L}(V, W)$;
- (b) $(\lambda T)^* = \overline{\lambda} T^* \quad \forall \lambda \in \mathbb{F}$;
- (c) $(T^*)^* = T$;
- (d) $(ST)^* = T^* S^* \quad \forall S \in \mathcal{L}(W, U)$ [here U , $\dim U \neq \infty$ is an inner product space over $\mathbb{F}$.]
- (e) $I_V^* = I_V$;
- (f) If T is invertible, then T^* is invertible and $(T^*)^{-1} = (T^{-1})^*$.

PROOF: Let $v \in V$, $w \in W$, $T \in \mathcal{L}(V, W)$ and $\lambda \in \mathbb{F}$. [Where V and W are inner product spaces.]

$$\begin{aligned}\text{(a)} \quad \langle v, (S + T)^*w \rangle &= \langle (S + T)v, w \rangle = \langle Sv + Tv, w \rangle \\ &= \langle Sv, w \rangle + \langle Tv, w \rangle = \langle v, S^*w \rangle + \langle v, T^*w \rangle = \langle v, S^*w + T^*w \rangle\end{aligned}$$

Thus $(S + T)^*w = S^*w + T^*w \implies (S + T)^* = S^* + T^*$, as desired.

$$\begin{aligned}\text{(b)} \quad \langle v, (\lambda T)^*w \rangle &= \langle (\lambda T)v, w \rangle = \lambda \langle Tv, w \rangle \\ &= \lambda \langle v, T^*w \rangle = \langle v, \overline{\lambda} T^*w \rangle\end{aligned}$$

Thus $(\lambda T)^* = \overline{\lambda} T^*$.

$$\text{(c)} \quad \langle v, (T^*)^*w \rangle = \langle T^*v, w \rangle = \overline{\langle w, T^*v \rangle} = \overline{\langle Tw, v \rangle} = \langle v, Tw \rangle. \text{ Thus } (T^*)^* = T.$$

(d) Suppose $S \in \mathcal{L}(W, U)$ and $u \in U$ [where U is an inner product space]. Then

$$\langle v, (ST)^*u \rangle = \langle (ST)v, u \rangle = \langle S(Tv), u \rangle = \langle Tv, S^*u \rangle = \langle v, T^*(S^*u) \rangle = \langle v, (T^*S^*)u \rangle.$$

Therefore, $(ST)^* = T^*S^*$.

$$\text{(e)} \quad \text{Let } v_1, v_2 \in V. \text{ Then } \langle v_1, I_V^*v_2 \rangle = \langle I_Vv_1, v_2 \rangle = \langle v_1, v_2 \rangle = \langle v_1, I_Vv_2 \rangle. \text{ Thus } I^* = I$$

(f) We have:

$$\begin{aligned}T^{-1}T &= I \iff (T^{-1}T)^* = I^* \iff T^*(T^{-1})^* = I, \\ TT^{-1} &= I \iff (TT^{-1})^* = I^* \iff (T^{-1})^*T^* = I.\end{aligned}$$

Hence, $(T^{-1})^* = (T^*)^{-1}$, as desired. □

THM 7.6 (*null space and range of T^**):

If $T \in \mathcal{L}(V, W)$, then

- (a) $\text{null } T^* = (\text{range } T)^\perp$;
- (b) $\text{range } T^* = (\text{null } T)^\perp$;
- (c) $\text{null } T = (\text{range } T^*)^\perp$;
- (d) $\text{range } T = (\text{null } T^*)^\perp$;

PROOF: We begin by proving (a). Let $w \in W$. Then

$$\begin{aligned} w \in \text{null } T^* &\iff T^*w = 0 \\ &\iff \langle v, T^*w \rangle = 0 \quad \forall v \in V \\ &\iff \langle Tv, w \rangle = 0 \quad \forall v \in V \\ &\iff w \in (\text{range } T)^\perp. \end{aligned}$$

Thus $\text{null } T^* = (\text{range } T)^\perp$, proving (a).

If we use 6.52 where $U = (U^\perp)^\perp$ and we take orthogonal complements on both sides in (a) we get

$$\begin{aligned} (\text{null } T^*)^\perp &= ((\text{range } T)^\perp)^\perp \\ (\text{null } T^*)^\perp &= \text{range } T, \end{aligned}$$

which proves (d).

Using 7.5 (c) where $(T^*)^* = T$, replacing T with T^* in (a) will get us

$$\begin{aligned} \text{null } ((T^*)^*) &= (\text{range } T^*)^\perp \\ \text{null } T &= (\text{range } T^*)^\perp \end{aligned}$$

which gives (c).

Finally, replacing T with T^* in (d) gives (b). $\text{range } T^* = (\text{null } (T^*)^*)^\perp \iff \text{range } T^* = (\text{null } T)^\perp$

□

DEF 7.7 (*conjugate transpose, A^**):

For $A \in \mathbb{F}^{m,n}$ we define its “conjugate transpose” $A^* \in \mathbb{F}^{n,m}$ like follows:

If $j \in \{1, \dots, n\}$ and $k \in \{1, \dots, m\}$ then

$$(A^*)_{j,k} := \overline{A_{k,j}}.$$

If a matrix has only real entries, then

$$A^* = A^t.$$

EXAMPLE 7.8:

$$\begin{pmatrix} 2 & 3+4i & 7 \\ 6 & 5 & 8i \end{pmatrix}^* = \begin{pmatrix} 2 & 6 \\ 3-4i & 5 \\ 7 & -8i \end{pmatrix}$$

THM 7.9 (*matrix of T^* equals conjugate transpose matrix of T*):

Let $T \in \mathcal{L}(V, W)$. Suppose $e_1, \dots, e_n$ is a orthonormal basis of V and $f_1, \dots, f_m$ is a orthonormal basis of W . Then

$$\mathcal{M}(T^*, (f_1, \dots, f_m), (e_1, \dots, e_n)) = (\mathcal{M}(T, (e_1, \dots, e_n), (f_1, \dots, f_m)))^*.$$

Or the short version:

$$\mathcal{M}(T^*) = (\mathcal{M}(T))^*.$$

PROOF: From 6.30 (c) we know that for any $v \in V$: $v = \langle v, e_1 \rangle e_1 + \dots + \langle v, e_n \rangle e_n$. This means $\forall k \in \{1, \dots, n\}$:

$$Te_k = \langle Te_k, f_1 \rangle f_1 + \dots + \langle Te_k, f_m \rangle f_m$$

So $\mathcal{M}(T)$ looks as follows

$$\mathcal{M}(T) = \begin{matrix} & e_1 & \cdots & e_k & \cdots & e_n \\ \begin{matrix} f_1 \\ \vdots \\ f_j \\ \vdots \\ f_m \end{matrix} & \left(\begin{array}{cccc} & & \langle T e_k, f_1 \rangle & \\ & & \vdots & \\ & & \langle T e_k, f_j \rangle & \\ & & \vdots & \\ & & \langle T e_k, f_m \rangle & \end{array} \right) \end{matrix}$$

Similarly, $\forall k \in \{1, \dots, m\}$:

$$T^* f_k = \langle T^* f_k, e_1 \rangle e_1 + \cdots + \langle T^* f_k, e_n \rangle e_n.$$

Since $\forall k \in \{1, \dots, m\}$ and $\forall j \in \{1, \dots, n\}$ we have

$$\langle T^* f_k, e_j \rangle = \langle f_k, T e_j \rangle = \overline{\langle T e_j, f_k \rangle},$$

Now $\mathcal{M}(T^*)$ looks as follows

$$\mathcal{M}(T^*) = \begin{matrix} & f_1 & \cdots & f_k & \cdots & f_m \\ \begin{matrix} e_1 \\ \vdots \\ e_j \\ \vdots \\ e_n \end{matrix} & \left(\begin{array}{cccc} & & \langle T^* f_k, e_1 \rangle & \\ & & \vdots & \\ & & \langle T^* f_k, e_j \rangle & \\ & & \vdots & \\ & & \langle T^* f_k, e_n \rangle & \end{array} \right) \end{matrix} = \begin{matrix} & f_1 & \cdots & f_k & \cdots & f_m \\ \begin{matrix} e_1 \\ \vdots \\ e_j \\ \vdots \\ e_n \end{matrix} & \left(\begin{array}{cccc} & & \overline{\langle T e_1, f_k \rangle} & \\ & & \vdots & \\ & & \overline{\langle T e_j, f_k \rangle} & \\ & & \vdots & \\ & & \overline{\langle T e_n, f_k \rangle} & \end{array} \right) \end{matrix}$$

Thus the entry in row j , column k , of $\mathcal{M}(T)$ equals the complex conjugate of the entry in row k , column j , of $\mathcal{M}(T)$. Thus $\mathcal{M}(T^*) = (\mathcal{M}(T))^*$. $\square$

SELF-ADJOINT OPERATORS (SUB-SECTION 7A2)

DEF 7.10 (*self-adjoint*):

An operator $T \in \mathcal{L}(V)$ is called “self-adjoint” if

$$T = T^* \iff \langle T v, w \rangle = \langle v, T w \rangle \quad \forall v, w \in V.$$

If $\mathbb{F} = \mathbb{R}$, then every eigenvalue is real, so the next result is interesting only when $\mathbb{F} = \mathbb{C}$.

THM 7.12 (*eigenvalues of self-adjoint operators*):

Every eigenvalue of a self-adjoint operator is real.

PROOF: Suppose T is a self-adjoint operator on V . Let λ be an eigenvalue of T , and let $v \in V$, $v \neq 0$ such that $T v = \lambda v$. Then

$$\lambda \|v\|^2 = \langle \lambda v, v \rangle = \langle T v, v \rangle = \langle v, T v \rangle = \langle v, \lambda v \rangle = \bar{\lambda} \|v\|^2.$$

Thus $\lambda = \bar{\lambda}$, which means that λ is real, as desired. $\square$

THM 7.13 (*$T v$ orthogonality to v*):

Suppose V is a complex inner product space [$\mathbb{F} = \mathbb{C}$] and $T \in \mathcal{L}(V)$. Then

$$\langle T v, v \rangle = 0 \quad \forall v \in V \iff T = 0.$$

PROOF: $\Rightarrow$ **DIRECTION:** If $u, w \in V$, then

- $\langle T(u + w), u + w \rangle = \langle T u, u \rangle + \langle T w, u \rangle + \langle T u, w \rangle + \langle T w, w \rangle$
- $\langle T(u - w), u - w \rangle = \langle T u, u \rangle - \langle T w, u \rangle - \langle T u, w \rangle + \langle T w, w \rangle$

- $\langle T(u + iw), u + iw \rangle = \langle Tu, u \rangle + i\langle Tw, u \rangle + \bar{i}\langle Tu, w \rangle + i\bar{i}\langle Tw, w \rangle$
 $= \langle Tu, u \rangle + i\langle Tw, u \rangle - i\langle Tu, w \rangle + \langle Tw, w \rangle$
- $\langle T(u - iw), u - iw \rangle = \langle Tu, u \rangle - i\langle Tw, u \rangle - \bar{i}\langle Tu, w \rangle + i\bar{i}\langle Tw, w \rangle$
 $= \langle Tu, u \rangle - i\langle Tw, u \rangle + i\langle Tu, w \rangle + \langle Tw, w \rangle$

That means that

- $\langle T(u + w), u + w \rangle - \langle T(u - w), u - w \rangle = 2\langle Tw, u \rangle + 2\langle Tu, w \rangle$
- $\langle T(u + iw), u + iw \rangle - \langle T(u - iw), u - iw \rangle = 2i\langle Tw, u \rangle - 2i\langle Tu, w \rangle$

Therefore it holds $\forall u, w \in V$:

$$\begin{aligned} & \frac{\langle T(u + w), u + w \rangle - \langle T(u - w), u - w \rangle}{4} + \frac{\langle T(u + iw), u + iw \rangle - \langle T(u - iw), u - iw \rangle}{4} \cdot i \\ &= \frac{1}{2}\langle Tw, u \rangle + \frac{1}{2}\langle Tu, w \rangle + \left(\frac{1}{2}i\langle Tw, u \rangle - \frac{1}{2}i\langle Tu, w \rangle \right) \cdot i \\ &= \frac{1}{2}\langle Tw, u \rangle + \frac{1}{2}\langle Tu, w \rangle - \frac{1}{2}\langle Tw, u \rangle + \frac{1}{2}\langle Tu, w \rangle \\ &= \langle Tu, w \rangle \end{aligned}$$

Note that each term in the numerator of the two big expressions which have the denominator 4 are of the form $\langle Tv, v \rangle$ for appropriate $v \in V$, for example $\langle T(u - iw), u - iw \rangle$, where $u - iw \in V$. Note that this two fractions added together are equal to $\langle Tu, w \rangle \quad \forall u, w \in V$ at the end.

Now suppose $\langle Tv, v \rangle = 0 \quad \forall v \in V$. Then the (big) equation above implies that $\langle Tu, w \rangle = 0 \quad \forall u, w \in V$. We can further conclude that $Tu = 0 \quad \forall u \in V$, because only $\vec{0}$ is always orthogonal to every vector (take $w = Tu$. Only if $w = Tu = \vec{0}$, the expression $\langle Tu, Tu \rangle$ can always be 0). Hence $T = 0$, as desired. Here the last 0 means the zero operator.

$\Leftarrow$ **DIRECTION:** If $T = 0$ then trivially $\langle Tv, v \rangle = 0 \quad \forall v \in V$. □

THM 7.14 (condition for T such that $\langle Tv, v \rangle$ is always real):

Suppose V is a complex inner product space and $T \in \mathcal{L}(V)$. Then

T is self-adjoint $\iff \langle Tv, v \rangle \in \mathbb{R} \quad \forall v \in V$.

PROOF: If $v \in V$, then

$$\langle T^*v, v \rangle = \overline{\langle v, T^*v \rangle} = \overline{\langle Tv, v \rangle}. \quad (7.1)$$

Now

$$\begin{aligned} T \text{ is self-adjoint} &\iff T - T^* = 0 \\ &\iff \langle (T - T^*)v, v \rangle = 0 \quad \forall v \in V \\ &\iff \langle Tv, v \rangle - \langle T^*v, v \rangle = 0 \quad \forall v \in V \\ &\iff \langle Tv, v \rangle - \overline{\langle Tv, v \rangle} = 0 \quad \forall v \in V \quad (\text{which means } \langle Tv, v \rangle = \overline{\langle Tv, v \rangle}) \\ &\iff \langle Tv, v \rangle \in \mathbb{R} \quad \forall v \in V, \end{aligned}$$

where we used 7.13 for the second equivalence as applied to $T - T^*$ and the fourth equivalence follows from (7.1) □

THM 7.16:

Suppose T is a self-adjoint operator on V . Then

$$\langle Tv, v \rangle = 0 \quad \forall v \in V \iff T = 0.$$

PROOF: We have already proved this (without the hypothesis that T is self-adjoint) when V is a complex inner product space. Thus we can assume that V is a real inner product space. If $w, u \in V$,

then

$$\langle Tw, u \rangle = \langle w, Tu \rangle = \overline{\langle Tu, w \rangle} = \langle Tu, w \rangle.$$

Thus we can compute

$$\begin{aligned} \frac{\langle T(u+w), u+w \rangle - \langle T(u-w), u-w \rangle}{4} &= \frac{1}{2} \langle Tw, u \rangle + \frac{1}{2} \langle Tu, w \rangle \\ &= \frac{1}{2} \langle Tu, w \rangle + \frac{1}{2} \langle Tu, w \rangle \\ &= \langle Tu, w \rangle \end{aligned} \quad (7.2)$$

Now suppose $\langle Tv, v \rangle = 0 \quad \forall v \in V$. Because each term on the numerator on the first expression of (7.2) is of the form $\langle Tv, v \rangle$ for appropriate v , this implies $\langle Tu, w \rangle = 0 \quad \forall u, w \in V$. Because only the zero vector is orthogonal to every vector, we have $Tu = 0 \quad \forall u \in V$ [take $w = Tu$].

Hence $T = 0$, as desired. $\square$

This theorem means that for a self-adjoint operator T , it can not happen that $\langle Tv, v \rangle = 0 \quad \forall v \in V$ other than the zero operator. But there are other operators for which this is true, for example a counterclockwise rotation of 90° around the origin; thus $T(x, y) = (-y, x)$. Then we have $\langle T(x, y), (x, y) \rangle = \langle (-y, x), (x, y) \rangle = -yx + xy = 0$.

NORMAL OPERATORS (SUB-SECTION 7A3)

DEF 7.18 (normal): • An operator on an inner product space is called “normal” if it commutes with its adjoint.

- In other words, $T \in \mathcal{L}(V)$ is normal if $TT^* = T^*T$.

Every self-adjoint operator is normal, because if $T^* = T$, then T commutes with T^* .

EXAMPLE 7.19 (an operator that is normal but not self-adjoint):

Let T be the operator on $\mathbb{F}^2$ whose matrix with respect to the standard basis is

$$\begin{pmatrix} 2 & -3 \\ 3 & 2 \end{pmatrix}$$

Thus $T(x_1, x_2) = (2x_1 - 3x_2, 3x_1 + 2x_2)$.

This operator T is not self-adjoint because $\mathcal{M}(T)_{2,1} = 3$ which is not equal to the complex conjugate of the entry in $\mathcal{M}(T)_{1,2} = -3$.

Nonetheless, for the matrix of TT^* we have

$$\begin{pmatrix} 2 & -3 \\ 3 & 2 \end{pmatrix} \begin{pmatrix} 2 & 3 \\ -3 & 2 \end{pmatrix} = \begin{pmatrix} 13 & 0 \\ 0 & 13 \end{pmatrix}$$

Similarly, for the matrix of T^*T we have

$$\begin{pmatrix} 2 & 3 \\ -3 & 2 \end{pmatrix} \begin{pmatrix} 2 & -3 \\ 3 & 2 \end{pmatrix} = \begin{pmatrix} 13 & 0 \\ 0 & 13 \end{pmatrix}$$

Because TT^* and T^*T have the same matrix, we see that $TT^* = T^*T$. Thus T is normal.

THM 7.20 (condition for T to be normal):

Let $T \in \mathcal{L}(V)$. Then

$$T \text{ is normal} \iff \|Tv\| = \|T^*v\| \quad \forall v \in V.$$

$$\begin{aligned}
 \text{PROOF: } T \text{ is normal} &\iff T^*T - TT^* = 0 \\
 &\iff \langle (T^*T - TT^*)v, v \rangle = 0 \quad \forall v \in V \\
 &\iff \langle T^*Tv, v \rangle = \langle TT^*v, v \rangle \\
 &\iff \langle Tv, Tv \rangle = \langle T^*v, T^*v \rangle \quad \forall v \in V \\
 &\iff \|Tv\|^2 = \|T^*v\|^2 \quad \forall v \in V \\
 &\iff \|Tv\| = \|T^*v\| \quad \forall v \in V,
 \end{aligned}$$

where the second equivalence follows from ????. The operator $T^*T - TT^*$ is self-adjoint for every operator $T \in \mathcal{L}(V)$. This is because $\forall v \in V$:

$$\begin{aligned}
 \langle (T^*T - TT^*)v, v \rangle &= \langle T^*Tv, v \rangle - \langle TT^*v, v \rangle \\
 &= \langle v, (T^*T)^*v \rangle - \langle v, (TT^*)^*v \rangle \\
 &= \langle v, T^*(T^*)^*v \rangle - \langle v, (T^*)^*T^*v \rangle \\
 &= \langle v, T^*Tv \rangle - \langle v, TT^*v \rangle \\
 &= \langle v, (T^*T - TT^*)v \rangle
 \end{aligned}$$

One can also just calculate $(T^*T - TT^*)^*$ directly using Theorem 7.5 and verify that $(T^*T - TT^*)^* = (T^*T)^* - (TT^*)^* = T^*T - TT^*$. $\square$

THM 7.21 (*range, null space, and eigenvectors of a normal operator*):

Suppose $T \in \mathcal{L}(V)$ is normal. Then

- (a) $\text{null } T = \text{null } T^*$;
- (b) $\text{range } T = \text{range } T^*$;
- (c) $V = \text{null } T \oplus \text{range } T$;
- (d) $T - \lambda I$ is normal $\forall \lambda \in \mathbb{F}$;
- (e) If $v \in V$ and $\lambda \in \mathbb{F}$, then $Tv = \lambda v \iff T^*v = \bar{\lambda}v$.

PROOF: Suppose $v \in V$, $\lambda \in \mathbb{F}$ and let $T \in \mathcal{L}(V)$ be a normal operator. Then

- (a) $v \in \text{null } T \iff \|Tv\| = 0 \xrightarrow{\text{Theorem 7.20}} \|T^*v\| = 0 \iff v \in \text{null } T^*$.
- (b) $\text{range } T \xrightarrow{\text{Theorem 7.6 (d)}} (\text{null } T^*)^\perp \xrightarrow{\text{(a)}} (\text{null } T)^\perp \xrightarrow{\text{Theorem 7.6 (b)}} \text{range } T^*$.
- (c) $V \xrightarrow{\text{Theorem 6.49}} (\text{null } T) \oplus (\text{null } T)^\perp \xrightarrow{\text{Theorem 7.6}} \text{null } T \oplus \text{range } T^* \xrightarrow{\text{(b)}} \text{null } T \oplus \text{range } T$
- (d) $(T - \lambda I)^*(T - \lambda I) = (T^* - \bar{\lambda}I)(T - \lambda I)$
 $= T^*T - \lambda T^* - \bar{\lambda}T + |\lambda|^2 I$
 $= TT^* - \bar{\lambda}T - \lambda T^* + |\lambda|^2 I$
 $= (T - \lambda I)(T^* - \bar{\lambda}I)$
 $= (T - \lambda I)(T - \lambda I)^*.$

Thus $T - \lambda I$ commutes with its adjoint. Hence $T - \lambda I$ is normal.

- (e) Part (d) tells us that $T - \lambda I$ is normal and we can use 7.20 to yield that

$$\|(T - \lambda I)v\| = \|(T - \lambda I)^*v\| = \|(T^* - \bar{\lambda}I)v\|.$$

Thus $\|(T - \lambda I)v\| = 0 \iff \|(T^* - \bar{\lambda}I)v\| = 0$. Hence $Tv = \lambda v \iff T^*v = \bar{\lambda}v$. $\square$

THM 7.22 (*orthogonal eigenvectors for normal operators*):

Suppose $T \in \mathcal{L}(V)$ is normal. Then eigenvectors of T corresponding to distinct eigenvalues are orthogonal.

PROOF: Suppose α, β are distinct eigenvalues of T , with corresponding eigenvectors $u, v \in V$. Thus

$Tu = \alpha u$ and $Tv = \beta v$. From 7.21 (e) we have $T^*v = \bar{\beta}v$. Now

$$\begin{aligned} (\alpha - \beta)\langle u, v \rangle &= \langle \alpha u, v \rangle - \langle u, \beta v \rangle \\ &= \langle Tu, v \rangle - \langle u, T^*v \rangle \\ &= 0. \end{aligned}$$

Because $\alpha \neq \beta \implies \alpha - \beta \neq 0$, the equation above implies that $\langle u, v \rangle = 0$. Thus u, v are orthogonal, as desired. $\square$

THM 7.23 (T is normal $\iff$ the real and imaginary parts of T commute):

Suppose $\mathbb{F} = \mathbb{C}$ and $T \in \mathcal{L}(V)$. Then T is normal $\iff$ there exist commuting self-adjoint operators A and B such that $T = A + iB$.

PROOF: $\implies$ **DIRECTION:** Let T be normal. Let

$$A := \frac{T + T^*}{2} \text{ and } B := \frac{T - T^*}{2i}.$$

So $T = A + iB$. For A we have $A^* = \left(\frac{T+T^*}{2}\right)^* = \frac{1}{2}(T^* + T) = A$. For B we get following computation:

$$\begin{aligned} B^* &= \overline{\left(\frac{1}{2i}\right)}(T - T^*)^* = \overline{\left(-\frac{i}{2}\right)}(T^* - T) \\ &= \left(\frac{i}{2}\right)(T^* - T) = \left(-\frac{1}{2i}\right)(T^* - T) \\ &= \frac{T - T^*}{2i} = B \end{aligned}$$

So $A = A^*$ and $B = B^*$, therefore A and B are self-adjoint. A quick computation shows that

$$\begin{aligned} AB - BA &= \frac{TT - TT^* + T^*T - T^*T^*}{4i} - \frac{TT + TT^* - T^*T - T^*T^*}{4i} \\ &= \frac{-2TT^* + 2T^*T}{4i} \\ &= \frac{T^*T - TT^*}{2i} \\ &= \frac{T^*T - T^*T}{2i} \quad \text{because } T \text{ is normal} \\ &= 0, \text{ and therefore } AB = BA. \end{aligned}$$

Thus the operators A and B commute, as desired.

$\Leftarrow$ **DIRECTION:** Now suppose there exist commuting self-adjoint operators A and B [$AB = BA$] such that

$$T = A + iB \implies T^* = A - iB$$

Now we can compute A , because $T + T^* = 2A$ and therefore we get the same equation for A as before, namely

$$A = \frac{T + T^*}{2}$$

Therefore, if we subtract the equation for T^* from the equation for A , we get the same equation for B :

$$\begin{aligned} A - A + iB &= \frac{T + T^*}{2} - T^* \\ B &= \frac{T - T^*}{2i} \end{aligned}$$

Since A and B commute, the expression from $AB - BA$ from the first part, which evaluates to $\frac{T^*T - TT^*}{2i}$, is equal to 0, because we assumed $AB = BA$. Therefore, $T^*T = TT^*$, which means

that T is normal, as desired. [We have not used the fact that A and B are self-adjoint in this direction of the proof.] $\square$

SPECTRAL THEOREM (SECTION 7B)

REAL SPECTRAL THEOREM (SUB-SECTION 7B1)

Let $x, b, c \in \mathbb{R}$ and $b^2 < 4c$. Then

$$x^2 + bx + c = \left(x + \frac{b}{2}\right)^2 + \left(c - \frac{b^2}{4}\right) > 0.$$

THM 7.24 (*invertible quadratic expressions*):

Suppose $T \in \mathcal{L}(V)$ is self-adjoint and $b, c \in \mathbb{R}$ are such that $b^2 < 4c$. Then

$$T^2 + bT + cI$$

is an invertible operator.

PROOF: Let $v \in V$, $v \neq 0$, $T \in \mathcal{L}(V)$ and let $b, c \in \mathbb{R}$ such that $b^2 < 4c$. Then

$$\begin{aligned} \langle (T^2 + bT + cI)v, v \rangle &= \langle T^2v, v \rangle + b\langle Tv, v \rangle + c\langle v, v \rangle \\ &= \langle Tv, Tv \rangle + b\langle Tv, v \rangle + c\|v\|^2 \\ &\geq \|Tv\|^2 - |b|\|Tv\|\|v\| + c\|v\|^2 \quad [\text{Cauchy-Schwarz inequality 6.14}] \\ &= \left(\|Tv\| - \frac{|b|\|v\|}{2}\right)^2 + \left(c - \frac{b^2}{4}\right)\|v\|^2 \\ &> 0, \end{aligned}$$

The last inequality implies that $(T^2 + bT + cI)v \neq 0$. Since $v \neq 0$, it follows that $T^2 + bT + cI$ is injective by 3.15, because only $(T^2 + bT + cI)0 = 0$. This also implies that it is invertible by 3.65.

$\square$

THM 7.27 (*minimal polynomial of self-adjoint operator*):

Suppose $T \in \mathcal{L}(V)$ is self-adjoint. Then the minimal polynomial of T equals $(z - \lambda_1) \cdots (z - \lambda_m)$ for some $\lambda_1, \dots, \lambda_m \in \mathbb{R}$.

PROOF: First suppose $\mathbb{F} = \mathbb{C}$. The zeros of the minimal polynomial of T are the eigenvalues of T by 5.27 (a). Because T is self-adjoint, all eigenvalues of T are real by 7.12. Thus the second version of the fundamental theorem of algebra 4.13 tells us that the minimal polynomial of T has the desired form.

Now suppose $\mathbb{F} = \mathbb{R}$. By 4.16 there exist $\lambda_1, \dots, \lambda_m \in \mathbb{R}$ and $b_1, \dots, b_N, c_1, \dots, c_N \in \mathbb{R}$ with $b_k^2 < 4c_k \forall k \in \{1, \dots, N\}$ such that the minimal polynomial of T equals

$$(z - \lambda_1) \cdots (z - \lambda_m) \cdot (z^2 + b_1z + c_1) \cdots (z^2 + b_Nz + c_N). \quad (7.3)$$

Note that it might be the case that $m = 0$ or $N = 0$, meaning that there are not terms of the corresponding form. If $N = 0$, we are done with this proof. Now

$$(T - \lambda_1 I) \cdots (T - \lambda_m I) \cdot (T^2 + b_1 T + c_1 I) \cdots (T^2 + b_N T + c_N I) = 0$$

If $N > 0$, then we could multiply both sides of the equation above on the right by the inverse $(T^2 + b_N T + c_N I)^{-1}$ of $T^2 + b_N T + c_N I$, which is an invertible operator by 7.24. We would then obtain a polynomial expression of T that equals 0. The corresponding polynomial would have a degree 2 less than the degree of (7.3), violating the minimality of the minimal polynomial. Thus we must have $N = 0$, which means that the minimal polynomial in (7.3) has the form $(z - \lambda_1) \cdots (z - \lambda_m)$, as desired. $\square$

The previous result along with 5.27 implies that every self-adjoint operator has an eigenvalue. The next result will show that self-adjoint operators have enough eigenvalues to form a basis. It gives a complete description of self-adjoint operators on a real inner product space and is one of the major theorems in linear algebra.

THM 7.28 (*real spectral theorem*):

Suppose $\mathbb{F} = \mathbb{R}$ and $T \in \mathcal{L}(V)$. Then the following are equivalent.

- (a) T is self-adjoint.
- (b) T has a diagonal matrix with respect to some orthonormal basis of V .
- (c) V has an orthonormal basis consisting of eigenvectors of T .

PROOF: **STEP 1:** First, suppose (a) holds, so T is self-adjoint. Our results on minimal polynomials, specifically

- Every orthonormal list extends to an orthonormal basis (Todo: 6.37)
- minimal polynomial of self-adjoint operator

imply that T has an upper-triangular matrix with respect to some orthonormal basis of V . With respect to this basis, the matrix $\mathcal{M}(T^*)$ of T^* is the transpose of the matrix $\mathcal{M}(T)$ of T :

$$\mathcal{M}(T^*) = \mathcal{M}(T)^t.$$

However, $T^* = T$. Thus $\mathcal{M}(T)^t = \mathcal{M}(T)$. Because $\mathcal{M}(T)$ is upper-triangular, this means that all entries of the matrix above and below the diagonal are 0. Hence $\mathcal{M}(T)$ is a diagonal matrix with respect to the orthonormal basis. Thus (a) implies (b).

STEP 2: Second, suppose (b) holds, so T has a diagonal matrix $\mathcal{M}(T)$ with respect to some orthonormal basis of V . That diagonal matrix $\mathcal{M}(T)$ equals its transpose (i.e. $\mathcal{M}(T) = \mathcal{M}(T)^t$). Thus, with respect to that basis, $T^* = T$, proving that (b) implies (a).

STEP 3: The equivalence of (b) and (c) follows from the definitions. Suppose (b) has a diagonal matrix whose entries are $\lambda_1, \dots, \lambda_m$ with an orthonormal basis $e_1, \dots, e_m$. Hence, $Te_k = \lambda_k e_k$ (for $k = 1, \dots, m$). Therefore, $e_1, \dots, e_m$ are the eigenvectors of T . $\square$

COMPLEX SPECTRAL THEOREM (SUB-SECTION 7B2)

THM 7.29 (*complex spectral theorem*):

Suppose $\mathbb{F} = \mathbb{C}$ and $T \in \mathcal{L}(V)$. Then the following are equivalent.

- (a) T is normal. ($TT^* = T^*T$)
- (b) T has a diagonal matrix with respect to some orthonormal basis of V .
- (c) V has an orthonormal basis consisting of eigenvectors of T .

PROOF: Since $\mathbb{F} = \mathbb{C}$, by Schur's Theorem, there is an orthonormal basis $e_1, \dots, e_n$ of V with respect to which T has an upper-triangular matrix. Thus, we can write

$$\mathcal{M}(T) = \mathcal{M}(T, (e_1, \dots, e_n)) = \begin{pmatrix} a_{1,1} & \cdots & a_{1,n} \\ & \ddots & \vdots \\ 0 & & a_{n,n} \end{pmatrix}. \quad (7.4)$$

We see that

$$\|Te_1\|^2 = \langle a_{1,1}e_1, a_{1,1}e_1 \rangle = a_{1,1}\overline{a_{1,1}}\langle e_1, e_1 \rangle = |a_{1,1}|^2$$

as well as

$$\begin{aligned}\|T^*e_1\|^2 &= \langle (a_{1,1}e_1 + \cdots + a_{1,n}e_n), (a_{1,1}e_1 + \cdots + a_{1,n}e_n) \rangle \\ &= |a_{1,1}|^2 + |a_{1,2}|^2 + \cdots + |a_{1,n}|^2.\end{aligned}$$

STEP 1: Now, suppose (a) holds. Because T is normal, $\|Te_1\|^2 = \|T^*e_1\|^2$, by Theorem 7.20. Thus,

$$|a_{1,1}|^2 = |a_{1,1}|^2 + |a_{1,2}|^2 + \cdots + |a_{1,n}|^2,$$

which implies that all entries in the first row of $\mathcal{M}(T)$, except possibly the first entry $a_{1,1}$, equal 0 (so $a_{1,2} = \cdots = a_{1,n} = 0$).

Now, (7.4) implies that (since $a_{1,2} = 0$)

$$\|Te_2\|^2 = |a_{1,2}|^2 + |a_{2,2}|^2 = |a_{2,2}|^2$$

and

$$\|T^*e_2\|^2 = |a_{2,2}|^2 + |a_{2,3}|^2 + \cdots + |a_{2,n}|^2.$$

Because T is normal, $\|Te_2\| = \|T^*e_2\|$, and therefore

$$|a_{2,2}|^2 = |a_{2,2}|^2 + |a_{2,3}|^2 + \cdots + |a_{2,n}|^2.$$

Thus, the equations above imply that all entries in the second row of $\mathcal{M}(T)$, except possibly for the diagonal entry $a_{2,2}$, equal 0. Continuing in this fashion, we see that all non-diagonal entries in the matrix (7.4) equal 0. Thus (b) holds, completing the proof that (a) $\implies$ (b).

STEP 2: Now suppose (b) holds, so T has a diagonal matrix $\mathcal{M}(T)$ with respect to some orthonormal basis of V . The matrix of T^* (with respect to this basis) is obtained by taking the conjugate transpose of $\mathcal{M}(T)$; hence T^* also has a diagonal matrix. Any two diagonal matrices commute; thus T commutes with T^* , which means that T is normal. In other words, (a) holds, completing the proof that (b) $\implies$ (a).

STEP 3: The equivalence of (b) and (c) follows from the definitions, see 5.55. $\square$

POSITIVE (SEMIDEFINITE) OPERATORS (SECTION 7C)

DEF 7.34 (*positive (semidefinite) operators*):

An operator $T \in \mathcal{L}(V)$ is called “*positive (semidefinite)*”, if T is self-adjoint and

$$\langle Tv, v \rangle \geq 0 \quad \forall v \in V.$$

One can either use the term “*positive operator*” or “*positive semidefinite operator*”, which means the same.

DEF 7.36 (*square root*):

An operator R is called “*square root*” of an operator T if $R^2 = T$.

THM 7.38 (*characterizations of positive operators*):

Let $T \in \mathcal{L}(V)$. Then the following are equivalent.

- (a) T is a positive operator.
- (b) T is self-adjoint and all eigenvalues of T are nonnegative
- (c) With respect to some orthonormal basis of V , the matrix $\mathcal{M}(T)$ of T is a diagonal matrix with only nonnegative numbers on the diagonal.
- (d) T has a positive square root.
- (e) T has a self-adjoint square root.
- (f) $T = R^*R$ for some $R \in \mathcal{L}(V)$.

PROOF: We will prove $(a) \Rightarrow (b) \Rightarrow (c) \Rightarrow (d) \Rightarrow (e) \Rightarrow (f) \Rightarrow (a)$

STEP 1: If (a) holds, that means T is self-adjoint and $\langle Tv, v \rangle \geq 0$ for all $v \in V$. Let $\lambda \in \mathbb{F}$ be an eigenvalue of T with eigenvector $v \neq 0$. Then

$$0 \leq \langle Tv, v \rangle = \langle \lambda v, v \rangle = \lambda \langle v, v \rangle = \lambda \|v\|^2 \implies \lambda \geq 0.$$

Thus (b) holds.

STEP 2: Suppose (b) holds. By the Spectral Theorem, there is an orthonormal basis $e_1, \dots, e_n$ of V consisting of eigenvectors of T . Let $\lambda_1, \dots, \lambda_n$ be the eigenvalues corresponding to $e_1, \dots, e_n$. By (b), each $\lambda_k \geq 0$. So $(b) \implies (c)$.

STEP 3: Now suppose (c) holds. Then there is an orthonormal basis $e_1, \dots, e_n$ of V such that $\mathcal{M}(T) = \text{diag}(\lambda_1, \dots, \lambda_n)$ with $\lambda_k \geq 0$. The linear map lemma Theorem 3.4 implies that $\exists R \in \mathcal{L}(V)$:

$$Re_k = \sqrt{\lambda_k} e_k \in V \quad \forall k \in \{1, \dots, n\}.$$

$$R^2 e_k = RRe_k = R\sqrt{\lambda_k} e_k = \sqrt{\lambda_k} Re_k = \sqrt{\lambda_k} \sqrt{\lambda_k} e_k = \lambda_k e_k = Te_k \quad \forall k \in \{1, \dots, n\}.$$

which implies that $R^2 = T$. Thus R is a positive (semidefinite) square root of T . Hence (d) holds, which shows $(c) \implies (d)$.

STEP 4: Every positive (semidefinite) operator is self-adjoint by the definition of a positive (semidefinite) operator. Thus $(d) \implies (e)$.

STEP 5: Now suppose (e) holds, meaning that there exists a self-adjoint operator $R = R^* \in \mathcal{L}(V)$ on V such that $T = R^2$. Then $T = R^*R$. Hence $(e) \implies (f)$.

STEP 6: Finally, suppose (f) holds. Let $R \in \mathcal{L}(V)$ be such that $T = R^*R$. Then $T^* = (R^*R)^* = R^*(R^*)^* = R^*R = T$. Hence T is self-adjoint. Note that $\forall v \in V$:

$$\langle Tv, v \rangle = \langle R^*Rv, v \rangle = \langle Rv, (R^*)^*v \rangle = \langle Rv, Rv \rangle \geq 0.$$

Thus T is positive (semidefinite), showing that $(f) \implies (a)$. □

THM 7.39 (uniqueness of square root):

Each positive operator on V has a unique positive square root.

DEF 7.40 (notation: $\sqrt{T}$):

For T a positive operator, $\sqrt{T}$ denotes the unique positive square root of T .

THM 7.41:

Suppose T is a positive operator on V and $v \in V$ such that $\langle Tv, v \rangle = 0$.

$$\implies Tv = 0.$$

PROOF: We have $0 = \langle Tv, v \rangle = \langle \sqrt{T}\sqrt{T}v, v \rangle = \langle \sqrt{T}v, \sqrt{T}v \rangle = \|\sqrt{T}v\|^2 \implies \sqrt{T}v = 0$. Thus $Tv = \sqrt{T}(\sqrt{T}v) = 0$, as desired. □

ISOMETRIES, UNITARY OPERATORS, AND MATRIX FACTORIZATION (SECTION 7D)

ISOMETRIES (SUB-SECTION 7D1)

DEF 7.42 (isometry):

$S \in \mathcal{L}(V, W)$ is called an “isometry” if

$$\|Sv\| = \|v\| \quad \forall v \in V.$$

In other words, a linear map is an isometry if it preserves norms.

If $S \in \mathcal{L}(V, W)$ is an isometry and $v \in V$ is such that $Sv = 0$, then

$$\|v\| = \|Sv\| = \|0\| = 0,$$

which implies that $v = 0$. Thus every isometry is injective.

(The Greek word **isos** means “equal”; the Greek word **metron** means “measure”. Thus **isometry** literally means “equal measure”.)

THM 7.49 (*characterizations of isometries*):

Suppose $S \in \mathcal{L}(V, W)$. Suppose $e_1, \dots, e_n$ is an orthonormal basis of V and $f_1, \dots, f_m$ is an orthonormal basis of W . Then the following are equivalent.

- (a) S is an isometry.
- (b) $S^*S = I$.
- (c) $\langle Su, Sv \rangle = \langle u, v \rangle \quad \forall u, v \in V$.
- (d) $Se_1, \dots, Se_n$ is an orthonormal list in W .
- (e) The columns of $\mathcal{M}(S, (e_1, \dots, e_n), (f_1, \dots, f_m))$ form an orthonormal list in $\mathbb{F}^m$ with respect to the Euclidean inner product.

PROOF: We will prove $(a) \Rightarrow (b) \Rightarrow (c) \Rightarrow (d) \Rightarrow (e) \Rightarrow (a)$

“First step:” First suppose (a) holds, so S is an isometry. If $v \in V$, then

$$\langle (I - S^*S)v, v \rangle = \langle v, v \rangle - \langle S^*Sv, v \rangle = \|v\|^2 - \langle Sv, Sv \rangle = \|v\|^2 - \|Sv\|^2 = 0.$$

Since $I - S^*S$ is a self adjoint operator because $(I - S^*S)^* = I - S^*S$, we can use 7.16 to conclude that $I - S^*S$ equals the zero operator. Thus $S^*S = I$, proving $(a) \Rightarrow (b)$.

“Second step:” Now suppose (b) holds, so $S^*S = I$. If $u, v \in V$ then

$$\langle Su, Sv \rangle = \langle S^*Su, v \rangle = \langle Iu, v \rangle = \langle u, v \rangle, \text{ proving } (b) \Rightarrow (c)$$

“Third step:” Now suppose that (c) holds, so $\langle Su, Sv \rangle = \langle u, v \rangle \quad \forall u, v \in V$. Thus if $j, k \in \{1, \dots, n\}$, then

$$\langle Se_j, Se_k \rangle = \langle e_j, e_k \rangle.$$

Hence $Se_1, \dots, Se_n$ is an orthonormal list in W , proving that $(c) \Rightarrow (d)$.

“Fourth step:” Suppose (d) holds, so $Se_1, \dots, Se_n$ is an orthonormal list in W . The k -th column of $A = \mathcal{M}(S, (e_1, \dots, e_n), (f_1, \dots, f_m))$ is the coordinate vector of Se_k with respect to the orthonormal basis $f_1, \dots, f_m$. By Theorem 6.30 (c), the Euclidean inner product of column j and column k equals $\langle Se_j, Se_k \rangle = \delta_{j,k}$. Thus the columns of A form an orthonormal list in $\mathbb{F}^m$, proving $(d) \Rightarrow (e)$.

“Fifth step:” Suppose (e) holds. Let $v \in V$ and write $v = a_1e_1 + \dots + a_n e_n$. Then $Sv = a_1Se_1 + \dots + a_n Se_n$. Since the columns of $\mathcal{M}(S)$ are orthonormal, $Se_1, \dots, Se_n$ is an orthonormal list in W . By Theorem 6.24,

$$\|Sv\|^2 = |a_1|^2 + \dots + |a_n|^2 = \|v\|^2 \Rightarrow \|Sv\| = \|v\|.$$

Thus S is an isometry, completing the proof that $(e) \Rightarrow (a)$. □

UNITARY OPERATORS (SUB-SECTION 7D2)

DEF 7.51 (*unitary operator*):

An operator $S \in \mathcal{L}(V)$ is called “unitary” if S is an invertible isometry.

Note: Every isometry is injective. Every injective operator on a finite-dimensional vector space is invertible. So “unitary” and “isometry” mean the same thing for operators on finite-dimensional

inner product spaces. But remember: A unitary operator maps a vector space to itself, while an isometry maps a vector space to another (possibly different) vector space.

EXAMPLE 7.52 (*rotation of $\mathbb{R}^2$*):

Suppose $\theta \in \mathbb{R}$ and S is the operator on $\mathbb{F}^2$ whose matrix $\mathcal{M}(S)$ with respect to the standard basis of $\mathbb{F}^2$ is

$$\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} = \mathcal{M}(S).$$

The two columns of this matrix form an orthonormal list in $\mathbb{F}^2$. Therefore, S is an isometry by the equivalence of (a) and (e) in Theorem 7.49. Thus S is a unitary operator.

If $\mathbb{F} = \mathbb{R}$, then S is the counterclockwise rotation by θ radians around the origin of $\mathbb{R}^2$. So it preserves norms.

THM 7.53 (*characterizations of unitary operators*):

Suppose $S \in \mathcal{L}(V)$. Suppose $e_1, \dots, e_n$ is an orthonormal basis of V . Then

- (a) S is a unitary operator.
- (b) $S^*S = SS^* = I$.
- (c) S is invertible and $S^{-1} = S^*$.
- (d) $Se_1, \dots, Se_n$ is an orthonormal basis of V .
- (e) The rows of $\mathcal{M}(S, (e_1, \dots, e_n))$ form an orthonormal basis of $\mathbb{F}^n$ with respect to the Euclidean inner product.
- (f) S^* is a unitary operator.

THM 7.54 (*eigenvalues of unitary operators*):

Suppose λ is an eigenvalue of a unitary operator. Then $|\lambda| = 1$.

THM 7.55 (*description of unitary operators on complex inner product spaces*):

Suppose $\mathbb{F} = \mathbb{C}$ and $S \in \mathcal{L}(V)$. Then the following are equivalent.

- (a) S is a unitary operator.
- (b) There is an orthonormal basis of V consisting of eigenvectors of S whose corresponding eigenvalues all have absolute value 1.

QR FACTORIZATION (SUB-SECTION 7D3)

DEF 7.56 (*unitary matrix*):

An n -by- n matrix is called “unitary” if its columns form an orthonormal list in $\mathbb{F}^n$.

THM 7.57 (*characterizations of unitary matrices*):

Let $Q \in \mathbb{F}^{n,n}$. Then the following are equivalent.

- (a) Q is a unitary matrix.
- (b) The rows of Q form an orthonormal list in $\mathbb{F}^n$.
- (c) $\|Qv\| = \|v\| \quad v \in \mathbb{F}^n$.
- (d) $Q^*Q = QQ^* = I$, the n -by- n matrix with 1’s on the diagonal and 0’s elsewhere.

THM 7.58 (*QR factorization*):

Suppose A is a square matrix with linearly independent columns. Then there exist unique matrices Q and R such that Q is unitary, R is upper triangular with only positive numbers on its diagonal, and $A = QR$.

CHOLESKY FACTORIZATION (SUB-SECTION 7D4)**THM 7.61** (*positive invertible operator*):A self-adjoint operator $T \in \mathcal{L}(V)$ is a positive invertible operator $\iff \langle Tv, v \rangle > 0 \quad \forall v \in V, v \neq 0$.**DEF 7.62** (*positive definite*):A matrix $B \in \mathbb{F}^{n,n}$ is called “positive definite” if $B^* = B$ and

$$\langle Bx, x \rangle > 0 \quad \forall x \in \mathbb{F}^n, x \neq 0.$$

THM 7.63 (*Cholesky factorization*):Suppose B is a positive definite matrix. Then there exists a unique upper triangular matrix R with only positive numbers on its diagonal such that

$$B = R^*R.$$

SINGULAR VALUE DECOMPOSITION (SECTION 7E)**SINGULAR VALUES** (SUB-SECTION 7E1)**THM 7.64** (*properties of T^*T*):Suppose $T \in \mathcal{L}(V, W)$. Then

- (a) T^*T is a positive operator on V ;
- (b) $\text{null } T^*T = \text{null } T$;
- (c) $\text{range } T^*T = \text{range } T^*$;
- (d) $\dim \text{range } T = \dim \text{range } T^* = \dim \text{range } T^*T$;

DEF 7.65 (*singular values*):Suppose $T \in \mathcal{L}(V, W)$. The “singular values” of T are the nonnegative square roots of the eigenvalues of T^*T , listed in decreasing order, each included as many times as the dimension of the corresponding eigenspace of T^*T .**THM 7.68** (*role of positive singular values*):Suppose that $T \in \mathcal{L}(V, W)$. Then

- (a) T is injective $\iff 0$ is not a singular value of T ;
- (b) the number of positive singular values of T equals $\dim \text{range } T$;
- (c) T is surjective $\iff$ number of positive singular values of T equals $\dim W$.

The table below compares eigenvalues with singular values.

LIST OF EIGENVALUES	LIST OF SINGULAR VALUES
context: vector spaces	context: inner product spaces
defined only for linear maps from a vector space to itself	defined for linear maps from an inner product space to a possibly different inner product space
can be arbitrary real numbers (if $\mathbb{F} = \mathbb{R}$) or complex numbers (if $\mathbb{F} = \mathbb{C}$)	are nonnegative numbers
can be the empty list if $\mathbb{F} = \mathbb{R}$	length of list equals dimension of domain
includes $0 \iff$ operator is not invertible	includes $0 \iff$ linear map is not injective
no standard order, especially if $\mathbb{F} = \mathbb{C}$	always listed in decreasing order

THM 7.69 (*isometries characterized through singular values*):

Suppose $S \in \mathcal{L}(V, W)$. Then

S is an isometry $\iff$ all singular values of S equal 1.

SVD FOR LINEAR MAPS AND FOR MATRICES (SUB-SECTION 7E2)

THM 7.70 (*singular value decomposition*):

Let $T \in \mathcal{L}(V, W)$ and the positive singular values of T are $s_1, \dots, s_m$. Then there exist two orthonormal lists $e_1, \dots, e_m$ in V and $f_1, \dots, f_m$ in W such that

$$Tv = s_1 \langle v, e_1 \rangle f_1 + \dots + s_m \langle v, e_m \rangle f_m \quad \forall v \in V.$$

DEF 7.74 (*diagonal matrix*):

An M -by- N matrix A is called a “diagonal matrix” if all entries of the matrix are 0 except possibly $A_{k,k}$ for $k \in \{1, \dots, \min\{M, N\}\}$.

The table below compares the spectral theorem with the singular value decomposition

SPECTRAL THEOREM	SINGULAR VALUE DECOMPOSITION
describes only self-adjoint operators (when $\mathbb{F} = \mathbb{R}$) or normal operators (when $\mathbb{F} = \mathbb{C}$)	describes arbitrary linear maps from an inner product space to a possibly different inner product space
produces a single orthonormal basis	produces two orthonormal lists, one for domain space and one for range space, that are not necessarily the same even when range space equals domain space
different proofs depending on whether $\mathbb{F} = \mathbb{R}$ or $\mathbb{F} = \mathbb{C}$	same proof works regardless of whether $\mathbb{F} = \mathbb{R}$ or $\mathbb{F} = \mathbb{C}$

THM 7.75 (*singular value decomposition of adjoint and pseudoinverse*):

Suppose $T \in \mathcal{L}(V, W)$ and the positive singular values of T are $s_1, \dots, s_m$. Suppose $e_1, \dots, e_m$ and $f_1, \dots, f_m$ are orthonormal lists in V and W such that we have the following SVD:

$$Tv = s_1 \langle v, e_1 \rangle f_1 + \dots + s_m \langle v, e_m \rangle f_m \quad \forall v \in V.$$

Then we have $\forall w \in W$:

- $T^*w = s_1 \langle w, f_1 \rangle e_1 + \dots + s_m \langle w, f_m \rangle e_m$
- $T^\dagger w = \frac{\langle w, f_1 \rangle}{s_1} e_1 + \dots + \frac{\langle w, f_m \rangle}{s_m} e_m$

THM 7.76 (*matrix version of SVD*):

Suppose A is a p -by- n matrix of rank $m \geq 1$. Then there exists a p -by- m matrix B with orthonormal columns, an m -by- m diagonal matrix D with positive numbers on the diagonal, and an n -by- m matrix C with orthonormal columns such that

$$A = BDC^*.$$

CONSEQUENCES OF SINGULAR VALUE DECOMPOSITION (SECTION 7F)

NORMS OF LINEAR MAPS (SUB-SECTION 7F1)

THM 7.82 (*upper bound for $\|Tv\|$*):

Suppose $T \in \mathcal{L}(V, W)$. Let s_1 be the largest singular value of T . Then

$$\|Tv\| \leq s_1 \|v\| \quad \forall v \in V.$$

DEF 7.86 (*norm of a linear map*):

Suppose $T \in \mathcal{L}(V, W)$. Then the “norm” of T , denoted by $\|T\|$, is defined by

$$\|T\| := \max\{\|Tv\| \mid v \in V \text{ and } \|v\| \leq 1\}.$$

THM 7.87 (*basic properties of norms of linear maps*):

Let $T \in \mathcal{L}(V, W)$. Then

- (a) $\|T\| \geq 0$;
- (b) $\|T\| = 0 \iff T = 0$;
- (c) $\|\lambda T\| = |\lambda| \|T\| \quad \forall \lambda \in \mathbb{F}$;
- (d) $\|S + T\| \leq \|S\| + \|T\| \quad \forall S \in \mathcal{L}(V, W)$;

THM 7.88 (*alternative formulas for $\|T\|$*):

Let $T \in \mathcal{L}(V, W)$. Then

- (a) $\|T\|$ = the largest singular value of T ;
- (b) $\|T\| = \max\{\|Tv\| \mid v \in V \text{ and } \|v\| = 1\}$;
- (c) $\|T\|$ = the smallest number c such that $\|Tv\| \leq c\|v\| \quad \forall v \in V$.

THM 7.89 (*norm of the adjoint*):

Suppose $T \in \mathcal{L}(V, W)$. Then $\|T^*\| = \|T\|$.

APPROXIMATION BY LINEAR MAPS WITH LOWER-DIMENSIONAL RANGE (SUB-SECTION 7F2)

THM 7.92 (*best approximation by linear map whose range has dimension $\leq k$*):

Let $T \in \mathcal{L}(V, W)$ and $s_1 \geq \dots \geq s_m$ are positive singular values of T . Suppose $1 \leq k < m$. Then

$$\min\{\|T - S\| \mid S \in \mathcal{L}(V, W) \text{ and } \dim \text{range } S \leq k\} = s_{k+1}.$$

Furthermore, if

$$Tv = s_1 \langle v, e_1 \rangle f_1 + \dots + s_m \langle v, e_m \rangle f_m$$

is a singular value decomposition of T and $T_k \in \mathcal{L}(V, W)$ is defined by

$$T_k v = s_1 \langle v, e_1 \rangle f_1 + \dots + s_k \langle v, e_k \rangle f_k \quad \forall v \in V,$$

then $\dim \text{range } T_k = k$ and $\|T - T_k\| = s_{k+1}$.

THM 7.93 (*polar decomposition*):

Suppose $T \in \mathcal{L}(V)$. Then there exists a unitary operator $S \in \mathcal{L}(V)$ such that

$$T = S\sqrt{T^*T}.$$

Furthermore, if T is invertible, then this unitary operator S is unique.

OPERATORS APPLIED TO ELLIPSOIDS AND PARALLELEPIPEDS (SUB-SECTION 7F3)

DEF 7.95 (*ball, B*):

The “ball” in V of radius 1 centered at 0, denoted by B , is defined by

$$B := \{v \in V \mid \|v\| < 1\}.$$